CONTENTS

Preface vii

The list of authors ix

I Free Algebras and Functional Representation
Stefano Aguzzoli, Simone Bova, and Brunella Gerla xiii

1 Free Algebras . . . xiii
2 MV-algebras and Related Structures . . . xviii
2.1 McNaughton's Theorem . . . xviii
2.2 Wajsberg Hoops . . . xxx
2.3 Finitely Valued Łukasiewicz Logics . . . xxxiii
2.4 Extensions of MV-algebras . . . xxxv
2.5 MV-algebras with Division Operators . . . xxxviii
3 Product Algebras . . . xxxix
3.1 Free Cancellative Hoops . . . xxxix
3.2 Free Product Algebras . . . xl
4 Gödel Algebras and Related Structures . . . xliii
4.1 Functional Representation of Gödel Algebras . . . xliii
4.2 Finite Spectral Duality for Gödel Algebras . . . xlvii
4.3 Gödel Hoops . . . lv
4.4 Finite-valued Gödel Algebras . . . lv
4.5 Nilpotent Minimum Algebras . . . lvi
4.6 Functional Representation of Free NM-algebras . . . lxi
5 BL-algebras and Related Structures . . . lxiv
5.1 Free $\mathrm{BL}_{\triangle}$-algebras . . . lxv
5.2 Free BL-algebras . . . lxviii
5.3 SBL-algebras, and Some Revisitations . . . lxxviii
5.4 Locally Finite Varieties of BL-algebras . . . lxxix
6 Pierce Representation . . . lxxxiv
7 Open Problems . . . lxxxviii
8 Historical and Bibliographical Remarks . . . lxxxix

Preface

some great text

Petr Cintula, Petr Hájek, Carles Noguera
the editors

The list of authors

STEFANO AGUZZOLI
Dipartimento di Scienze dell'Informazione
University of Milano
via Comelico 39/41
I-20135 Milano MI, Italy
Email: aguzzoli@dsi.unimi.it

MATTHIAS BAAZ
Institute for Discrete Mathematics and Geometry (E104)
Vienna University of Technology
Wiedner Hauptstrasse 8-10 A-1040 Wien, Austria
Email: baaz@logic.at

LIBOR BĚHOUNEK
Institute of Computer Science
Academy of Sciences of the Czech Republic
Pod Vodárenskou věží 2
182 07 Prague 8, Czech Republic
Email: behounek@cs.cas.cz

SIMONE BOVA
Department of Mathematics
Vanderbilt University
1326 Stevenson Center
Nashville, TN 37240-001
Email: simone.bova@vanderbilt.edu

MANUELA BUSANICHE
Instituto de Matemática Aplicada del Litoral
FIQ, CONICET-UNL
Guemes 3450
S3000GLN-Santa Fe, Argentina
Email: manuelabusaniche@yahoo.com.ar

PETR CINTULA
Institute of Computer Science
Academy of Sciences of the Czech Republic
Pod Vodárenskou věží 2
182 07 Prague 8, Czech Republic
Email: cintula@cs.cas.cz

ANTONIO DI NOLA
Department of Mathematics and Computer Science
Universita degli Studi di Salerno
Via Ponte Don Melillo
84084 Fisciano (SA) Italy
Email: adinola@unisa.it

FRANCESC ESTEVA
Artificial Intelligence Research Institute (IIIA), CSIC
Campus de la Universitat Autònoma de Barcelona s/n
08193 Bellaterra, Catalonia, Spain
Email: esteva@iiia.csic.es

BRUNELLA GERLA
Dipartimento di Informatica e Comunicazione
Universita dell'Insubria
Via Mazzini, 5
21100 Varese, Italy
Email: brunella.gerla@uninsubria.it

LLUÍS GODO
Artificial Intelligence Research Institute (IIIA), CSIC
Campus de la Universitat Autònoma de Barcelona s/n
08193 Bellaterra, Catalonia, Spain
Email: godo@iiia.csic.es

PETR HÁJEK
Institute of Computer Science
Academy of Sciences of the Czech Republic
Pod Vodárenskou věží 2
182 07 Prague 8, Czech Republic
Email: hajek@cs.cas.cz

ZUZANA HANIKOVÁ
Institute of Computer Science
Academy of Sciences of the Czech Republic
Pod Vodárenskou věží 2
182 07 Prague 8, Czech Republic
Email: zuzana@cs.cas.cz

ROSTISLAV HORČÍK
Institute of Computer Science
Academy of Sciences of the Czech Republic
Pod Vodárenskou věží 2
182 07 Prague 8, Czech Republic
Email: horcik@cs.cas.cz

IOANA LEUSTEAN
Faculty of Mathematics and Computer Science
University of Bucharest
Str. Academiei 14, sector 1,
C.P. 010014, Bucharest, Romania
Email: ioana@funinf.cs.unibuc.ro

GEORGE METCALFE
Mathematical Institute
University of Bern
Sidlerstrasse 5 und Alpeneggstrasse 22
CH-3012 Bern, Switzerland
Email: george.metcalfe@math.unibe.ch

FRANCO MONTAGNA
Department of Mathematics and Computer Science,
Pian dei Mantellini 44,
53100 Siena, Italy
Email: montagna@unisi.it

CARLES NOGUERA
Artificial Intelligence Research Institute (IIIA), CSIC
Campus de la Universitat Autònoma de Barcelona s/n
08193 Bellaterra, Catalonia, Spain
Email: cnoguera@iiia.csic.es

NORBERT PREINING
Research Center for Integrated Science
Japan Advanced Institute for Science and Technology
1-1 Asahidai, Nomi
Ishikawa 923-1292 Japan
Email: preining@logic.at

Chapter I: Free Algebras and Functional Representation

STEFANO AGUZZOLI, SIMONE BOVA, AND BRUNELLA GERLA

In this chapter we deal with concrete representations of free algebras in varieties that constitute the equivalent algebraic semantics of some prominent schematic extensions of BL and MTL. The chapter is organised as follows. In the first section we recall the notion of free algebra and stress its importance for logic. In the second section we shall deal with free MV-algebras, and related structures. In the third one, the subject are free Product algebras, while the fourth is concerned with Gödel algebras and other subvarieties of WNM-algebras. The fifth section is concerned with free BL-algebras and related structures, while the sixth section deals with Pierce representations of free algebras in some subvarieties of MTL-algebras. We conclude the chapter with a section on open problems, and a final one on historical and bibliographical remarks.

1 Free Algebras

In this section we recall some basic notions about free algebras in a variety, and their relationship with logic.

DEFINITION 1.0.1. *Let $\mathbb{K}$ be a class of algebras of one signature, and let $\boldsymbol{A}$ be an algebra of the same signature. Let $X \subseteq A$. Then $\boldsymbol{A}$ has the* universal mapping property for $\mathbb{K}$ over X *if for every $\boldsymbol{B} \in \mathbb{K}$ and any map $f\colon X \to B$ there is a homomorphism $h\colon \boldsymbol{A} \to \boldsymbol{B}$ extending f, that is $h(x) = f(x)$ for all $x \in X$.*

$\boldsymbol{A}$ is free for $\mathbb{K}$ over X *if $\boldsymbol{A}$ is generated by X and $\boldsymbol{A}$ has the universal mapping property for $\mathbb{K}$ over X. $\boldsymbol{A}$ is* free in $\mathbb{K}$ over X *if, in addition, $\boldsymbol{A} \in \mathbb{K}$. In this case X is called a set of* (free) generators *for $\boldsymbol{A}$. If $\mathbb{K}$ is a variety we shall call $\boldsymbol{A}$ the* free $\mathbb{K}$-algebra over X.

Note that if $\mathbb{K}$ is a variety then every free $\mathbb{K}$-algebra is in $\mathbb{K}$.

Since a homomorphism $\boldsymbol{A} \to \boldsymbol{B}$ is completely determined by how it maps the generators of $\boldsymbol{A}$, it follows that the homomorphism h extending f in Definition 1.0.1 is necessarily unique.

In the following lemma we collect some well-known properties of algebras free in a class.

LEMMA 1.0.2. *Assume $\boldsymbol{A}_1$ and $\boldsymbol{A}_2$ are free in a class of algebras $\mathbb{K}$, respectively over X_1 and X_2. If $|X_1| = |X_2|$ then $\boldsymbol{A}_1 \cong \boldsymbol{A}_2$. This fact allows to speak of $\boldsymbol{A}_1$ (or $\boldsymbol{A}_2$) as* the algebra free in $\mathbb{K}$ over $|X_1|$ many generators.

The free $\mathbb{V}$-algebra over κ many generators exists for any cardinal κ and any variety $\mathbb{V}$ containing non-trivial members.

In the sequel of this chapter we shall only consider varieties containing non-trivial members. For any variety $\mathbb{V}$, we shall denote $\boldsymbol{F}_{\mathbb{V}}^{\kappa}$ the free $\mathbb{V}$-algebra over κ generators.

DEFINITION 1.0.3. *Let $\mathbb{K}$ be a class of algebras and $\boldsymbol{A}$ be an algebra of the same signature. We define the congruence $\Theta_{\boldsymbol{A}}(\mathbb{K})$ as*

$$\Theta_{\boldsymbol{A}}(\mathbb{K}) = \bigcap \{\theta \mid \theta \in Con(\boldsymbol{A}),\ \boldsymbol{A}/\theta \in \mathbf{S}(\mathbb{K})\}.$$

LEMMA 1.0.4. *If $\boldsymbol{A}$ is free for $\mathbb{K}$ over X then the algebra $\boldsymbol{A}/\Theta_{\boldsymbol{A}}(\mathbb{K})$ is free for $\mathbb{K}$ over the set $X/\Theta_{\boldsymbol{A}}(\mathbb{K}) = \{x/\Theta_{\boldsymbol{A}}(\mathbb{K}) \mid x \in X\}$. Moreover $\boldsymbol{A}/\Theta_{\boldsymbol{A}}(\mathbb{K}) \in \mathbf{SP}(\mathbb{K})$.*

Lemma 1.0.4 allows to recover free algebras in a variety $\mathbb{V}$ from the *absolutely free algebras* in the class of all algebras of the same signature as $\mathbb{V}$.

DEFINITION 1.0.5. *Let $\mathbb{K}$ be the class of all algebras of some fixed signature. Then an algebra free in $\mathbb{K}$ over a set X is called* absolutely free *over X.*

LEMMA 1.0.6. *Let $\mathbb{K}$ be the class of all algebras of some fixed signature and let $\mathbb{H} \subseteq \mathbb{K}$. If $\boldsymbol{A}$ is absolutely free in $\mathbb{K}$ over X then $\boldsymbol{A}$ is free for $\mathbb{H}$ over X.*

Let $\boldsymbol{A}$ be an algebra, and φ a term on the signature of $\boldsymbol{A}$ whose individual variables are in $\{x_1, \ldots, x_n\}$. Then, throughout the chapter, for any $\langle a_1, \ldots, a_n\rangle \in \boldsymbol{A}^n$ we denote

$$\varphi^{\boldsymbol{A}}(a_1, \ldots, a_n)$$

the element of $\boldsymbol{A}$ obtained by substituting uniformly over φ each occurrence of x_i with a_i for all $i \in \{1, 2, \ldots, n\}$, and interpreting each operation symbol in φ with the corresponding operation of $\boldsymbol{A}$. The function mapping each $\langle a_1, \ldots, a_n\rangle \in \boldsymbol{A}^n$ to $\varphi^{\boldsymbol{A}}(a_1, \ldots, a_n) \in \boldsymbol{A}$ will be denoted $\varphi^{\boldsymbol{A}}$.

DEFINITION 1.0.7. *Let $\mathbb{K}$ be the class of all algebras of some fixed signature, and let $\mathbb{K}_0$ be the set of constants in this signature. Let further X be a set disjoint from $\mathbb{K}_0$. Then the* term algebra *$\boldsymbol{T}_{\mathbb{K}}(X)$ is the algebra whose universe is the smallest set $T(X)$ such that it contains the set of terms $X \cup \mathbb{K}_0$ and for any function symbol f in the signature, if f is n-ary, then for all $t_1, t_2, \ldots, t_n \in T(X)$, the term $f(t_1, t_2, \ldots, t_n)$ is in $T(X)$. The operations in $\boldsymbol{T}_{\mathbb{K}}(X)$ are defined as follows. For any n-ary function symbol f in the signature, and all $t_1, \ldots, t_n \in T(X)$,*

$$f^{\boldsymbol{T}_{\mathbb{K}}(X)}(t_1, \ldots, t_n) = f(t_1, \ldots, t_n)\,.$$

LEMMA 1.0.8. *Let $\mathbb{K}$ be the class of all algebras of some fixed signature. Then $\boldsymbol{T}_{\mathbb{K}}(X)$ is absolutely free in $\mathbb{K}$ over X. Hence, if $\mathbb{H} \subseteq \mathbb{K}$ then $\boldsymbol{T}_{\mathbb{K}}(X)/\Theta_{\boldsymbol{T}_{\mathbb{K}}(X)}(\mathbb{H})$ is free for $\mathbb{H}$ over $X/\Theta_{\boldsymbol{T}_{\mathbb{K}}(X)}(\mathbb{H})$.*

Definitions 1.0.3,1.0.5,1.0.7 and Lemmas 1.0.4,1.0.6,1.0.8 show that free algebras in a variety over κ generators do not satisfy any additional equations, written on the set of variables $\{v_\alpha\}_{\alpha \leq \kappa}$, but those holding in every algebra in the variety. More precisely,

every algebra $\boldsymbol{A}$ in a variety $\mathbb{V}$ is presented by a pair $\langle X, E\rangle$, where X is a set of generators, and E is a set of $\mathbb{V}$-equations whose variables are elements of X. An algebra is free in $\mathbb{V}$ over κ generators if it is presented by

$$\langle \{v_\alpha\}_{\alpha\leq\kappa}, \emptyset\rangle\,.$$

Moreover, the following holds.

LEMMA 1.0.9. *Each algebra in a variety $\mathbb{V}$ belongs to $\mathbf{H}(\boldsymbol{F}^\kappa_{\mathbb{V}})$ for some κ.*

We recall that for any algebraizable (in the sense defined in Chapter I) extension L of BL (or, *mutatis mutandis*, of MTL), the Lindenbaum algebra of the pure logic L over a set of variables V is the algebra obtained by quotienting the absolutely free algebra $\boldsymbol{T}_{\mathbb{K}}(V)$ (where $\mathbb{K}$ is the class of all algebras whose signature is the language of L) by the congruence $\equiv$ given by $\varphi\equiv\psi$ iff L proves both $\varphi\to\psi$ and $\psi\to\varphi$. See [20] for further background on algebraic logic and Lindenbaum algebras. The importance for logic of free algebras resides in their connection with Lindenbaum algebras, as stressed in the following Lemma.

LEMMA 1.0.10. *Let L be a logic and $\mathbb{V}$ be its equivalent algebraic semantics. Then the Lindenbaum algebra of L is isomorphic to $\boldsymbol{F}^\omega_{\mathbb{V}}$, the free $\mathbb{V}$-algebra over denumerably many generators. Moreover, the Lindenbaum algebra of the L-formulas written over the set $\{v_i\}_{i=1}^n$ of propositional variables is isomorphic to $\boldsymbol{F}^n_{\mathbb{V}}$, for each integer $n\geq 0$.*

Thus, knowledge of the fine structure of free algebras in a variety $\mathbb{V}$ often translates to knowledge of the fine structure of the logic whose algebraic semantics is given by $\mathbb{V}$. When available, a concrete, intrinsic representation of free algebras is then a very powerful tool for the study of the associated logic. Unfortunately, finding concrete representations of free algebras is usually a highly non-trivial task. In this chapter we shall deal with such representations, and in particular with those that represent free algebras as algebras of functions over powers of the real unit interval $[0,1]$. We call this kind of results a *functional representation*. A useful byproduct of concrete representation theorems are normal forms: as a matter of fact a main ingredient in these theorems consists in finding a uniform way to represent elements of the free algebras, that is, a uniform way to pick a representative element from each one of the equivalence classes of formulas which constitute the Lindenbaum algebras of the logic at hand.

A most useful tool for building concrete representation of free algebras is given by the following Lemma.

DEFINITION 1.0.11. *An algebra $\boldsymbol{A}$ is* generic *for a variety $\mathbb{V}$ if*

$$\mathbb{V}=\mathbf{HSP}(\{\boldsymbol{A}\})\,.$$

Note that $\boldsymbol{F}^\omega_{\mathbb{V}}$ is generic for $\mathbb{V}$.

LEMMA 1.0.12. *Let $\boldsymbol{A}$ be generic for a variety $\mathbb{V}$. Then, for each cardinal κ the free $\mathbb{V}$-algebra over κ generators is isomorphic to the subalgebra of $\boldsymbol{A}^{\boldsymbol{A}^\kappa}$ generated by the projections $\langle a_1, a_2, \ldots, a_\alpha, \ldots\rangle \in \boldsymbol{A}^\kappa \mapsto a_\alpha$, for each $\alpha \leq \kappa$. Further, let $\boldsymbol{A}$ be generic for the subvariety of $\mathbb{V}$ generated by all κ-generated algebras in $\mathbb{V}$. Then again, $\boldsymbol{F}^\kappa_{\mathbb{V}}$ is isomorphic to the subalgebra of $\boldsymbol{A}^{\boldsymbol{A}^\kappa}$ generated by the projections.*

Since each algebraic term is a finite string, for each infinite cardinal κ the free $\mathbb{V}$-algebra over κ generators is the algebra of all functions $f\colon \boldsymbol{A}^\kappa \to \boldsymbol{A}$ such that there exist an integer $m \geq 0$ and ordinals $\alpha_1 < \alpha_2 < \cdots < \alpha_m < \kappa$ and a function $g \in \boldsymbol{F}^m_{\mathbb{V}}$ such that for each $\langle a_1, a_2, \ldots, a_\alpha, \ldots\rangle \in \boldsymbol{A}^\kappa$,

$$f(a_1, a_2, \ldots, a_\alpha, \ldots) = g(a_{\alpha_1}, a_{\alpha_2}, \ldots, a_{\alpha_m})\,.$$

This observation justifies the fact that in the rest of the chapter we shall deal only with finitely generated free algebras, unless we explicitly state this is not the case.

Lemma 1.0.12 is really useful in the algebraic study of schematic extensions of BL or MTL, in particular for those extensions L enjoying *canonical* standard completeness, that is, the variety which constitutes the algebraic semantics of L contains, up to isomorphism, a unique standard algebra. In this case, if we single out one such standard algebra $\boldsymbol{A} = \langle [0,1], \odot, \to, \wedge, 0\rangle$, then the free n-generated algebra is the algebra of all functions $f\colon [0,1]^n \to [0,1]$ constructible from the projections by means of pointwise defined operations: that is, the zero of the algebra is the constant function 0 and for each binary operation $* \in \{\odot, \to, \wedge\}$,

$$(f * g)(a_1, \ldots, a_n) = f(a_1, \ldots, a_n) * g(a_1, \ldots, a_n)\,.$$

It is worth to stress that even the availability of the aforementioned form of standard completeness, or of a standard algebra that is generic for the variety under consideration, does not render the task of describing the concrete structure of the free algebras a trivial one. Good examples supporting this statement are the concrete representation of free MV-algebras (see Section 2.1) and of free BL-algebras (see Section 5.2).

As is well known, each variety as a category is complete and co-complete. This fact allows to express each free algebra as a suitable co-power of the singly generated free algebra. As a matter of fact, if $\{\boldsymbol{A}_i\}_{i\in I}$ is a family of algebras in a variety $\mathbb{V}$, such that $\{\boldsymbol{A}_i\}$ is presented by $\langle X_i, E_i\rangle$ for all $i \in I$, then their coproduct $\coprod_{i\in I} \boldsymbol{A}_i$ is the algebra in $\mathbb{V}$ presented by $\langle \biguplus_{i\in I} X_i, \biguplus_{i\in I} E_i\rangle$, where $\uplus$ denotes disjoint union, and each set of equations E_j uses only variables from the image of the injection of X_j into $\biguplus_{i\in I} X_i$. Recall now that $\boldsymbol{F}^1_{\mathbb{V}}$ is presented by $\langle \{v\}, \emptyset\rangle$. Then the following holds.

LEMMA 1.0.13. *In each variety $\mathbb{V}$, the κ-generated free $\mathbb{V}$-algebra is the co-power of κ copies of the singly generated free $\mathbb{V}$-algebra:*

$$\boldsymbol{F}^\kappa_{\mathbb{V}} = \coprod^{\kappa} \boldsymbol{F}^1_{\mathbb{V}}\,.$$

The preceding Lemma together with Lemma 1.0.12, are the most useful tools to build free algebras that we shall use in this chapter. In particular Lemma 1.0.13 is used in combination with finite spectral dualities. If a dual category $\mathbb{V}^{\mathrm{op}}$ to a variety $\mathbb{V}$ is available, we can try to work on the dual side, and express coproducts, which are notoriously hard to compute, as the dual of products in the dual category. If products in $\mathbb{V}^{\mathrm{op}}$ are easily handled, then what is left to do is just to characterise the singly generated free algebra in $\mathbb{V}$ and its dual. This approach is amazingly simple to apply for some locally finite subvarieties of MTL, as the objects dual to algebras turn out to be combinatorially defined finite partially ordered sets.

We close this section with a few words on free Boolean algebras, and functional (in)completeness.

As the algebra $\{0,1\}$ is generic for the variety of Boolean algebras, the free n-generated Boolean algebra is the algebra of functions $f\colon \{0,1\}^n \to \{0,1\}$ generated by the projections, equipped with pointwise defined operations. As is well-known this algebra is the algebra of *all* functions $f\colon \{0,1\}^n \to \{0,1\}$. One way to prove this fact is to realize each such function, also called a *truth table*, either as a disjunction of minterms or a conjunction of maxterms, where a minterm is a conjunction of variables and negated variables, and dually, a maxterm is a disjunction of variables and negated variables. This fact is referred to as the *functional completeness* of Boolean logic. A trivial argument on cardinalities shows that functional completeness is not enjoyed by standard complete logics. However, those logics whose equivalent semantics is a locally finite variety are amenable to the definition of conjunctive and disjunctive normal forms, for suitably defined notions of generalised minterms and maxterms. This is shown to be the case for Gödel logics (see Section 4), and for logics that are schematic extension of WNM (Section 4.5). A similar approach is not conceivable for logics whose equivalent semantics is not locally finite, as it is the case for Łukasiewicz, Product, BL, SBL and MTL logics, among others. In these cases a more sophisticated theory is needed in order to develop normal-form like results. An instance of this kind of approach is given by the theory of *Schauder hats* for MV-algebras, the algebraic semantics of Łukasiewicz logics.

As a last remark we point out that for any schematic extension L of BL, if we let $\mathbb{V}$ denote the variety constituting the algebraic semantics of L then

$$\boldsymbol{F}^0_{\mathbb{V}} \cong \{0,1\}\,.$$

In the sequel we shall then focus on n-generated free algebras for integers $n > 0$, only.

We fix some notation we use throughout this Chapter. Apart from the conjunction connective, that we denote $\&$, we shall blur the distinction between a connective and the algebraic operation interpreting it in standard and free MV-algebras and related structures. Then, unless otherwise stated, we use $\odot$ to denote the Łukasiewicz t-norm and the monoidal conjunction operation in free MV-algebras and related structures. For what regards the other operations, unless otherwise stated, we shall use $\oplus$ for the monoidal disjunction, $\to$ for the residuum of $\odot$, $\neg$ for the negation, $\vee$ and $\wedge$ for the lattice operations, and $\ominus$ for truncated difference, that is, $x \ominus y = x \odot \neg y$. Further, for any

element a in an MV-algebra we denote ma and a^m, the element $a \oplus a \oplus \cdots \oplus a$ and $a \odot a \odot \cdots \odot a$, respectively, where in both expressions, a occurs m many times. We shall use the same convention for MV-terms φ and write $m\varphi$ and φ^m for, respectively, the m-fold monoidal conjunction and disjunction of φ with itself.

For each set A, each subset $K = \{j_1, j_2, \ldots, j_k\} \subseteq \{1, 2, \ldots, n\}$, and each $\langle a_1, \ldots, a_n\rangle \in A^n$, we denote $\pi_K(a_1, \ldots, a_n) = \langle a_{j_1}, a_{j_2}, \ldots, a_{j_k}\rangle$ the *K-projection* of $\langle a_1, \ldots, a_n\rangle$. We shall sometimes write π_k instead of $\pi_{\{k\}}$.

2 MV-algebras and Related Structures

We recall that the variety $\mathbb{MV}$ of MV-algebras constitutes the equivalent algebraic semantics of Łukasiewicz logic. In this section we introduce McNaughton's result about the functional representation of free MV-algebras. Then we deal with representation theorems for related logics, such as the logic of Wajsberg hoops, finitely valued Łukasiewicz logics and extensions/expansions of Łukasiewicz logic with division operators.

We start introducing some structures whose elements are polynomials in n indeterminates. As we shall see, these polynomials, the linear ones in particular, play a major role in the functional representation of free MV-algebras and related structures.

DEFINITION 2.0.14. *Let $\mathsf{Z}[x_1, \ldots, x_n]$ be the domain of polynomials in n indeterminates and integer coefficients, $\mathsf{Q}(x_1, \ldots, x_n)$ its field of fractions and $\mathsf{Z}_1[x_1, \ldots, x_n]$ its submodule of polynomials of degree at most* 1.

Note that a generic member $p \in \mathsf{Z}_1[x_1, \ldots, x_n]$ has the form

$$p(x_1, \ldots, x_n) = b + \sum_{i=1}^{n} a_i x_i$$

for some $a_1, \ldots, a_n, b \in \mathsf{Z}$.

2.1 McNaughton's Theorem

Łukasiewicz logic enjoys standard completeness. In particular, the variety $\mathbb{MV}$ is generated by the standard MV-algebra

$$\langle [0,1], \oplus, \neg, 0\rangle\,,$$

where $x \oplus y = \min\{1, x + y\}$ and $\neg x = 1 - x$, for all $x, y \in [0,1]$. Hence, by Lemma 1.0.12, the free n-generated MV-algebra $\boldsymbol{F}^n_{\mathbb{MV}}$ is the subalgebra of $[0,1]^{[0,1]^n}$ generated by the projections, equipped with pointwise defined operations, that is, for any $\langle t_1, \ldots, t_n\rangle \in [0,1]^n$:

$$(f \oplus g)(t_1, \ldots, t_n) = \min\{1, f(t_1, \ldots, t_n) + g(t_1, \ldots, t_n)\}\,,$$

$$(\neg f)(t_1, \ldots, t_n) = 1 - f(t_1, \ldots, t_n)\,, \qquad 0(t_1, \ldots, t_n) = 0\,.$$

However, this result sheds no light on the intrinsic form of elements in $\boldsymbol{F}^n_{\mathbb{MV}}$. McNaughton's representation Theorem states that elements of $\boldsymbol{F}^n_{\mathbb{MV}}$ are a special type of functions.

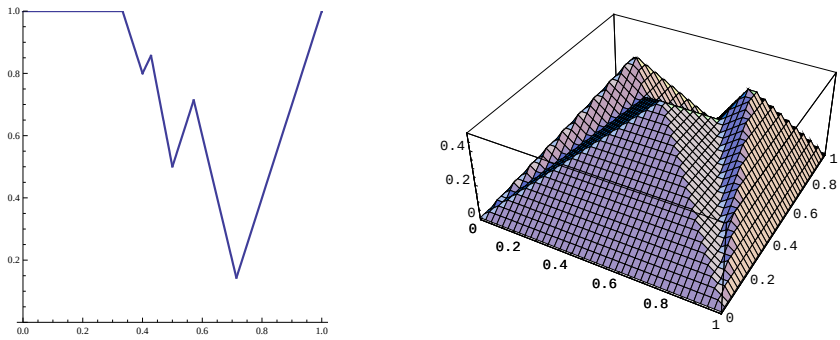
DEFINITION 2.1.1. *For each integer $n \geq 0$, a function $f\colon [0,1]^n \to [0,1]$ is a* McNaughton function *if f is continuous (in the usual topology) and it is piecewise linear with integer coefficients, that is, there exists a finite set $\{p_1, p_2, \ldots, p_u\}$ of linear polynomials in $\mathsf{Z}_1[x_1, \ldots, x_n]$ (called the* linear components *of f), with each p_i of the form*

$$p_i(t_1, \ldots, t_n) = b_i + \sum_{j=1}^{n} a_{ij} t_j \qquad \textit{for all } \langle t_1, \ldots, t_n \rangle \in [0,1]^n\,,$$

for some $a_{i1}, \ldots, a_{in}, b_i \in \mathsf{Z}$, such that, for each $\langle t_1, \ldots, t_n \rangle \in [0,1]^n$, there exists $i \in \{1, 2, \ldots, u\}$ with

$$f(t_1, \ldots, t_n) = p_i(t_1, \ldots, t_n)\,.$$

THEOREM 2.1.2. *For each integer $n \geq 0$, the free n-generated MV-algebra $\boldsymbol{F}^n_{\mathbb{MV}}$ is isomorphic to the algebra of all McNaughton functions $f\colon [0,1]^n \to [0,1]$, equipped with pointwise defined operations.*



(a) $f_1 \in \boldsymbol{F}^1_{\mathbb{MV}}$. (b) $f_2 \in \boldsymbol{F}^2_{\mathbb{MV}}$.

Figure 1. Examples of McNaughton functions.

The proof of the theorem splits in the proofs of the two containments. Showing that each element of $\boldsymbol{F}^n_{\mathbb{MV}}$ is a McNaughton function is rather straightforward.

LEMMA 2.1.3. *For each integer $n \geq 0$, the elements of $\boldsymbol{F}^n_{\mathbb{MV}}$ are n-variate McNaughton functions.*

Proof. By induction on the complexity of terms. □

Proving the other containment, that is that each McNaughton function $f\colon [0,1]^n \to [0,1]$ belongs to $\boldsymbol{F}^n_{\mathbb{MV}}$, is much harder. In the literature there are several proofs of this fact. The proof we are going to give relies on the machinery of regular triangulations, Farey mediants and Schauder hats.

We shall assume familiarity with the basic concepts of convex geometry, in particular with the notions of compact, convex polyhedron (an intersection of finitely many halfspaces which is bounded, that is, it is included in a ball with finite radius) and with the lattices of faces of such a polyhedron. Throughout the chapter "polyhedron" always stands for compact convex polyhedron. We refer to [44] for background on convex and polyhedral geometry.

A *polyhedral complex* Σ is a finite set of polyhedra closed under taking faces and such that any pair of them intersect in a common face. The *support* $|\Sigma|$ of Σ is the set-theoretic union of all its polyhedra.

By a *rational point* we mean a point $\mathbf{v} \in ([0,1] \cap \mathsf{Q})^n$ for some integer $n \geq 0$. Given a rational point $\mathbf{v} = \langle v_1, \ldots, v_n \rangle$, the *denominator* $\mathrm{den}(\mathbf{v})$ of $\mathbf{v}$ is the least common denominator of the components $v_1, \ldots, v_n$. The *homogeneous coordinates* of $\mathbf{v}$ are the components of the point $\mathrm{den}(\mathbf{v})(\mathbf{v}, 1) \in \mathsf{Z}^{n+1}$.

The following property of McNaughton functions is best formulated in terms of polyhedral complexes.

LEMMA 2.1.4. *For any McNaughton function $f\colon [0,1]^n \to [0,1]$, there is a polyhedral complex Σ such that $|\Sigma| = [0,1]^n$ and f is linear over each polyhedron in Σ. Moreover each vertex of a polyhedron in Σ is a rational point.*

Proof. Let $f\colon [0,1]^n \to [0,1]$ be a McNaughton function. Add to the set of linear components of f all linear polynomials of the form 0, $x_i - 1$ and x_i, for all $i \in \{1, \ldots, n\}$. Display $p_1, p_2, \ldots, p_k\colon [0,1]^n \to [0,1]$ the set of linear polynomials so obtained. For each pair $\langle i, j \rangle \in \{1, \ldots, k\}^2$, the set of points $H_{i,j} = \{\mathbf{x} \in \mathsf{R}^n \mid p_i(\mathbf{x}) = p_j(\mathbf{x})\}$ is a, possibly empty, affine linear space. In particular, if $i \neq j$ and $H_{i,j} \neq \emptyset$ then $H_{i,j}$ is $(n-1)$-dimensional. Each point $\mathbf{x} \in [0,1]^n$ satisfies a finite set $D_{\mathbf{x}}$ of conditions of the form $p_i(\mathbf{x}) \leq p_j(\mathbf{x})$, hence it belongs to an n-dimensional polyhedron $P_{\mathbf{x}} \subseteq [0,1]^n$ arising as the intersection of finitely many half-spaces, each one of them bounded by an hyperplane of the form $H_{i,j}$. It is then clear that $[0,1]^n$ is subdivided into a finite collection of polyhedra $P_1, \ldots, P_h$ each one of them of the form $P_{\mathbf{x}}$ for some $\mathbf{x} \in [0,1]^n$. In particular each intersection $P_{r,s} = P_r \cap P_s$ is a (possibly empty) face of both P_r and P_s as each point in $P_{r,s}$ satisfies all conditions $p_i(\mathbf{x}) \leq p_j(\mathbf{x})$ determining P_r and P_s, some of them in the stronger form $p_i(\mathbf{x}) = p_j(\mathbf{x})$. The collection of polyhedra obtained taking all faces of each polyhedron $P_1, \ldots, P_h$ is a polyhedral complex Σ such that $|\Sigma| = [0,1]^n$. It remains to show that f is linear over each P_r. Now, fix $\mathbf{x}$ in the interior of some polyhedron P_r. Then there is p_t such that $f(\mathbf{x}) = p_t(\mathbf{x})$, and for any other $\mathbf{y}$ in the interior of P_r, the line segment l connecting $\mathbf{x}$ to $\mathbf{y}$ does not intersect any space $H_{i,j}$. Since f is continuous and piecewise-linear, $f = p_t$ all over the line segment l, and hence, all over the interior of P_r. By continuity, f coincides with p_t on the whole of P_r. Each vertex of Σ is the solution of a system of equations of the form $p_i(\mathbf{x}) = p_j(\mathbf{x})$, hence is trivially a rational point. $\square$

To synthesize the MV-term corresponding to a given n-variate McNaughton function $f\colon [0,1]^n \to [0,1]$ we need to further subdivide $[0,1]^n$.

DEFINITION 2.1.5. *For each integer $n \geq 0$, an n-*simplex *S is the convex hull of $n+1$ many affinely independent points in R^m for some integer $m \geq n$, called the* vertices *of*

S. That is, displaying the set of vertices of S as $\mathrm{Vert}(S) = \{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_{n+1}\}$,

$$S = \left\{ \lambda_1 \mathbf{v}_1 + \cdots + \lambda_{n+1}\mathbf{v}_{n+1} \mid 0 \leq \lambda_i \in \mathsf{R}, \sum_{i=1}^{n+1} \lambda_i = 1 \right\}.$$

For instance, a (set containing exactly one) point is a 0-simplex, a line segment is a 1-simplex, a triangle is a 2-simplex, and so on. By definition the empty set is the only -1-simplex.

A *face* of an n-simplex S is the convex hull of a subset of the vertices of S. Hence $\emptyset$ is face of every simplex.

A *rational n-simplex* is a simplex whose vertices are rational points. Let S be a rational n-simplex S with $\mathrm{Vert}(S) = \{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_{n+1}\}$. Its associated matrix M_S is the $(n+1) \times (n+1)$ integer-entry matrix whose ith row is given by the homogeneous coordinates of the vertex $\mathbf{v}_i$. Then S is *regular* (or, *unimodular*) if the absolute value of the determinant of M_S is 1. For any $-1 \leq m \leq n$, a rational m-simplex in $[0,1]^n$ is *regular* if it is a face of a regular n-simplex.

DEFINITION 2.1.6. *A set U of rational simplices is a* rational triangulation *of* $[0,1]^n$ *if*

1. *If $S, T \in U$ then $S \cap T$ is face of both S and T;*
2. *If $S \in U$ and F is face of S then $F \in U$;*
3. $\bigcup_{S \in U} S = [0,1]^n$.

If, in addition, each simplex in U is regular, then U is a rational regular triangulation *of* $[0,1]^n$.

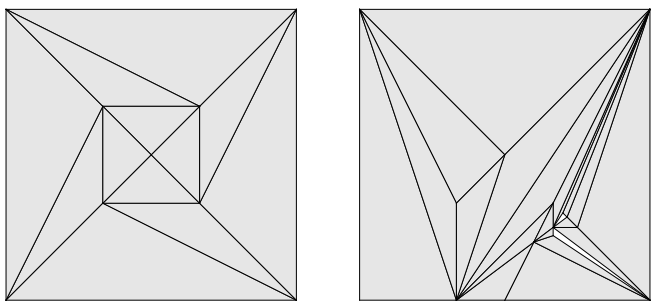


Figure 2. Two examples of rational regular triangulations of $[0,1]^2$.

In the rest of the chapter we shall usually omit the qualification "rational", as we shall deal with rational simplices and rational regular triangulations only.

We observe the following.

LEMMA 2.1.7. *For each polyhedral complex Σ there exists a triangulation U_Σ such that each polyhedron in Σ is union of simplices in U_Σ.*

Proof. By [66, Prop. 2.9], each polyhedron P in Σ can be subdivided into simplices without introducing new vertices, that is, one can find a finite set S_P of simplices such that $\bigcup_{S \in S_P} S = P$, each two simplices in S_P intersect in a common face, and the set of all vertices of all simplices in S_P coincides with the set of vertices of P. Then $U_\Sigma = \bigcup_{P \in \Sigma} S_P$ is the desired triangulation. □

If U_1, U_2 are triangulations of $[0,1]^n$ and each $S \in U_1$ is contained in some $T \in U_2$, then U_1 is a *refinement* of U_2, in symbols $U_1 \prec U_2$.

Let S be a regular 1-simplex with vertex-set $\mathrm{Vert}(S) = \{\mathbf{v}, \mathbf{w}\}$. Let $\langle v_1, \ldots, v_{n+1} \rangle$, $\langle w_1, \ldots, w_{n+1} \rangle \in \mathsf{Z}^{n+1}$ be the homogeneous coordinates of $\mathbf{v}$ and $\mathbf{w}$, respectively. Then the *Farey mediant* of $\mathbf{v}$ and $\mathbf{w}$ is the point $\mathbf{v} +_F \mathbf{w}$ whose homogeneous coordinates are

$$\langle v_1 + w_1, \ldots, v_{n+1} + w_{n+1} \rangle .$$

LEMMA 2.1.8. *Let S be a regular n-simplex with vertex-set given by* $\mathrm{Vert}(S) = \{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_{n+1}\}$. *Let further S_i be the n-simplex with* $\mathrm{Vert}(S_i) = (\mathrm{Vert}(S) \setminus \{\mathbf{v}_i\}) \cup \{\mathbf{v}_1 +_F \mathbf{v}_2\}$, *for each $i \in \{1, 2\}$. Then both S_1 and S_2 are regular.*

Proof. It follows directly from the definition of Farey mediant and the multilinear properties of the determinant. □

DEFINITION 2.1.9. *Let U be a regular triangulation and $S \in U$ be a 1-simplex with* $\mathrm{Vert}(S) = \{\mathbf{v}_1, \mathbf{v}_2\}$. *Then the* (edge) starring *of U* along *S, in symbols $U * S$, is the set of simplices obtained as follows.*

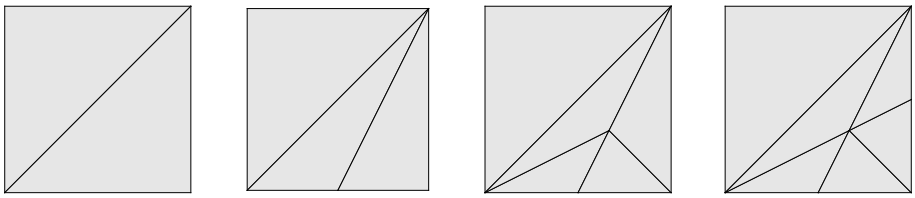
1. *Put in $U * S$ all simplices of U not containing S;*
2. *Replace each simplex $T \in U$ such that S is face of T by the simplices T_1, T_2 with* $\mathrm{Vert}(T_i) = (\mathrm{Vert}(T) \setminus \{\mathbf{v}_{3-i}\}) \cup \{\mathbf{v}_1 +_F \mathbf{v}_2\}$ *for each $i \in \{1, 2\}$;*
3. *Close $U * S$ under taking faces.*

LEMMA 2.1.10. *For each regular triangulation U and 1-simplex $S \in U$, the set of simplices $U * S$ is a regular triangulation refining U.*

Proof. Immediate, from Lemma 2.1.8. □

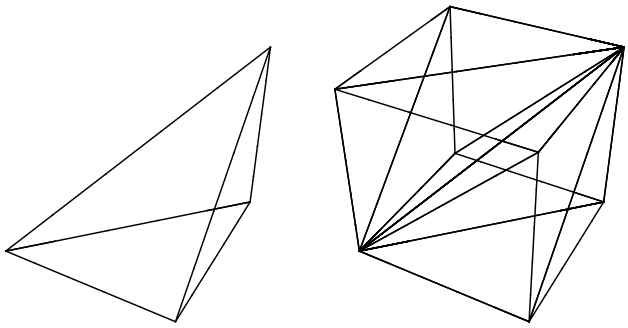
DEFINITION 2.1.11. *Let T be the n-simplex whose vertex-set is $\{\mathbf{v}_j\}_{j=0}^n$, such that $\pi_{\{i\}}(\mathbf{v}_j) = 0$ if $i + j \leq n$, $\pi_{\{i\}}(\mathbf{v}_j) = 1$, otherwise. Let Sym_n be the group of all permutations of the set $\{1, 2, \ldots, n\}$. For each $\sigma \in \mathrm{Sym}_n$ let T_σ be the simplex whose ith vertex is such that its jth component is $\pi_{\{\sigma(j)\}}(\mathbf{v}_i)$. Let F_σ be the set of all faces of T_σ. Then let*

$$U_0^n = \bigcup_{\sigma \in \mathrm{Sym_n}} F_\sigma .$$



(a) U (b) $U' = U * S$ (c) $U'' = U' * S'$ (d) $U''' = U'' * S''$

Figure 3. A regular triangulation U of $[0,1]^2$, and three successive starrings along 1-simplices S, S', S'', where $\mathrm{Vert}(S) = \{\langle 0,0\rangle, \langle 1,0\rangle\}$, $\mathrm{Vert}(S') = \{\langle 1/2,0\rangle, \langle 1,1\rangle\}$, $\mathrm{Vert}(S'') = \{\langle 1,0\rangle, \langle 1,1\rangle\}$.



(a) T (b) U_0^3

Figure 4. The 3-simplex T and U_0^3 as in Definition 2.1.11.

LEMMA 2.1.12. *Let H be an affine hyperplane of the form $\{\mathbf{x} \in \mathsf{R}^n \mid \sum_{i=1}^n a_i x_i = b\}$ for $a_1, a_2, \ldots, a_n, b \in \mathsf{Z}$. Let further H^- and H^+ be the two closed n-dimensional halfspaces bounded by H. Then there exists a finite sequence*

$$U_0^n = U_0 \succ U_1 \succ \cdots \succ U_u$$

of regular triangulations of $[0,1]^n$ such that for each $i \in \{1, \ldots, u\}$, the triangulation U_i is obtained by starring U_{i-1} along one of its 1-simplices, and each simplex of U_u is entirely contained either in H^- or H^+.

Proof. Let U be a regular triangulation of $[0,1]^n$. We display the homogeneous coordinates of each vertex $\mathbf{u}$ of a 1-simplex of U as $\langle u_1, u_2, \ldots, u_{n+1}\rangle$. Let $f\colon \mathsf{Z}^{n+1} \to \mathsf{Z}$ be the map $\langle x_1, \ldots, x_{n+1}\rangle \mapsto a_1x_1 + \cdots a_nx_n - bx_{n+1}$. For each regular 1-simplex S with vertex-set $\mathrm{Vert}\,(S) = \{\mathbf{u}, \mathbf{v}\}$, let us define the *discrepancy* $\mathrm{disc}\,(S)$ of S (with respect to H) as

$$\mathrm{disc}\,(S) = \begin{cases} 0 & \text{if } 0 \le f(\mathbf{u}), f(\mathbf{v}) \text{ or } 0 \ge f(\mathbf{u}), f(\mathbf{v}), \\ |f(\mathbf{u})| + |f(\mathbf{v})| & \text{otherwise.} \end{cases}$$

We further set $d = \max\{\mathrm{disc}\,(S) \mid S \in U, S \text{ 1-simplex}\}$ and $s = |\{S \mid S \in U, S \text{ 1-simplex}, \mathrm{disc}\,(S) = d\}|$. We proceed by induction on the lexicographically ordered pairs $\langle d, s\rangle$. The base of the induction follows immediately by noting that, by the definition of f, if $d = 0$ then each 1-simplex S of U already lies either in H^- or in H^+. For the inductive step, pick a 1-simplex S of U such that $\mathrm{disc}\,(S) = d$, and let $\mathrm{Vert}\,(S) = \{\mathbf{u}, \mathbf{v}\}$. Let $U' = U * S$. As $S \notin U'$, it suffices to show that each 1-simplex $R \in U' \setminus U$ is such that $\mathrm{disc}\,(R) < d$. If we accomplish this task then by induction we are done, since U' has either a number $s' < s$ of 1-simplices of discrepancy d, or the maximum discrepancy of 1-simplices in U' is $d' < d$, for some d'. Each such 1-simplex R has as vertex-set $\mathrm{Vert}\,(R) = \{\mathbf{u} +_F \mathbf{v}, \mathbf{p}\}$ for some 0-simplex $\mathbf{p}$ of U. Notice that by linearity of f and the definition of Farey mediant the value of $f(\mathbf{u} +_F \mathbf{v})$ is intermediate between $f(\mathbf{u})$ and $f(\mathbf{v})$. Now, if $f(\mathbf{u} +_F \mathbf{v}) = 0$ then $\mathrm{disc}\,(R) = 0$ and we are through. Otherwise, assume without loss of generality that $f(\mathbf{u}) < 0 < f(\mathbf{u} +_F \mathbf{v}) < f(\mathbf{v})$. If $f(\mathbf{p}) \geq 0$ then $\mathrm{disc}\,(R) = 0$ and the lemma is settled. We are left with the case $f(\mathbf{p}) < 0$. To conclude the proof we observe that $\mathrm{disc}\,(R) = |f(\mathbf{u} +_F \mathbf{v})| + |f(\mathbf{p})| < |f(\mathbf{v})| + |f(\mathbf{p})|$ and that $\{\mathbf{v}, \mathbf{p}\}$ is, by Definition 2.1.9, the vertex-set of a 1-simplex $T \in U$. Since $|f(\mathbf{v})| + |f(\mathbf{p})| = \mathrm{disc}\,(T) \leq d$, we have shown $\mathrm{disc}\,(R) < d$ as desired. □

LEMMA 2.1.13. *For each integer $n \geq 0$ the set of simplices U_0^n is a regular triangulation of $[0,1]^n$. Further, there exists an infinite sequence*

$$U_0^n = U_0 \succ U_1 \succ U_2 \succ \cdots$$

of regular triangulations such that

1. *for each integer $i \geq 0$, U_{i+1} is obtained from U_i by starring along a 1-simplex of U_i;*
2. *for each triangulation U there exists an index i such that $U_i \prec U$.*

Proof. For each $\sigma \in \mathrm{Sym}_n$, the n-simplex T_σ is regular by construction; moreover, T_σ is the set of points $\mathbf{x} \in [0,1]^n$ satisfying the following set of inequalities:

$$0 \leq x_{\sigma(1)},\, x_{\sigma(1)} \leq x_{\sigma(2)},\, x_{\sigma(2)} \leq x_{\sigma(3)},\, \cdots,\, x_{\sigma(n-1)} \leq x_{\sigma(n)},\, x_{\sigma(n)} \leq 1\,. \qquad (1)$$

It immediately follows that $\bigcup_{\sigma \in \mathrm{Sym_n}} T_\sigma = [0,1]^n$. Notice that $\mathbf{x}$ belongs to the interior of T_σ iff (1) holds with all $\leq$ replaced by $<$. By the same token each m-dimensional face of T_σ is the set of points satisfying (1) with $n+1-m$ many $\leq$ replaced by $=$. It follows that for each $\sigma, \tau \in \mathrm{Sym}_n$, the simplices T_σ and T_τ intersects in a common face. The same applies for the intersection of proper faces of T_σ and T_τ. Then U_0^n is a regular triangulation of $[0,1]^n$. This settles the first statement.

Let us now fix an enumeration $H_1, H_2, \ldots,$ of all affine hyperplanes with integer coefficients. By Lemma 2.1.12 there exists a sequence:

$$\begin{gathered} U_0 \succ U_1 \succ \cdots \succ U_{u_1} \\ U_{u_1} \succ U_{u_1+1} \succ \cdots \succ U_{u_1+u_2} \\ \cdots \\ U_{u_1+\cdots+u_i} \succ U_{u_1+\cdots+u_i+1} \succ \cdots \succ U_{u_1+\cdots+u_{i+1}} \\ \cdots \end{gathered}$$

such that

1. $U_0 = U_0^n$;
2. U_j is obtained by starring U_{j-1} along one of its 1-simplices;
3. For each $k \geq 1$, each simplex of $U_{u_1+\cdots+u_k}$ is entirely contained either in H_k^- or H_k^+.

Now, each simplex of U, being a rational polyhedron, is the intersection of a finite number of halfspaces of the form H_t^- or H_t^+. Hence, the sequence we have constructed eventually refines U. □

U_0^n and the sequence $U_0 \succ U_1 \succ U_2 \succ \cdots$ are respectively called the *fundamental partition* and the *fundamental sequence* of $[0,1]^n$. Note that the set of 0-simplices contained in U_0^n is the set

$$\{\{\mathbf{v}\} \mid \mathbf{v} \in \{0,1\}^n\}\,.$$

We stress that the fundamental partition is by no means canonical for each $n > 1$: the partition is fixed by the choice of the enumeration of rational hyperplanes in the proof of Lemma 2.1.13. On the other hand, the case $n = 1$ is completely canonical, as the fundamental sequence of $[0,1]$ is the one given by Farey sequences.

We are now ready to introduce the notion of *Schauder hat*.

DEFINITION 2.1.14. *Let U be a regular triangulation of $[0,1]^n$, and let $\mathbf{v}$ be a vertex of some simplex in U (note that the 0-simplex $\{\mathbf{v}\}$ belongs to U). Then the Schauder hat with* apex $\mathbf{v}$ *with respect to U is the continuous function $h_{U,\mathbf{v}}\colon [0,1]^n \to [0,1]$ determined by the following constraints.*

1. *$h_{U,\mathbf{v}}(\mathbf{v}) = 1/\mathrm{den}(\mathbf{v})$;*
2. *$h_{U,\mathbf{v}}$ is constantly 0 over each simplex $T \in U$ such that $\mathbf{v} \notin T$;*
3. *$h_{U,\mathbf{v}}$ is linear over each simplex $T \in U$.*

The set $H_U = \{h_{U,\mathbf{v}} \mid \{\mathbf{v}\} \in U\}$ is called the Schauder set *of U.*

LEMMA 2.1.15. *For each regular triangulation U of $[0,1]^n$ and any vertex $\mathbf{v}$ of some simplex in U, the Schauder hat $h_{U,\mathbf{v}}$ is a McNaughton function.*

Proof. It is sufficient to prove that the linear function $h \restriction T$, defined as the restriction of $h_{U,\mathbf{v}}$ over an n-simplex T of U having $\mathbf{v}$ among its vertex, has integer coefficients. Display the vertices of T as $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n, \mathbf{v}_{n+1} = \mathbf{v}$, and express $h \restriction T(\mathbf{x})$ as $a_1x_1 + \cdots a_nx_n + b$. Then conditions (1) and (2) of Definition 2.1.14 are to the effect that the $(n+1)$-tuple of reals $\mathbf{a} = \langle a_1, \ldots, a_n, b\rangle$ is the only solution to the linear system expressed in matricial form by $M\mathbf{a} = \mathbf{d}$, where M is the $(n+1)\times(n+1)$ integer-entry matrix whose ith row is the expression in homogeneous coordinates of $\mathbf{v}_i$, while $d_i = 0$ for all $i \in \{1, \ldots, n\}$, and $d_{n+1} = \mathrm{den}(\mathbf{v})(1/\mathrm{den}(\mathbf{v})) = 1$. Notice that $M = M_T$, and then, by regularity of T, M^{-1} is an $(n+1)\times(n+1)$ integer-entry matrix. It immediately follows that $\mathbf{a} = M^{-1}\mathbf{d}$ is a $(n+1)$-tuple of integers, thus setting the lemma. □

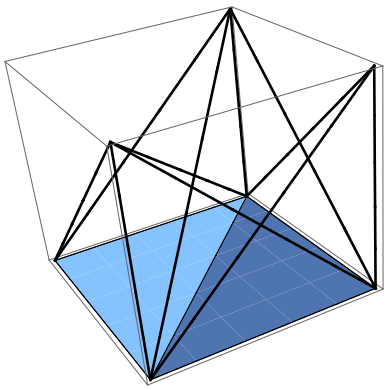


Figure 5. The Schauder hats of apices $\langle 0,0\rangle$, $\langle 1,0\rangle$ and $\langle 1,1\rangle$, with respect to U_0^2.

LEMMA 2.1.16. *For each* $\mathbf{v} \in \{0,1\}^n$ *let* $f_{\mathbf{v}} \colon [0,1]^n \to [0,1]$ *be the function*

$$\left(\bigwedge_{v_i=0} \neg x_i \ \& \bigwedge_{v_i=1} x_i \right)^{\boldsymbol{F}^n_{\mathbb{MV}}} .$$

Then $f_{\mathbf{v}} = h_{U_0^n,\mathbf{v}}$*, and hence the set* $\{f_{\mathbf{v}} \mid \{\mathbf{v}\} \in U_0^n\}$ *is the Schauder set of* U_0^n*.*

Proof. Somewhat lengthy, but straightforward. □

Notice that the formula for the Schauder hat $f_{\mathbf{v}}$ in Lemma 2.1.16 coincides with the Boolean minterm corresponding to the truth-value assignment $x_i \mapsto v_i$.

LEMMA 2.1.17. *Let* U *be a regular triangulation of* $[0,1]^n$ *and let* $S \in U$ *be a* 1*-simplex with vertices* $\mathrm{Vert}(S) = \{\mathbf{v}_1, \mathbf{v}_2\}$*. Then the Schauder set of the starring of* U *along* S *is*

$$H_{U*S} = \{h_{U*S,\mathbf{v}} \mid \{\mathbf{v}\} \in U\} \cup \{h_{U*S,\mathbf{v}_1+_F\mathbf{v}_2}\},$$

where

1. $h_{U*S,\mathbf{w}} = h_{U,\mathbf{w}}$ *for each* $\{\mathbf{w}\} \in U$ *such that* $\mathbf{v}_1 \neq \mathbf{w} \neq \mathbf{v}_2$*;*
2. $h_{U*S,\mathbf{v}_1} = h_{U,\mathbf{v}_1} \ominus h_{U,\mathbf{v}_2}$ *and* $h_{U*S,\mathbf{v}_2} = h_{U,\mathbf{v}_2} \ominus h_{U,\mathbf{v}_1}$*;*
3. $h_{U*S,\mathbf{v}_1+_F\mathbf{v}_2} = h_{U,\mathbf{v}_1} \wedge h_{U,\mathbf{v}_2}$.

Proof. Let $\mathbf{w} = \mathbf{v}_1 +_F \mathbf{v}_2$. We first note that $h_{U,\mathbf{v}_1}(\mathbf{w}) = h_{U,\mathbf{v}_2}(\mathbf{w}) = 1/\mathrm{den}(\mathbf{w})$. This is the case since the homogeneous expression of $\mathbf{w}$ is just the sum in Z^{n+1} of the homogeneous expressions of $\mathbf{v}_1$ and $\mathbf{v}_2$. Since for each $i \in \{1,2\}$ we have that $\mathrm{den}(\mathbf{v}_i)$ is the $(n+1)$-component of the homogeneous expression of $\mathbf{v}_i$, it follows that the last component of the sums of the homogeneous expressions of $\mathbf{v}_1$ and $\mathbf{v}_2$ is just $\mathrm{den}(\mathbf{v}_1) + \mathrm{den}(\mathbf{v}_2)$, that is, $\mathrm{den}(\mathbf{w})$. From $h_{U,\mathbf{v}_i}(\mathbf{v}_i) = 1/\mathrm{den}(\mathbf{v}_i)$ we then have $h_{U,\mathbf{v}_1}(\mathbf{w}) = h_{U,\mathbf{v}_2}(\mathbf{w}) = 1/\mathrm{den}(\mathbf{w})$.

Consider $\mathbf{t} \notin \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{w}\}$. Then $h_{U,\mathbf{t}}(\mathbf{v}_1) = h_{U,\mathbf{t}}(\mathbf{v}_2) = 0$, hence, by linearity, $h_{U*S,\mathbf{t}}(\mathbf{w}) = h_{U,\mathbf{t}}(\mathbf{w}) = 0$. As $U * S$ is a refinement of U, $h_{U*S,\mathbf{t}} = h_{U,\mathbf{t}}$ is linear over each simplex of $U * S$.

Observe now that $h_{U*S,\mathbf{w}}(\mathbf{w}) = \min\{1/\mathrm{den}(\mathbf{w}), 1/\mathrm{den}(\mathbf{w})\} = 1/\mathrm{den}(\mathbf{w})$, and $h_{U*S,\mathbf{w}}(\mathbf{v}_i) = \min\{1/\mathrm{den}(\mathbf{v}_i), 0\} = 0$ for each $i \in \{1, 2\}$. Clearly $h_{U*S,\mathbf{w}}(\mathbf{t}) = 0$ for all $\mathbf{t} \notin \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{w}\}$. Let T be any n-simplex in U. If S is not a face of T, then at least one in $\mathbf{v}_1$,$\mathbf{v}_2$ does not belong to T, hence $h_{U*S,\mathbf{w}}$ is constantly 0 on T. If S is a face of T then $U * S$ contains the two simplices T_1, T_2 replacing T by starring. Note $h_{U,\mathbf{v}_1}$ coincides with $h_{U,\mathbf{v}_2}$ in all vertices of $T_i \setminus \{\mathbf{v}_i\}$, while $h_{U,\mathbf{v}_{2-i}}(\mathbf{v}_i) = 0$ and $h_{U,\mathbf{v}_i}(\mathbf{v}_i) = 1/\mathrm{den}(\mathbf{v}_i)$. Hence, by linearity of both hats $h_{U,\mathbf{v}_i}$ on S, we have $h_{U,\mathbf{v}_{2-i}} \leq h_{U,\mathbf{v}_i}$ on T_i and hence $h_{U*S,\mathbf{w}} = h_{U,\mathbf{v}_{2-i}}$ on T_i.

Note $h_{U*S,\mathbf{v}_1}(\mathbf{v}_1) = h_{U,\mathbf{v}_1}(\mathbf{v}_1) = 1/\mathrm{den}(\mathbf{v}_1)$ while $h_{U*S,\mathbf{v}_1}(\mathbf{w}) = h_{U*S,\mathbf{v}_1}(\mathbf{v}_2) = 0$; moreover $h_{U*S,\mathbf{v}_1}(\mathbf{t}) = 0$ for each vertex $\mathbf{t} \notin \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{w}\}$. It remains to show that $h_{U*S,\mathbf{v}_1}$ is linear over each n-simplex T of $U * S$ (The case for $h_{U*S,\mathbf{v}_2}$ is dealt with analogously). If S is not a face of T, then at least one in $\mathbf{v}_1$,$\mathbf{v}_2$ does not belong to T. If $\mathbf{v}_1 \notin T$, then $h_{U*S,\mathbf{v}_1} = h_{U,\mathbf{v}_1} = 0$ on T. If $\mathbf{v}_2 \notin T$, then $h_{U,\mathbf{v}_2} = 0$ on T, and hence $h_{U*S,\mathbf{v}_1} = h_{U,\mathbf{v}_1}$ on T. In both cases $h_{U*S,\mathbf{v}_1}$ is linear on T. Assume now $U * S$ contains the two simplices T_1, T_2 that replace T by starring. Recall that $h_{U,\mathbf{v}_{2-i}} \leq h_{U,\mathbf{v}_i}$ over T_i, hence $h_{U*S,\mathbf{v}_i} = h_{U,\mathbf{v}_i} - h_{U,\mathbf{v}_{2-i}}$ over T_i and $h_{U*S,\mathbf{v}_i} = 0$ over T_{2-i}. This concludes the proof. □

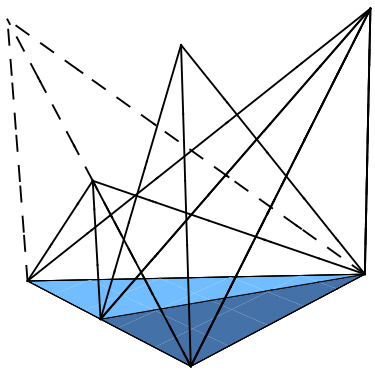


Figure 6. The Schauder hats of apices $\langle 1/2, 0\rangle$, $\langle 1, 0\rangle$ and $\langle 0, 1\rangle$, with respect to $U_0^2 * S$, where S is the 1-simplex of vertices $\mathrm{Vert}(S) = \{\langle 0, 0\rangle, \langle 1, 0\rangle\}$. Note the hat with apex $\langle 1/2, 0\rangle$ is obtained by starring along S.

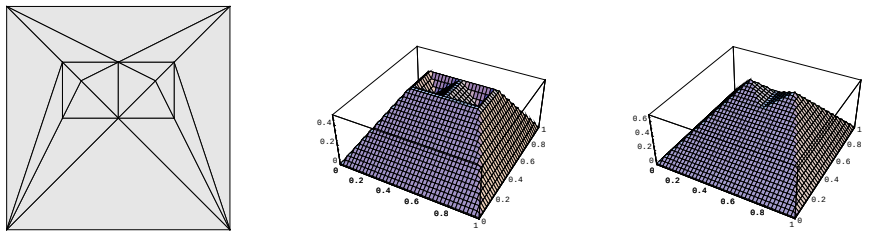
Lemma 2.1.15, Lemma 2.1.16 and Lemma 2.1.17 show that for each Schauder hat h in the Schauder set of any regular partition in the fundamental sequence of $[0, 1]^n$ one constructs an MV-term φ such that

$$\varphi^{\boldsymbol{F}^n_{\mathrm{MV}}} = h\,.$$

The next step consists in showing that each n-variate McNaughton function $f\colon [0, 1]^n \to [0, 1]$ can be realised as a suitable finite combination of Schauder hats.

LEMMA 2.1.18. *For any McNaughton function* $f\colon [0,1]^n \to [0,1]$ *there exists an index* i *such that* f *is linear over each simplex in the* i*th member* U_i *of the fundamental sequence of* $[0,1]^n$.

Proof. By Lemma 2.1.4 there exists a polyhedral complex Σ such that $|\Sigma| = [0,1]^n$ and f is linear over each polyhedron in Σ. By Lemma 2.1.7 there exists a triangulation U_Σ refining Σ. By Lemma 2.1.13 there exists an index i such that the i-th member U_i of the fundamental sequence refines U_Σ, and hence f is linear over each simplex of U_i. □



(a) U (b) f (c) g

Figure 7. A regular triangulation U of $[0,1]^2$ and two McNaughton functions f and g that are linear over each simplex of U.

We are now ready to prove the hard side of McNaughton's representation theorem.

LEMMA 2.1.19. *For any McNaughton function* $f\colon [0,1]^n \to [0,1]$ *there exists an MV-term* φ *such that*

$$\varphi^{\boldsymbol{F}^n_{\mathbb{MV}}} = f\,.$$

In particular, let the index i *be determined as in Lemma 2.1.18. Then*

$$\varphi = \bigoplus_{\{\mathbf{v}\}\in U_i} m_{\mathbf{v}}\varphi_{\mathbf{v}}\,,$$

where $\varphi_{\mathbf{v}}^{\boldsymbol{F}^n_{\mathbb{MV}}} = h_{U_i,\mathbf{v}}$ *and* $m_{\mathbf{v}} = \mathrm{den}(\mathbf{v})f(\mathbf{v})$ *for each* $\{\mathbf{v}\} \in U_i$.

Proof. First notice that since each linear component of f has integer coefficients then $f(\mathbf{p})$ is an integer multiple $m_{\mathbf{p}}$ of $1/\mathrm{den}(\mathbf{p})$ for each $\mathbf{p} \in ([0,1] \cap \mathsf{Q})^n$.

Now, take U_i as in Lemma 2.1.18 and observe that $f(\mathbf{v}) = m_{\mathbf{v}}h_{U_i,\mathbf{v}}(\mathbf{v})$ while $m_{\mathbf{v}}h_{U_i,\mathbf{v}}(\mathbf{w}) = 0$ for each pair $\mathbf{w} \neq \mathbf{v}$ of U_i. Then $\varphi^{\boldsymbol{F}^n_{\mathbb{MV}}}$ coincides with f at every vertex of U_i. By linearity of both f and of each $h_{U_i,\mathbf{v}}$ over each simplex of U_i, we conclude that $\varphi^{\boldsymbol{F}^n_{\mathbb{MV}}}$ coincides with f over the whole of $[0,1]^n$. □

THEOREM 2.1.20. *For each integer* $n \geq 0$*, the free* n*-generated MV-algebra* $\boldsymbol{F}^n_{\mathbb{MV}}$ *is isomorphic to the algebra of all McNaughton functions* $f\colon [0,1]^n \to [0,1]$*, equipped with pointwise defined operations.*

Proof. By Lemma 2.1.3 and Lemma 2.1.19. □

Lemma 2.1.19 can be seen as a *normal form* lemma for Łukasiewicz logic: each formula φ can be equivalently expressed as a $\oplus$-disjunction of Schauder hat formulas. In this light, Schauder hats plays the role of generalised *semantical minterms*, and the corresponding Schauder hat formula are generalised *syntactical* minterms.

Alternative proofs of Theorem 2.1.20 use different "building blocks". In particular, they often use the following Lemma to find MV-terms expressing truncated linear functions (with integer coefficients).

LEMMA 2.1.21. *Let* $f\colon [0,1]^n \to \mathsf{R}$ *be an* integer linear polynomial*, that is, a function of the form*

$$f(x_1,\ldots,x_n) = \sum_{i=1}^{n} a_i x_i + b\,,$$

for $a_1,\ldots,a_n,b \in \mathsf{Z}$*. Then there is an algorithm that outputs an MV-term* φ *such that* $\varphi^{\boldsymbol{F}^n_{\mathrm{MV}}} = \max\{0,\min\{1,f\}\}$.

Proof. Let us denote $f^{\#}$ the function $\max\{0,\min\{1,f\}\}$. The proof is by induction on $m = \sum_{i=1}^{n} |a_i|$. If $m = 0$ then $f^{\#}$ is either constantly 0 or 1, then we can take as φ either the term $\bot$ or $\top$, respectively. Assume now $m > 0$ and let a_j be such that $|a_j| = \max_{i=1}^{n} |a_i|$. Suppose first $a_j > 0$. Let $g = f - x_j$. By induction hypothesis we have terms ψ_1 and ψ_2 such that $\psi_1^{\boldsymbol{F}^n_{\mathrm{MV}}} = g^{\#}$ and $\psi_2^{\boldsymbol{F}^n_{\mathrm{MV}}} = (g+1)^{\#}$. We claim that for each $\langle x_1,\ldots,x_n\rangle \in [0,1]^n$ the following identity holds.

$$(g + x_j)^{\#} = (g^{\#} \oplus x_j) \odot (g+1)^{\#}\,. \tag{2}$$

We run a case analysis on the value of $g(x_1,\ldots,x_n)$. If $g(x_1,\ldots,x_n) > 1$ then both sides of (2) evaluate to 1; similarly if $g(x_1,\ldots,x_n) < -1$ both sides evaluate to 0. If $0 \le g(x_1,\ldots,x_n) \le 1$ then $g(x_1,\ldots,x_n) = g^{\#}(x_1,\ldots,x_n)$ and $(g+1)^{\#}(x_1,\ldots,x_n) = 1$. Hence, the right hand side of (2) evaluates to $g(x_1,\ldots,x_n) \oplus x_j = \min\{1, g(x_1,\ldots,x_n) + x_j\}$. As $0 \le g(x_1,\ldots,x_n)$, this value coincides with $(g(x_1,\ldots,x_n) + x_j)^{\#}$. If $-1 \le g(x_1,\ldots,x_n) \le 0$ then $g^{\#}(x_1,\ldots,x_n) = 0$ and $(g+1)^{\#}(x_1,\ldots,x_n) = g(x_1,\ldots,x_n) + 1$. Now, as $g(x_1,\ldots,x_n) \le 0$, we have $(g(x_1,\ldots,x_n) + x_j)^{\#} = \max\{0, g(x_1,\ldots,x_n) + x_j\} = x_j \odot (g(x_1,\ldots,x_n) + 1)$, that is the value taken by the right hand side of (2). The claim is settled. Then, in our current assumptions, as $f^{\#} = (g + x_j)^{\#}$, we have that the term

$$\varphi = (\psi_1 \oplus x_j) \mathbin{\&} \psi_2$$

is such that $\varphi^{\boldsymbol{F}^n_{\mathrm{MV}}} = f^{\#}$. To conclude the proof it remains to consider the case $a_j < 0$. This is easily settled as we can use the preceding case to find a term ψ such that $\psi^{\boldsymbol{F}^n_{\mathrm{MV}}} = (1-f)^{\#}$, and then conclude that $(\neg\psi)^{\boldsymbol{F}^n_{\mathrm{MV}}} = 1 - (1-f)^{\#} = f^{\#}$. $\square$

And then each McNaughton function $f\colon [0,1]^n \to [0,1]$ is constructed as an arrangement of integer linear polynomials. In particular, one can show that there exists a finite set of truncated integer linear polynomials $\{f_{i,j} \mid i \in I, J \in J_i\}$ such that

$$f = \bigvee_{i\in I} \bigwedge_{j\in J_i} f_{i,j}\,.$$

In this case, we have a disjunctive normal form whose minterms are finite conjunctions of truncated hyperplanes terms as given by Lemma 2.1.21: each such term can then be seen as a generalised *literal*, as in Boolean logic literals are either variables or negated variables.

We give here some details explaining how to construct the $\bigvee$-$\bigwedge$ combination of truncated integer linear polynomials which equals a given n-variate McNaughton function f. First one lists the distinct linear components of f as $p_1, p_2, \ldots, p_k$ and considers the partition of $[0,1]^n$ given by the collection of all full-dimensional polyhedra P_σ indexed by a permutation σ of $\{1, 2, \ldots, k\}$ such that for all $\mathbf{t} \in P_\sigma$ it holds $p_{\sigma(1)}(\mathbf{t}) \geq p_{\sigma(2)}(\mathbf{t}) \geq \cdots \geq p_{\sigma(k)}(\mathbf{t})$. Note f over such a P_σ coincides with $p_{\sigma(i_\sigma)}$ for some index i_σ. Clearly, $\bigwedge_{j=1}^{i_\sigma} p_{\sigma(j)}$ coincides with f all over P_σ. It can be proved, using an analytic-geometric argument that $\left(\bigwedge_{j=1}^{i_\sigma} p_{\sigma(j)}\right)(\mathbf{t}) \leq f(\mathbf{t})$ for all $\mathbf{t} \in [0,1]^n \setminus P_\sigma$. Let Σ denote the set of all permutations σ of $\{1, 2, \ldots, k\}$ such that P_σ is full-dimensional. One then concludes that $\bigwedge_{\sigma \in \Sigma} \bigvee_{j=1}^{i_\sigma} p_{\sigma(j)} = f$, as desired.

As is well known, each MV-algebra $\langle A, \oplus, \neg, 0\rangle$ is isomorphic to its *order dual* $\langle A, \odot, \neg, 1\rangle$, via the map $a \mapsto \neg a$. It follows that both kinds of disjunctive normal forms, that is, $\oplus$-disjunction of Schauder hats, or $\bigvee\bigwedge$-combinations of truncated hyperplanes, can be dualised, obtaining conjunctive normal forms either as $\odot$-conjunctions of Schauder co-hats, or $\bigwedge\bigvee$-combinations of truncated hyperplanes.

When dealing with Wajsberg hoops, which are the topic of the next Subsection, and their relations with MV-algebras, the order dual of an MV-algebra turns out to be a useful tool to work with.

2.2 Wajsberg Hoops

We recall that the variety of Wajsberg hoops consists of the $\bot$-free subreducts of Wajsberg algebras, which in turns constitutes a variety term-wise equivalent to $\mathbb{MV}$.

The variety $\mathbb{WH}$ of Wajsberg hoops is generated by the standard algebra

$$\langle [0,1], \odot, \rightarrow, 1\rangle\,,$$

where $x \odot y = \max\{0, x + y - 1\}$ and $x \rightarrow y = \min\{1, 1 - x + y\}$. Notice that $\min\{1, x + y\} = (x \rightarrow (x \odot y)) \rightarrow y$. Hence the operation $\oplus$ is definable as $x \oplus y = (x \rightarrow (x \odot y)) \rightarrow y$.

A slight adaptation of McNaughton's Theorem yields the functional representation for the n-generated free Wajsberg hoop, as a subalgebra of $[0,1]^{[0,1]^n}$. To accomplish this task we shall work on order duals of MV-algebras, and we have first to dualise some notions of the theory of Schauder hats introduced in the previous Subsection.

In particular, given a regular triangulation U of $[0,1]^n$ and one of its vertices $\mathbf{v}$, the *Schauder co-hat* $k_{U,\mathbf{v}}$ with apex $\mathbf{v}$ is just the negation of the corresponding Schauder hat.

$$k_{U,\mathbf{v}} = 1 - h_{U,\mathbf{v}}\,.$$

Clearly, *Schauder co-hats* can be defined replacing items 1. and 2. of Definition 2.1.14 with:

1. $k_{U,\mathbf{v}}(\mathbf{v}) = (\mathrm{den}(\mathbf{v}) - 1)/\mathrm{den}(\mathbf{v})$;

2. $k_{U,\mathbf{v}}(\mathbf{v})$ is constantly 1 over each simplex $T \in U$ such that $\mathbf{v} \notin T$.

The set $K_U = \{k_{U,\mathbf{v}} \mid \{\mathbf{v}\} \in U\}$ is the *Schauder co-set* of U. The dual of Lemma 2.1.16 states that $\{g_{\mathbf{v}} \mid \{\mathbf{v}\} \in U_0^n\}$ is the Schauder co-set of U_0^n, where $g_{\mathbf{v}} \colon [0,1]^n \to [0,1]$ is the function

$$\left(\bigvee_{v_i=0} x_i \oplus \bigvee_{v_i=1} \neg x_i \right).$$

We shall also dualise Lemma 2.1.17 as follows.

LEMMA 2.2.1. *Let U be a regular triangulation of $[0,1]^n$ and let $S \in U$ be a 1-simplex with vertices $\mathrm{Vert}(S) = \{\mathbf{v}_1, \mathbf{v}_2\}$. Then the Schauder co-set of the starring of U along S is*

$$K_{U*S} = \{k_{U*S,\mathbf{v}} \mid \{\mathbf{v}\} \in U\} \cup \{k_{U*S,\mathbf{v}_1+_F\mathbf{v}_2}\},$$

where

1. $k_{U*S,\mathbf{w}} = k_{U,\mathbf{w}}$ *for each* $\{\mathbf{w}\} \in U$ *such that* $\mathbf{v}_1 \neq \mathbf{w} \neq \mathbf{v}_2$*;*
2. $k_{U*S,\mathbf{v}_1} = k_{U,\mathbf{v}_2} \to k_{U,\mathbf{v}_1}$ *and* $k_{U*S,\mathbf{v}_2} = k_{U,\mathbf{v}_1} \to k_{U,\mathbf{v}_2}$*;*
3. $k_{U*S,\mathbf{v}_1+_F\mathbf{v}_2} = k_{U,\mathbf{v}_1} \vee k_{U,\mathbf{v}_2}$.

The proof proceeds straightforwardly once noticed that the dualising map $a \mapsto \neg a$, is such that $0 \mapsto 1$, $a \wedge b \mapsto \neg a \vee \neg b$, $a \ominus b \mapsto \neg b \to \neg a$.

LEMMA 2.2.2. *Let U be a regular triangulation of $[0,1]^n$, and let $\{m_{\mathbf{v}} \mid, \{\mathbf{v}\} \in U\}$ be a collection of integers such that $0 \leq m_{\mathbf{v}} \leq \mathrm{den}(\mathbf{v})$. Then*

$$\bigoplus_{\{\mathbf{v}\}\in U} m_{\mathbf{v}} h_{U,\mathbf{v}} = \bigodot_{\{\mathbf{v}\}\in U} k_{U,\mathbf{v}}^{l_{\mathbf{v}}},$$

for $l_{\mathbf{v}} = \mathrm{den}(\mathbf{v}) - m_{\mathbf{v}}$ for each $\{\mathbf{v}\} \in U$.

Proof. By Lemma 2.1.19, $\bigoplus_{\{\mathbf{v}\}\in U} m_{\mathbf{v}} h_{U,\mathbf{v}}$ is a McNaughton function in $\boldsymbol{F}^n_{\mathbb{MV}}$. By linearity, and the definition of each $l_{\mathbf{v}}$, the real-valued function

$$\bigoplus_{\{\mathbf{v}\}\in U} m_{\mathbf{v}} h_{U,\mathbf{v}} + \bigoplus_{\{\mathbf{v}\}\in U} l_{\mathbf{v}} h_{U,\mathbf{v}}$$

is the constant 1. Hence,

$$\bigoplus_{\{\mathbf{v}\}\in U} m_{\mathbf{v}} h_{U,\mathbf{v}} = \neg \bigoplus_{\{\mathbf{v}\}\in U} l_{\mathbf{v}} h_{U,\mathbf{v}}.$$

Now, by De Morgan laws,

$$\neg \bigoplus_{\{\mathbf{v}\}\in U} l_{\mathbf{v}} h_{U,\mathbf{v}} = \bigodot_{\{\mathbf{v}\}\in U} (\neg h_{U,\mathbf{v}})^{l_{\mathbf{v}}} = \bigodot_{\{\mathbf{v}\}\in U} k_{U,\mathbf{v}}^{l_{\mathbf{v}}}.$$

□

THEOREM 2.2.3. *For each integer $n \geq 0$, the free n-generated Wajsberg hoop $\boldsymbol{F}^n_{\mathbb{WH}}$ is the algebra of all McNaughton functions $f\colon [0,1]^n \to [0,1]$ such that*

$$f(1,1,\ldots,1) = 1\,,$$

equipped with pointwise defined operations.

Proof. An easy induction shows that $\varphi^{\boldsymbol{F}^n_{\mathbb{WH}}}(1,1,\ldots,1) = 1$ for each term φ. There remains to prove the converse inclusion. Take a McNaughton functions $f\colon [0,1]^n \to [0,1]$ such that $f(1,1,\ldots,1) = 1$. Lemma 2.1.18 provides an index i such that f is linear over each simplex of U_i. By Lemma 2.1.19 and Lemma 2.2.2,

$$f = \bigoplus_{\{\mathbf{v}\}\in U_i} m_{\mathbf{v}} h_{U_i\mathbf{v}} = \bigodot_{\{\mathbf{v}\}\in U_i} k^{l_{\mathbf{v}}}_{U_i,\mathbf{v}}\,, \tag{3}$$

for a suitable collection $\{l_{\mathbf{v}} \mid \{\mathbf{v}\} \in U_i\}$ of integers. By definition of starring of Schauder co-hats and Lemma 2.2.1, there exists a set of MV-terms $\{\varphi_{\mathbf{v}} \mid \{\mathbf{v}\} \in U_i\}$ such that $k_{U,\mathbf{v}} = (\varphi_{\mathbf{v}})^{\boldsymbol{F}^n_{\mathbb{MV}}}$. That is, $(\varphi_{\mathbf{v}})^{F^n_{\mathbb{MV}}}$ is a Schauder co-hat obtained by finitely many starrings from $K_{U^n_0}$.

It suffices to show that each n-variate Schauder co-hat $k_{U,\mathbf{v}}$, *actually occurring* in the rightmost part of (3) is the interpretation of a Wajsberg hoop term in $F^n_{\mathbb{WH}}$, that is, for each $\mathbf{v}$ such that $l_{\mathbf{v}} > 0$, there exists a Wajsberg hoop term $\psi_{\mathbf{v}}$ such that

$$\psi_{\mathbf{v}}^{\boldsymbol{F}^n_{\mathbb{WH}}} = \varphi_{\mathbf{v}}^{\boldsymbol{F}^n_{\mathbb{MV}}}\,.$$

If this is the case, by (3) it follows that $\left(\&_{\{\mathbf{v}\}\in U}\psi^{l_{\mathbf{v}}}_{\mathbf{v}}\right)^{\boldsymbol{F}^n_{\mathbb{WH}}} = f$, and hence $f \in \boldsymbol{F}^n_{\mathbb{WH}}$.

Notice now that the condition $f(1,1,\ldots,1) = 1$ implies $l_{\langle 1,1,\ldots,1\rangle} = 0$, that is the Schauder co-hat with apex $\langle 1,1,\ldots,1\rangle$ does not occur in the expression of f.

Then we only need to prove the following claim: for each integer $j > 0$ and each vertex $\mathbf{v} \neq \langle 1,1,\ldots,1\rangle$, $\{\mathbf{v}\} \in U_j$, there exists a Wajsberg hoop term $\psi_{\mathbf{v}}$ such that $\psi_{\mathbf{v}}^{\boldsymbol{F}^n_{\mathbb{WH}}} = \varphi_{\mathbf{v}}^{\boldsymbol{F}^n_{\mathbb{MV}}}$. The claim is proved by induction on j. It clearly holds for the base case $j = 0$, as, for each vertex $\mathbf{v} \neq \langle 1,1,\ldots,1\rangle$ of $[0,1]^n$ the following equivalence holds

$$\bigvee_{v_i=0} x_i \oplus \bigvee_{v_i=1} \neg x_i \equiv_{\mathbb{WH}} \bigvee_{v_j=0, v_k=1} x_k \to x_j\,.$$

Let now U_{j+1} be obtained from U_j by starring along a 1-simplex of vertices $\mathbf{v}_1$ and $\mathbf{v}_2$. By induction, each $k_{U_j,\mathbf{v}}$ with $\mathbf{v} \neq \langle 1,1,\ldots,1\rangle$ coincides with $\psi_{j,\mathbf{v}}^{F^n_{\mathbb{WH}}}$ for some Wajsberg hoop term $\psi_{j,\mathbf{v}}$. It remains to show that each one of the three Schauder co-hats $k_{U_j,\mathbf{v}_1} \vee k_{U_j,\mathbf{v}_2}$, $k_{U_j,\mathbf{v}_2} \to k_{U_j,\mathbf{v}_1}$ and $k_{U_j,\mathbf{v}_1} \to k_{U_j,\mathbf{v}_2}$ either is a co-hat with apex $\langle 1,1,\ldots,1\rangle$ or is expressible as the interpretation in $\boldsymbol{F}^n_{\mathbb{WH}}$ of a Wajsberg hoop term. If $\mathbf{v}_1 \neq \langle 1,1,\ldots,1\rangle \neq \mathbf{v}_2$ the conclusion trivially follows. Assume, without loss of generality, that $\mathbf{v}_1 = \langle 1,1,\ldots,1\rangle$ (then $\mathbf{v}_2 \neq \langle 1,1,\ldots,1\rangle$). Then both $k_{U_j,\mathbf{v}_1}$ and $k_{U_{j+1},\mathbf{v}_1} = k_{U_j,\mathbf{v}_2} \to k_{U_j,\mathbf{v}_1}$ have apex $\langle 1,1,\ldots,1\rangle$. Now, $g = \neg k_{U_j,\mathbf{v}_1}$ is an n-variate McNaughton function such that $g(1,1\ldots,1) = 1$ that is linear over each simplex of

U_j and is such that $g(\mathbf{v}) = 0$ for all vertices $\langle 1, 1, \ldots, 1\rangle \neq \mathbf{v}$ of U_j. Hence, letting $I = \{\mathbf{v} \mid \{\mathbf{v}\} \in U_j\} \setminus \{\langle 1, 1, \ldots, 1\rangle\}$, it holds that

$$g = \bigodot_{\mathbf{v}\in I} k_{U_j,\mathbf{v}}^{\mathrm{den}(\mathbf{v})} .$$

By induction, each $k_{U_j,\mathbf{v}}$ with $\mathbf{v} \in I$ is such that there is $\psi_{j,\mathbf{v}}$ with $\psi_{j,\mathbf{v}}^{\boldsymbol{F}^n_{\mathbb{WH}}} = k_{U_j,\mathbf{v}}$. Hence there exists a Wajsberg hoop term ϑ such that $\vartheta^{\boldsymbol{F}^n_{\mathbb{WH}}} = \neg k_{U_j,\mathbf{v}_1}$. We conclude the proof by noting that $k_{U_j,\mathbf{v}_1} \to k_{U_j,\mathbf{v}_2} = \neg k_{U_j,\mathbf{v}_1} \oplus k_{U_j,\mathbf{v}_2}$ and $k_{U_j,\mathbf{v}_1} \vee k_{U_j,\mathbf{v}_2} = (\neg k_{U_j,\mathbf{v}_1} \oplus k_{U_j,\mathbf{v}_2}) \to k_{U_j,\mathbf{v}_2}$. □

We can use Theorem 2.2.3 to express free MV-algebras via Wajsberg hoops, attaching one additional bit of information to each element of $\boldsymbol{F}^n_{\mathbb{WH}}$.

THEOREM 2.2.4. *Let WH_n be the universe of $\boldsymbol{F}^n_{\mathbb{WH}}$. Then*

$$\boldsymbol{F}^n_{\mathbb{MV}} \cong \langle WH_n \times \{0, 1\}, \boxplus, \neg, \bot\rangle ,$$

where

$$\bot = \langle \top, 0\rangle , \qquad \neg\langle f, b\rangle = \langle f, 1 - b\rangle ,$$

and

$$\langle f, b\rangle \boxplus (g, c) = \begin{cases} \langle f \odot g, 0\rangle & \textit{if } b = 0 = c , \\ \langle f \to g, 1\rangle & \textit{if } b = 0, c = 1 , \\ \langle g \to f, 1\rangle & \textit{if } b = 1, c = 0 , \\ \langle f \oplus g, 1\rangle & \textit{if } b = 1 = c . \end{cases}$$

Proof. Let φ be an MV-term. By using the following equivalences:

$$\varphi \,\&\, \neg\psi \equiv_{\mathbb{MV}} \neg(\varphi \to \psi), \quad \neg\varphi \,\&\, \neg\psi \equiv_{\mathbb{MV}} \neg(\varphi \oplus \psi),$$

and

$$\varphi \to \neg\psi \equiv_{\mathbb{MV}} \neg(\varphi \,\&\, \psi), \quad \neg\varphi \to \psi \equiv_{\mathbb{MV}} \varphi \oplus \psi, \quad \neg\varphi \to \neg\psi \equiv_{\mathbb{MV}} \psi \to \varphi,$$

one can construct a Wajsberg hoop term ψ such that either $\varphi \equiv_{\mathbb{MV}} \psi$ (if it holds $\varphi^{\boldsymbol{F}^n_{\mathbb{MV}}}(1, 1, \ldots, 1) = 1$) or $\varphi \equiv_{\mathbb{MV}} \neg\psi$ (if $\varphi^{\boldsymbol{F}^n_{\mathbb{MV}}}(1, 1, \ldots, 1) = 0$). It is now straightforward to check that the map defined for each $f \in WH_n$, by $\langle f, 0\rangle \mapsto \neg f$, $\langle f, 1\rangle \mapsto f$, is an isomorphism of MV-algebras. □

2.3 Finitely Valued Łukasiewicz Logics

For each $k > 0$, the subvariety $\mathbb{MV}_k$ that constitutes the equivalent algebraic semantics of the $k + 1$-valued Łukasiewicz logic $Ł_k$ is axiomatised via Grigolia's axioms:

- $x^k = x^{k+1}$;
- $(k + 1)x^h = (h(x^{h-1}))^{k+1}$, for every integer $h \in \{2, \ldots, k - 1\}$ that does not divide k.

The variety $\mathbb{MV}_k$ is generated by the $(k+1)$-element MV-chain $\mathbf{Ł}_k$. By Lemma 1.0.12, $\boldsymbol{F}^n_{\mathbb{MV}_k}$ is the subalgebra of $\mathbf{Ł}_k^{\mathbf{Ł}_k^n}$ generated by the projections. A moment's reflection shows that this algebra coincides with the algebra of restrictions of McNaughton functions in $\boldsymbol{F}^n_{\mathbb{MV}}$ to the set

$$D^n_k = \left\{0, \frac{1}{k}, \frac{2}{k}, \ldots, \frac{k-1}{k}, 1\right\}^n .$$

It is easy to see the following.

LEMMA 2.3.1. *For each integer $n > 0$ and each integer $k > 0$,*

$$\boldsymbol{F}^n_{\mathbb{MV}_k} \cong \prod_{\mathbf{v} \in D^n_k} \mathbf{Ł}_{\mathrm{den}(\mathbf{v})} .$$

It remains to compute the denominator of elements in D^n_k. For each integer $d \geq 0$ we denote $\langle \mathrm{Div}(d), | \rangle$ the lattice of (nonnegative) divisors of d, where the partial order is given by $a|b$ if a divides b.

LEMMA 2.3.2. *For each pair of positive integers n, k the set D^n_k contains only rational points whose denominator divides k. More precisely, for every integer d that divides k, the set D^n_k contains exactly $\alpha(n, d)$ many points of denominator d, where*

- $\alpha(0, 1) = 1$ *and* $\alpha(0, d) = 0$ *if* $d > 1$*;*
- $$\alpha(n, d) = (d+1)^n + \sum_{\emptyset \neq X \subseteq \mathrm{CoAtDiv}(d)} (-1)^{|X|} (\gcd(X) + 1)^n ,$$

  where $\mathrm{CoAtDiv}(d)$ *is the set of coatoms in the lattice of divisors of* d*.*

Proof. Trivially, in $[0, 1]^n$ there are exactly $(d+1)^n$ points whose denominator is a divisor of d. We then have to subtract from this set the points whose denominator is a proper divisor of d. We shall use an instance of the combinatorial principle of inclusion/exclusion stating that for a collection $\{A_i\}_{i \in I}$ of sets,

$$\left| \bigcup_{i \in I} A_i \right| = \sum_{S \subseteq I} (-1)^{|S|-1} \left| \bigcap_{i \in S} A_i \right| . \tag{4}$$

For each nonempty subset X of coatoms of $\mathrm{Div}(d)$ we let L_X be the universe of the MV-algebra $\mathbf{Ł}^n_{\gcd(X)}$. Notice that L_X is the set of all rational points in $[0, 1]^n$ whose denominator divide each element of X. Then

$$\alpha(n, d) = (d+1)^n - \left| \bigcup_{x \in \mathrm{CoAtDiv}(d)} L_{\{x\}} \right| .$$

Consider now two nonempty subsets X, Y of coatoms in $\mathrm{Div}(d)$. Notice that

1. $|L_X \cup L_Y| = |L_X| + |L_Y| - |L_X \cap L_Y|$;
2. $L_X \cap L_Y = L_{X \cup Y}$.

Those two statements together with (4) immediately yield

$$\left| \bigcup_{x \in \mathrm{CoAtDiv}(d)} L_{\{x\}} \right| = \sum_{\emptyset \neq X \subseteq \mathrm{CoAtDiv}(d)} (-1)^{|X|-1} |L_X| \,,$$

that in turns, since $|L_X| = (\gcd(X)+1)^n$, gives the desired expression for $\alpha(n,d)$. $\square$

We can now improve on the description of Lemma 2.3.1.

THEOREM 2.3.3. *For each integer $n > 0$ and each integer $k > 0$,*

$$\boldsymbol{F}^n_{\mathbb{MV}_k} \cong \prod_{d \in \mathrm{Div}(k)} \boldsymbol{Ł}_d^{\alpha(n,d)} \,.$$

As $\mathbb{MV}_k$ is a locally finite variety, we can turn Theorem 2.3.3 into a combinatorial description of the prime spectrum of $\boldsymbol{F}^n_{\mathbb{MV}_k}$, and work on the dual space given by the spectrum. This kind of dual representation is exploited in Section 4.2, to deal with Gödel logic and other schematic extensions of WNM. It is further explored in Section 5.4 for locally finite subvarieties of $\mathbb{BL}$, where, as an application, we obtain explicitly the description of the prime spectrum of $\boldsymbol{F}^n_{\mathbb{MV}_k}$.

2.4 Extensions of MV-algebras

In this section we deal with logics $\mathrm{MV}\Delta$, $\mathrm{PMV}\Delta$, ŁΠ and ŁΠ$\frac{1}{2}$. We recall that the corresponding varieties are generated respectively by the standard algebras

$$\langle [0,1], \odot, \rightarrow_\odot, 1, \Delta \rangle \,, \qquad \langle [0,1], \odot, \cdot, \rightarrow_\odot, 1, \Delta \rangle \,,$$
$$\langle [0,1], \odot, \cdot, \rightarrow_\odot, \rightarrow_\cdot, 1 \rangle \,, \qquad \langle [0,1], \odot, \cdot, \rightarrow_\odot, \rightarrow_\cdot, 1, 1/2 \rangle \,,$$

where $\odot$ is Łukasiewicz t-norm, $\cdot$ the product of reals, and Δ is Baaz's projection operator.

Recall the algebras of polynomials introduced in Definition 2.0.14.

DEFINITION 2.4.1. *A subset S of $[0,1]^n$ is* Q-semialgebraic *if it is a boolean combination of sets of the form*

$$\{\langle x_1, \ldots, x_n \rangle \in [0,1]^n \mid P(\langle x_1, \ldots, x_n \rangle) > 0\}$$

for $P \in \mathsf{Z}[x_1, \ldots, x_n]$. If every $P \in \mathsf{Z}_1[x_1, \ldots, x_n]$ then S is linear Q-semialgebraic.

DEFINITION 2.4.2. *Let $\mathbb{E}$ be an equational class among $\mathbb{MV}\Delta$, $\mathbb{PMV}\Delta$ and ŁΠ$\frac{1}{2}$. An E-*hat over $[0,1]^n$ *is a function $h : [0,1]^n \to [0,1]$ such that there exist a* Q-*semialgebraic set S and a rational function $P/Q \in \mathsf{Q}(x_1, \ldots, x_n)$ such that:*

- *Q has no zeros on S, $h = P/Q$ on S and $h = 0$ on $[0,1]^n \setminus S$;*

- *if* $\mathbb{E} = \mathbb{PMV}\Delta$*, then* $Q = 1$*;*
- *if* $\mathbb{E} = \mathbb{MV}\Delta$*, then* $Q = 1$*,* $P \in \mathsf{Z}_1[x_1, \ldots, x_n]$ *and* S *is linear* Q*-semialgebraic.*

If h is the hat determined by S and P/Q as above, then we write $h = \langle S, P/Q\rangle$. An E*-function over* $[0,1]^n$ is a finite sum f of E-hats

$$f = \langle S_1, P_1/Q_1\rangle + \cdots + \langle S_r, P_r/Q_r\rangle$$

subject to the condition that $S_i \cap S_j = \emptyset$, for $i \neq j$.

We denote by $B_n(E)$ the set of all E-functions over $[0,1]^n$. A direct inspection shows that $B_n(E)$ equipped with the pointwise operations is an E-algebra that contains the projection functions.

LEMMA 2.4.3. *For any* $P \in \mathsf{Z}[x_1, \ldots, x_n]$ *let* $P^{\#} : [0,1]^n \to [0,1]$ *be defined by*

$$P^{\#}(v) = \min\{1, \max\{P(v), 0\}\}\,.$$

Then $P^{\#} \in \boldsymbol{F}^n_{\mathbb{PMV}\Delta}$*. Further, if* $P \in \mathsf{Z}_1[x_1, \ldots, x_n]$ *then* $P^{\#} \in \boldsymbol{F}^n_{\mathbb{MV}}$*.*

Proof. The proof is analogous to the proof of Lemma 2.1.21. Indeed let

$$P = k + \sum_{\langle i_0, i_1, \ldots, i_n\rangle \in \mathsf{N}^{n+1}} k_{i_0} x_1^{i_1} \cdot \ldots \cdot x_n^{i_n}$$

where $k, k_{i_0} \in \mathsf{Z}$ and $k_{i_0} \neq 0$ only for finitely many i_0's. The proof proceeds by induction on $K = \sum_{\langle i_0, i_1, \ldots, i_n\rangle \in \mathsf{N}^{n+1}} |k_{i_0}|$. If $K = 0$ then P is either equal to 0 or to 1 and the claim is settled. If $K \geq 1$ suppose there is $i \in I$ with $k_{i_0} \geq 1$. Then letting $Q = P - x_1^{i_1} \cdot \ldots \cdot x_n^{i_n}$ by induction hypothesis $Q^{\#}$ and $(Q+1)^{\#}$ are in $\boldsymbol{F}^n_{\mathbb{PMV}\Delta}$. It is now easy to check that

$$P^{\#} = (Q^{\#} \oplus x_1^{i_1} \cdot \ldots \cdot x_n^{i_n}) \odot (Q+1)^{\#}\,.$$

The last case is when $K \geq 0$ and for every $i \in I$, $k_{i_0} < 0$. Then the argument above can be applied to $1 - P$ obtaining $(1-P)^{\#} \in \boldsymbol{F}^n_{\mathbb{PMV}\Delta}$. The claim follows by noticing that $P^{\#} = \neg(1-P)^{\#}$. □

THEOREM 2.4.4. *Let* $\mathbb{E}$ *be any of* $\mathbb{MV}\Delta$*,* $\mathbb{PMV}\Delta$ *and* $\text{Ł}\Pi\frac{1}{2}$*. Then* $\boldsymbol{F}^n_{\mathbb{E}} \cong \boldsymbol{B}_n(E)$*.*

Proof. Since $\boldsymbol{B}_n(E)$ contains the projections and it is closed under the E-operations, we only have to show that each E-function is contained in $\boldsymbol{F}^n_{\mathbb{E}}$. Each E-function f has the form

$$f = \langle S_1, P_1/Q_1\rangle + \cdots + \langle S_r, P_r/Q_r\rangle$$

where S_i are pairwise disjoint algebraic sets, hence it can be written as

$$f = \langle S_1, P_1/Q_1\rangle \oplus \cdots \oplus \langle S_r, P_r/Q_r\rangle\,.$$

So in order to prove the theorem, it is enough to prove that each E-hat $\langle S_i, P_i/Q_i\rangle$ is in $\boldsymbol{F}^n_{\mathbb{E}}$.

Let $P \in \mathsf{Z}[x_1, \ldots, x_n]$ (resp. in $\mathsf{Z}_1[x_1, \ldots, x_n]$). Then $1 - \Delta(1 - P^{\#})$ is the characteristic function of the set $\{P > 0\} \subseteq [0,1]^n$. Hence the characteristic function of any Q-semialgebraic set (resp. linear Q-semialgebraic set) is an element of $\boldsymbol{F}^n_{\mathbb{PMV}\Delta}$ (resp. of $\boldsymbol{F}^n_{\mathbb{MV}\Delta}$).

Let $\langle S, P/Q\rangle$ be a $\mathbb{PMV}\Delta$-hat. Then $Q = 1$ and letting s be the characteristic function of S, we have $\langle S, P/Q\rangle = s \wedge P^{\#} \in \boldsymbol{F}^n_{\mathbb{PMV}\Delta}$. The case for $\mathbb{MV}\Delta$ proceeds analogously.

If $\mathbb{E} = \text{ŁΠ}\frac{1}{2}$ we can suppose $P \geq 0$ and $Q > 0$ on S. Further, write P and Q as

$$P = \sum_{i \in I} a_i \lambda_i - \sum_{j \in J} b_j \mu_j$$
$$Q = \sum_{r \in R} c_r \pi_r - \sum_{t \in T} d_t \rho_t$$

where $a_i, b_j, c_r, d_t \in \mathsf{N}$ and $\lambda_i, \mu_i, \pi_r, \rho_t$ are monic monomials in $x_1, \ldots, x_n$. Then let m be such that $\sum_{I,J,R,T} a_i + b_j + c_r + d_t \leq 2^m$ and

$$p = \bigoplus_{i \in I} a_i \cdot (1/2)^m \cdot \lambda_i \ominus \bigoplus_{j \in J} b_j \cdot (1/2)^m \cdot \mu_j =$$
$$1/2^m \cdot \left(\bigoplus_{i \in I} a_i \cdot \lambda_i \ominus \bigoplus_{j \in J} b_j \cdot \mu_j \right)$$
$$q = \bigoplus_{r \in R} c_r \cdot (1/2)^m \cdot \pi_i \ominus \bigoplus_{t \in T} d_t \cdot (1/2)^m \cdot \rho_j =$$
$$1/2^m \cdot \left(\bigoplus_{r \in R} c_r \cdot \pi_i \ominus \bigoplus_{t \in T} d_t \cdot \rho_j \right).$$

Then $p/q \in \boldsymbol{F}^n_{\text{ŁΠ}\frac{1}{2}}$ and $\langle S, P/Q\rangle = s \wedge (p/q) \in \boldsymbol{F}^n_{\text{ŁΠ}\frac{1}{2}}$. □

Let $\boldsymbol{B}_n(\text{ŁΠ})$ be the subalgebra of $\boldsymbol{B}_n(\text{ŁΠ}\frac{1}{2})$ made of functions mapping $\{0,1\}^n$ to $\{0,1\}$.

THEOREM 2.4.5. $\boldsymbol{F}^n_{\text{ŁΠ}} = \boldsymbol{B}_n(\text{ŁΠ})$.

Proof. Let $b \in \boldsymbol{B}_n(\text{ŁΠ})$, then by Theorem 2.4.4 there exists a term t in the language of $\text{ŁΠ}\frac{1}{2}$ that coincides with b when interpreted in $[0,1]$. Consider the term

$$p = \bigvee_{1 \leq i \leq n} \neg p_i$$

where

$$p_i = ((x_i \wedge \neg x_i) \oplus (x_i \wedge \neg x_i)) \rightarrow. (x_i \wedge \neg x_i).$$

It is easy to check that for every $v \in [0,1]^n$,

$$p(v) = \begin{cases} 1/2 & \text{if } v \notin \{0,1\}^n \\ 0 & \text{otherwise.} \end{cases}$$

Then substitute every occurrence in t of the constant $1/2$ with the term p. We obtain a term t' that interpreted in $[0,1]^n$ coincides with t except than on the vertices $\{0,1\}^n$. But each vertex is a Q-semialgebraic set, hence its characteristic function is a term over the language of $\mathbb{PMV}\Delta$ hence of ŁΠ. The result follows by considering a suitable boolean combination of t' and the characteristic functions of vertices. □

Note that in this treatment the variety $\mathbb{PMV}$ generated by the standard algebra $\langle [0,1], \odot, \cdot, \rightarrow_{\odot}, 1\rangle$ is not considered. Indeed, a description of the free algebra in this variety would settle the Pierce-Birkhoff conjecture in semialgebraic geometry [52].

2.5 MV-algebras with Division Operators

DEFINITION 2.5.1. *A* DMV-algebra $\boldsymbol{A} = \langle A, \oplus, \neg, \{\delta_n\}_{n\in\mathsf{N}}, 0, 1\rangle$ *is an algebraic structure such that* $\boldsymbol{A}^* = \langle A, \oplus, \neg, 0, 1\rangle$ *is an MV-algebra and the following hold for every* $x \in A$ *and* $n \in \mathsf{N}$*:*

(D1n) $n\,\delta_n x = x$

(D2n) $\delta_n x \odot (n-1)\,\delta_n x = 0$

The variety $\mathbb{DMV}$ of DMV-algebras is generated by

$$\langle [0,1], \oplus, \neg, \{_/n\}_{n\in\mathsf{N}}, 0\rangle\,,$$

hence the free DMV-algebra $\boldsymbol{F}^n_{\mathbb{DMV}}$ over n generators is isomorphic to the DMV-algebra of functions from $[0,1]^n$ to $[0,1]$ generated by the projections.

THEOREM 2.5.2. $\boldsymbol{F}^n_{\mathbb{DMV}}$ *is the algebra of continuous piecewise linear functions with rational coefficients.*

Proof. A direct inspection shows that every element of $\boldsymbol{F}^n_{\mathbb{DMV}}$ is a continuous piecewise linear function, where each piece has rational coefficients.

Let $f : [0,1]^n \rightarrow [0,1]$ be a continuous piecewise linear function with rational coefficients. Further, let s be an integer such that $s \cdot f : \mathbf{t} \in [0,1]^n \mapsto s \cdot f(\mathbf{t}) \in [0,s]$ is a continuous function with integer coefficients (for example s is the least common multiple of the denominators of the coefficients of pieces of f).

For every $i = 0, \ldots, s-1$, let

$$f_i : \mathbf{t} \in [0,1]^n \mapsto ((s \cdot f(\mathbf{t}) - i) \wedge 1) \vee 0 \in [0,1].$$

Since f_i are continuous functions with integer coefficients there exist MV-terms ψ_i such that $f_i = \psi_i^{\boldsymbol{F}^n_{\mathbb{MV}}}$. For any function $g : [0,1]^n \rightarrow [0,1]$ we define

$$\begin{aligned} Supp(g) &= \{\mathbf{t} \in [0,1]^n \mid g(\mathbf{t}) > 0\} \\ Supp^{<1}(g) &= \{\mathbf{t} \in [0,1]^n \mid 0 < g(\mathbf{t}) < 1\}. \end{aligned}$$

We have, for every $i = 1, \ldots, s-1$,

$$Supp^{<1}(f_i) \subseteq Supp(f_i) \subseteq Supp(f_{i-1}).$$

Indeed

$$Supp(f_i) = \{\mathbf{t} \in [0,1]^n \mid s \cdot f(\mathbf{t}) > i\} \subseteq \{\mathbf{t} \in [0,1]^n \mid s \cdot f(\mathbf{t}) > i-1\}.$$

Further, for any $i \neq j$, $Supp^{<1}(f_i) \cap Supp^{<1}(f_j) = \emptyset$.

Then

$$f(\mathbf{t}) = \varphi^{\boldsymbol{F}^n_{\mathbb{DMV}}}(\mathbf{t}) \qquad \text{where} \quad \varphi = \bigoplus_{i=0}^{s-1} \delta_s \psi_i.$$

Indeed, suppose that $\mathbf{t} \in [0,1]^n$ is such that $f(\mathbf{t}) = 0$. Then for every $i = 0, \ldots, s-1$, $f_i(\mathbf{t}) = 0$ whence $\psi_i^{\boldsymbol{F}^n_{\mathbb{DMV}}}(\mathbf{t}) = 0$ and $\varphi^{\boldsymbol{F}^n_{\mathbb{DMV}}}(\mathbf{t}) = 0 = f(\mathbf{t})$.

If $f(\mathbf{t}) = 1$ then for every $i = 0 \ldots, s-1$, $\varphi_i^{\boldsymbol{F}^n_{\mathbb{DMV}}}(\mathbf{t}) = 1$ whence $\delta_s \varphi_i^{\boldsymbol{F}^n_{\mathbb{DMV}}}(\mathbf{t}) = 1/s$ and $\varphi^{\boldsymbol{F}^n_{\mathbb{DMV}}}(\mathbf{t}) = 1$.

Suppose now that there exists $i \in \{0, \ldots, s-1\}$ such that $i < s \cdot f(\mathbf{t}) \leq i+1$. Then $\mathbf{t} \in Supp(f_i)$ and $f(\mathbf{t}) = (f_i(\mathbf{t}) + i)/s$. For every $j > i$, we have $s \cdot f(\mathbf{t}) - j \leq s \cdot f(\mathbf{t}) - i - 1 < 0$ whence $f_j(\mathbf{t}) = \psi_j^{\boldsymbol{F}^n_{\mathbb{DMV}}}(\mathbf{t}) = 0$. Further, for every $j < i$, we have $s \cdot f(\mathbf{t}) - j \geq s \cdot f(\mathbf{t}) - i + 1 > 1$ whence $f_j(\mathbf{t}) = 1$ and $(\delta_s \psi_j)^{\boldsymbol{F}^n_{\mathbb{DMV}}}(\mathbf{t}) = 1/s$.

Hence

$$\begin{aligned} \varphi^{\boldsymbol{F}^n_{\mathbb{DMV}}}(\mathbf{t}) &= \left(\bigoplus_{j=0}^{i-1} \delta_s \psi_j \oplus \delta_s \psi_i \oplus \bigoplus_{j=i+1}^{s-1} \delta_s \psi_j \right)^{\boldsymbol{F}^n_{\mathbb{DMV}}} (\mathbf{t}) = \\ &= \sum_{j=0}^{i-1} \frac{1}{s} + \frac{f_i(\mathbf{t})}{s} + 0 = f(\mathbf{t}) \,. \end{aligned}$$

□

3 Product Algebras

In this section we shall deal with free Product algebras and their representations. A first step towards these representation consists in giving a representation of free algebras in a variety deeply related to Product algebras, that is, the variety of cancellative hoops.

3.1 Free Cancellative Hoops

The variety of cancellative hoops is generated by

$$(0,1]^- = \langle (0,1], \cdot, \rightarrow_\cdot, 1 \rangle \,,$$

hence by Lemma 1.0.12 the free cancellative hoop over n generators is the cancellative hoop of functions from $(0,1]^n$ to $(0,1]$ that is generated by the projection functions.

A *monomial function* of order n is a function $f : (0,1]^n \to (0,1]$ that can be written as

$$f(x_1, \ldots, x_n) = 1 \wedge \prod_{i=1}^{n} x_i^{m_i},$$

where $m_i \in \mathsf{Z}$ for each $i = 1, \ldots, n$.

THEOREM 3.1.1. *The free cancellative hoop $\boldsymbol{F}^n_{\mathbb{CH}}$ over n generators is isomorphic to the hoop of piecewise monomial functions, that is functions $f : (0,1]^n \to (0,1]$ such that there is a finite family $\{f_{pq}\}$ of monomial functions with*

$$f = \bigvee_p \bigwedge_q f_{pq}\,. \tag{5}$$

Proof. A structural induction shows that every n-variate term over the language of cancellative hoops, when interpreted as a function over $[0,1]^n$, is a piecewise monomial function.

We prove that if f is a function of the form (5) then it can be written starting from projection functions using the operations of cancellative hoops. Indeed, let

$$f(t_1,\ldots,t_n) = 1 \wedge \prod_{i=1}^n t_i^{m_i},$$

and let $H = \{i \mid m_i > 0\}$ and $K = \{i \mid m_i < 0\}$. If $K = \emptyset$, then $f(t_1,\ldots,t_n) = \prod_{i=1}^n t_i^{m_i}$ with all $m_i \geq 0$, hence the term τ_f given by

$$\tau_f = \underbrace{x_1 \& \cdots \& x_1}_{m_1 \text{ times}} \& \cdots \& \underbrace{x_n \& \cdots \& x_n}_{m_n \text{ times}}$$

is such that $\tau_f^{\boldsymbol{F}^n_{\mathbb{CH}}} = f$. (We are denoting by x_i the i-th propositional variable.) If $H = \emptyset$ then $f = 1$ hence we can set $\tau_f = \top$. Otherwise, if $H \neq \emptyset$ and $K \neq \emptyset$, then letting

$$g = \prod_{i \in H} t_i^{m_i} \qquad h = \prod_{i \in K} t_i^{-m_i}\,,$$

we have $\tau_f = \tau_h \to \tau_g$.

If f has the form (5), then it is a boolean combination of monomial functions and the claim follows from the fact that boolean operators can be defined in cancellative hoops. □

3.2 Free Product Algebras

Recall that the variety $\mathbb{P}$ of product algebras is generated by

$$\langle [0,1], \cdot, \to_\cdot, 0 \rangle$$

where $x \cdot y = xy$ is the usual product of real numbers, and $x \to_\cdot y$ is its residuum: if $x > y$ then $x \to_\cdot y = y/x$.

For any product algebra $\boldsymbol{A}$, we let $\mathbf{B}(\boldsymbol{A}) = \{\neg\neg x \mid x \in A\}$. $\mathbf{B}(\boldsymbol{A})$ is the greatest Boolean algebra that is a subalgebra of $\boldsymbol{A}$.

PROPOSITION 3.2.1. *For each cardinal $\kappa > 0$, $\mathbf{B}(\boldsymbol{F}^\kappa_{\mathbb{P}})$ is the free Boolean algebra over κ generators. If Y is a set of generators of $\boldsymbol{F}^\kappa_{\mathbb{P}}$, then $\neg\neg Y$ is a set of free generators of $\mathbf{B}(\boldsymbol{F}^\kappa_{\mathbb{P}}) \cong \boldsymbol{F}^\kappa_{\mathbb{B}}$.*

Proof. Let $\boldsymbol{B}$ be any boolean algebra and $f : \neg\neg Y \to \boldsymbol{B}$. Since $\boldsymbol{B}$ is a product algebra, then there is a unique $\overline{f} : \boldsymbol{F}^n_{\mathbb{P}} \to \boldsymbol{B}$ such that for every $y \in Y$, $f(\neg\neg y) = \overline{f}(y)$. Since $\mathbf{B}(\boldsymbol{F}^n_{\mathbb{P}}) \subseteq \boldsymbol{F}^n_{\mathbb{P}}$, then the map $\overline{\overline{f}}$ defined by $\overline{\overline{f}}(t) = \overline{f}(t)$ for each $t \in \mathbf{B}(\boldsymbol{F}^n_{\mathbb{P}})$ uniquely extend f to a homomorphism of boolean algebras. □

We can then identify $\mathbf{B}(\boldsymbol{F}^n_{\mathbb{P}})$ with $\boldsymbol{F}^n_{\mathbb{B}}$. It is not difficult to check that for any atom z of $\boldsymbol{F}^n_{\mathbb{B}} = \mathbf{B}(\boldsymbol{F}^n_{\mathbb{P}})$, the interval $[\bot, z] \subseteq \boldsymbol{F}^n_{\mathbb{P}}$ equipped with the operations $\cdot$ and

$$x \to_z y = z \wedge (x \to y)\,,$$

is a product algebra.

PROPOSITION 3.2.2. *For each finite cardinal $n \geq 1$, let $\mathrm{at}(\mathbf{B}(\boldsymbol{F}^n_{\mathbb{P}})) \subseteq \boldsymbol{F}^n_{\mathbb{P}}$ be the set of atoms of $\mathbf{B}(\boldsymbol{F}^n_{\mathbb{P}})$. Then*

$$\boldsymbol{F}^n_{\mathbb{P}} \cong \prod_{z \in \mathrm{at}(\mathbf{B}(\boldsymbol{F}^n_{\mathbb{P}}))} [\bot, z]\,.$$

Proof. By Proposition 3.2.1, $\mathrm{at}(\mathbf{B}(\boldsymbol{F}^n_{\mathbb{P}})) = \mathrm{at}(\boldsymbol{F}^n_{\mathbb{B}})$. Display the set of atoms $\mathrm{at}(\boldsymbol{F}^n_{\mathbb{B}})$ as $\{z_1, \ldots, z_k\}$. The map $\varphi : f \in \boldsymbol{F}^n_{\mathbb{P}} \to (f \wedge z_1, \ldots, f \wedge z_k) \in [\bot, z_1] \times \cdots \times [\bot, z_k]$ is an isomorphism. Indeed for each $\langle y_1, \ldots, y_k\rangle \in \prod_{z \in \mathrm{at}(\boldsymbol{F}^n_{\mathbb{B}})}[\bot, z]$ it holds that $\langle y_1, \ldots, y_k\rangle = \varphi(y_1 \vee \cdots \vee y_k)$, since $z_i \wedge z_j = \bot$ for $i \neq j$. If $\varphi(f) = \varphi(g)$ then for every $i = 1, \ldots, k$, $f \wedge z_i = g \wedge z_i$ hence $\bigvee(f \wedge z_i) = \bigvee(g \wedge z_i)$ and this implies $f = g$ since $\bigvee_{i=1}^k z_i = \top$. □

Since the generators of $\boldsymbol{F}^n_{\mathbb{B}}$ are $\neg\neg\pi_i$ where π_i are the free generators of $\boldsymbol{F}^n_{\mathbb{P}}$, and since $\neg\neg\neg x = \neg\neg x$, then each atom of $\boldsymbol{F}^n_{\mathbb{B}}$ has the form

$$p_\epsilon = \bigwedge_{i=1}^{n} \neg^{\epsilon_i}\pi_i\,,$$

where $\epsilon = (\epsilon_1, \ldots, \epsilon_k) \in \{1, 2\}^k$ and $\neg^1 = \neg$ while $\neg^2 = \neg\neg$.

Since $\boldsymbol{F}^n_{\mathbb{B}} \cong \mathbf{B}(\boldsymbol{F}^n_{\mathbb{P}})$, then elements of $\boldsymbol{F}^n_{\mathbb{B}}$ can be seen as functions from $[0, 1]^n \to [0, 1]$. Let $\#\epsilon = |\{i \mid \epsilon_i = 2\}|$ and $G_\epsilon = \{\langle t_1, \ldots, t_n\rangle \in [0, 1]^n \mid t_i > 0 \text{ iff } \epsilon_i = 2\}$. The function p_ϵ is equal to 1 over G_ϵ and it is 0 otherwise, hence each function $f \in [\bot, p_\epsilon]$ is equal to 0 outside G_ϵ.

We need the following technical lemma that can be shown by direct inspection of the operations on $[0, 1]$.

LEMMA 3.2.3. *If $f : [0, 1]^n \to [0, 1]$ belongs to $\boldsymbol{F}^n_{\mathbb{P}}$, then:*

- *If there is $\langle t_1, \ldots, t_n\rangle \in G_\epsilon$ such that $f(t_1, \ldots, t_n) = 0$ then $f(s_1, \ldots, s_n) = 0$ for every $\langle s_1, \ldots, s_n\rangle \in G_\epsilon$;*
- $f(0, \cdots, 0) \in \{0, 1\}$.

PROPOSITION 3.2.4. *For every $\epsilon \in \{1, 2\}^n$, $(\bot, p_\epsilon]$ is isomorphic to the free cancellative hoop over $\#\epsilon$ generators.*

Proof. If $\#\epsilon = 0$, then $G_\epsilon = \{(0,\dots,0)\}$ and $(\bot, p_\epsilon] = \{\top\}$ that is the free cancellative hoop over 0 generators. Let $\#\epsilon = k > 0$ and $f \in (\bot, p_\epsilon]$. Then, by Lemma 3.2.3, the function f is never equal to 0 over G_ϵ, hence the function $\widetilde{f}(s_1,\dots,s_k) = f(t_{i_1},\dots,t_{i_n})$ where $t_{i_j} = s_j$ for $j = 1,\dots,k$ and $t_i = 0$ otherwise, is a function from $(0,1]^k \to (0,1]$. Let

$$\varphi_\epsilon : f \in (\bot, p_\epsilon] \to \widetilde{f} \in (0,1]^{(0,1]^k}.$$

It is easy to check that $\varphi_\epsilon((\bot, p_\epsilon])$ contains the projections and it is closed under hoops operations, hence it contains the free cancellative hoop over k generators. In order to prove the other inclusion, let $f \in (\bot, p_\epsilon]$. Since $f \in \boldsymbol{F}^n_{\mathbb{P}}$ then there exists a term τ in the language of product algebras such that $f = \tau^{\boldsymbol{F}^n_{\mathbb{P}}} \leq p_\epsilon$. The claim follows by noticing that there exists a term $\widetilde{\tau}$ with k variables in the language of cancellative hoops, such that $\widetilde{\tau}^{\boldsymbol{F}^n_{\mathbb{CH}}} = \widetilde{f}$. Indeed we can proceed by structural induction: if τ is a variable, then it can only be equal to a x_{i_j} hence $\widetilde{\tau} = x_{i_j}$. If $\tau = \tau_1 \cdot \tau_2$, recalling that $f > 0$, then it can only be $\tau_1^{\boldsymbol{F}^n_{\mathbb{P}}} > 0$ and $\tau_2^{\boldsymbol{F}^n_{\mathbb{P}}} > 0$, hence the claim follows by induction hypothesis. If $\tau = \tau_1 \to \tau_2$ then we have to consider the cases in which either $\tau_1^{\boldsymbol{F}^n_{\mathbb{P}}} = 0$ or $\tau_2^{\boldsymbol{F}^n_{\mathbb{P}}} = 0$. But if $\tau_1^{\boldsymbol{F}^n_{\mathbb{P}}} = 0$ then $\tau^{\boldsymbol{F}^n_{\mathbb{P}}} = 1$ hence we can set $\widetilde{\tau} = \top$. On the other hand, if $\tau_2^{\boldsymbol{F}^n_{\mathbb{P}}} = 0$ since $f > 0$, it can only be $\tau_1^{\boldsymbol{F}^n_{\mathbb{P}}} = 0$. □

Noticing that $\{G_\epsilon \mid \epsilon \in \{1,2\}^n\}$ is a partition of $[0,1]^n$, by Theorem 3.1.1 and Proposition 3.2.4, we have a representation of functions of $\boldsymbol{F}^n_{\mathbb{P}}$.

THEOREM 3.2.5. *The free Product algebra $\boldsymbol{F}^n_{\mathbb{P}}$ is isomorphic to the algebra of functions $f : [0,1]^n \to [0,1]$ such that for every $\epsilon \in \{1,2\}^n$, with $\#\epsilon = k$, the restriction of f to G_ϵ either is equal to 0 or is a k-variate piecewise monomial function.*

Let $\boldsymbol{G} = \langle G, +, -, \leq, 0\rangle$ be any abelian lattice ordered group (ℓ-group) and let $\bot$ be an element not belonging to $\boldsymbol{G}$. Let G^- be the negative cone of $\boldsymbol{G}$. On the set $G^- \cup \{\bot\}$ we define the operations $\odot$ and $\to$ as follows:

$$x \odot y = \begin{cases} x + y & \text{if } x, y \in G^-, \\ \bot & \text{otherwise,} \end{cases}$$

$$x \to y = \begin{cases} 0 \wedge (y - x) & \text{if } x, y \in G^-, \\ 0 & \text{if } x = \bot \\ \bot & \text{if } x \in G^- \text{ and } y = \bot. \end{cases}$$

Then $\langle G^- \cup \{\bot\}, \odot, \to, \bot\rangle$ is a Product algebra that will be denoted by $\mathfrak{B}(\boldsymbol{G})$. $\mathfrak{B}$ can be extended to a functor from the category of ℓ-groups to the category of Product algebras.

The free cancellative hoop $\boldsymbol{F}^n_{\mathbb{CH}}$ is a negatively ordered cancellative monoid, hence there is a unique (up to isomorphism) ℓ-group $\boldsymbol{G}_n$ such that $\boldsymbol{G}_n^- \cong \boldsymbol{F}^n_{\mathbb{CH}}$. Let $\boldsymbol{G}_0 = \{0\}$.

THEOREM 3.2.6. *For each integer $n \geq 1$, $\boldsymbol{F}^n_{\mathbb{P}} \cong \prod_{k=0}^n \mathfrak{B}(\boldsymbol{G}_k)^{\binom{n}{k}}$.*

Proof. For any k there are $\binom{n}{k}$ atoms p_ϵ in $\boldsymbol{F}^n_{\mathbb{B}}$ such that $\#\epsilon = k$. The claim then follows by Theorem 3.2.5, noticing that $\mathfrak{B}(\boldsymbol{G}_k) \cong [0, p_\epsilon]$. □

4 Gödel Algebras and Related Structures

In this section we deal with Gödel algebras. We first introduce a functional representation for free finitely generated Gödel algebras, built up, as we have seen in the previous sections for other varieties, from the generic Gödel algebra $[0,1]$ using Lemma 1.0.12. Then we expound representations of finite Gödel algebras via spectral duality. This last approach is very fruitful for describing free algebras in locally finite varieties. In particular we shall introduce a category of finite combinatorial objects which is dually equivalent to the category of finite Gödel algebras. Then we shall identify the object dual to the free singly generated Gödel algebra, and describe how to effectively compute products in the dual category. An application of Lemma 1.0.13 will provide us with concrete representations of all finitely generated free Gödel algebras.

We end this Section by providing another instance of the spectral duality approach, applied to the category of Nilpotent Minimum algebras, the algebraic semantics of Nilpotent Minimum logic. Both Gödel and of Nilpotent Minimum logics are schematic extensions of the logic WNM, which in turns is obtained extending MTL with the axiom $\neg(\varphi\&\psi) \vee ((\varphi \wedge \psi) \to (\varphi\&\psi))$. In particular, Gödel logic is given by extending WNM with idempotency of conjunction: $\varphi \to (\varphi\&\varphi)$, while Nilpotent Minimum logic is WNM plus involutiveness of negation: $\neg\neg\varphi \to \varphi$. These two logics are also extensions of important non-classical logics by way of the same axiom scheme: as is well known, Gödel logic is Intuitionistic logic plus prelinearity: $(\varphi \to \psi) \vee (\psi \to \varphi)$; Nilpotent Minimum logic is obtained by adding prelinearity to Constructive Logic with Strong Negation. On the algebraic side, Gödel algebras are exactly the prelinear Heyting algebras, while Nilpotent Minimum algebras, or NM-algebras, are the prelinear Nelson algebras.

4.1 Functional Representation of Gödel Algebras

Recall that the variety $\mathbb{G}$ of Gödel algebras is generated by any Gödel chain with infinitely many elements. In particular, the Gödel chain $[0,1]$ is generic for $\mathbb{G}$. (a chain can be equipped with a unique structure of Gödel algebra).

Let $\sim$ be the binary relation on $[0,1]^n$ defined in the following way: given two n-tuples $\mathbf{u} = \langle u_1, \ldots, u_n\rangle, \mathbf{v} = \langle v_1, \ldots, v_n\rangle \in [0,1]^n$ we set $\mathbf{u} \sim \mathbf{v}$ if and only if there is a permutation σ of $\{1, \ldots, n\}$ and a map $\prec_i$: $\{0, \ldots, n\} \to \{<, =\}$ such that

$$0 \prec_0 u_{\sigma(1)} \prec_1 \cdots \prec_{n-1} u_{\sigma(n)} \prec_n 1 \text{ iff } 0 \prec_0 v_{\sigma(1)} \prec_1 \cdots \prec_{n-1} v_{\sigma(n)} \prec_n 1\,.$$

The relation $\sim$ is an equivalence relation, we denote by $[\mathbf{u}]$ the equivalence class of $\mathbf{u}$. The quotient set $[0,1]^n/\sim$ is hence a partition of $[0,1]^n$.

A routine proof by structural induction shows that

LEMMA 4.1.1. *If φ is a Gödel formula with n variables, then it holds $\varphi^{\boldsymbol{F}^n_{\mathbb{G}}}(\mathbf{u}) \in \{0, u_1, \ldots, u_n, 1\}$ for every $\mathbf{u} = \langle u_1, \ldots, u_n\rangle \in [0,1]^n$. Further, let $\mathbf{u} \sim \mathbf{v}$. If $\varphi^{\boldsymbol{F}^n_{\mathbb{G}}}(\mathbf{u}) \in \{0,1\}$, then $\varphi^{\boldsymbol{F}^n_{\mathbb{G}}}(\mathbf{u}) = \varphi^{\boldsymbol{F}^n_{\mathbb{G}}}(\mathbf{v})$, while if $\varphi^{\boldsymbol{F}^n_{\mathbb{G}}}(\mathbf{u}) = u_i$ then $\varphi^{\boldsymbol{F}^n_{\mathbb{G}}}(\mathbf{v}) = v_i$.*

Hence the restriction of $\varphi^{\boldsymbol{F}^n_{\mathbb{G}}}$ to each equivalence class $[\mathbf{u}]$ is either the constant function 0 or 1, or it coincides with a projection function.

With each class $[\mathbf{u}]$, where

$$0 \prec_0 u_{\sigma(1)} \prec_1 \cdots \prec_{n-1} u_{\sigma(n)} \prec_n 1\,,$$

we associate a unique *ordered partition* $\rho_{\mathbf{u}} = Q_1 < \cdots < Q_h$ (i.e., a partition equipped with a total order among its blocks) of the set $\{\bot, x_1, \ldots, x_n, \top\}$ in the following way:

- $\bot \in Q_1$; $\top \in Q_h$;
- if $\prec_i$ is $=$ then $x_{\sigma(i)}$ and $x_{\sigma(i+1)}$ belong to the same Q_j;
- if $\prec_i$ is $<$ and $x_{\sigma(i)} \in Q_j$ then $x_{\sigma(i+1)} \in Q_{j+1}$.

Note that if $[\mathbf{u}] \neq [\mathbf{v}]$ then $\rho_{\mathbf{u}} \neq \rho_{\mathbf{v}}$.

By a *Gödel n-partition* we mean an ordered partition $\rho = Q_1 < \cdots < Q_h$ of $\{\bot, x_1, \ldots, x_n, \top\}$ such that $h > 1$, $\bot \in Q_1$ and $\top \in Q_h$. We can hence establish a bijection between Gödel n-partitions and equivalence classes $[\mathbf{u}] \in [0,1]^n/\sim$. If $\rho = \rho_{\mathbf{u}}$ is a Gödel n-partition, we denote by $[\rho]$ the associated equivalence class $[\mathbf{u}]$.

We can define a partial order on Gödel n-partitions, setting $\rho_1 \prec \rho_2$ if $\rho_1 = Q_1 < \cdots < Q_{h-1} < Q_h$ and $\rho_2 = Q_1 < \cdots < Q_{h-1} < R_h < \cdots < R_k$ with $R_h \neq Q_h$. The relation $\prec$, which is irreflexive and transitive, induces a partial order $\preceq$ on the set of all Gödel n-partitions.

Consider a Gödel n-partition $\rho = Q_1 < \cdots < Q_h$. We define the *$G_n$-region* of $[0,1]^n$ associated with ρ as

$$C_\rho = \bigcup_{\rho' \preceq \rho} [\rho'].$$

We denote by P_n the set of Gödel n-partitions and by M_n the set of all Gödel n-partitions that are maximal with respect to the partial order $\preceq$.

EXAMPLE 4.1.2. *Consider the Gödel* 2*-partitions*

- $\pi = \{\bot\} < \{x_1\} < \{x_2\} < \{\top\}\,,$
- $\sigma = \{\bot\} < \{x_1\} < \{x_2, \top\}\,,$
- $\tau = \{\bot\} < \{x_1, x_2, \top\}\,.$

Then $\tau \prec \sigma \prec \pi$ and

- $[\pi] = \{(t_1, t_2) \mid 0 < t_1 < t_2 < 1\}\,,$
- $[\sigma] = \{(t_1, 1) \mid 0 < t_1 < 1\}\,,$
- $[\tau] = \{(1, 1)\}\,.$

The corresponding G_2-regions of $[0,1]^2$ are

- $C_\pi = \{(t_1, t_2) \mid 0 < t_1 < t_2 \leq 1\} \cup \{(1,1)\} = [\pi] \cup [\sigma] \cup [\tau]\,,$
- $C_\sigma = \{(t_1, 1) \mid 0 < t_1 \leq 1\} = [\sigma] \cup [\tau]\,,$

- $C_\tau = \{(1,1)\} = [\tau]$.

We next discuss minterms for Gödel logic. In the Boolean case, a minterm is a Boolean function that takes value 1 at a prescribed point of $\{0,1\}^n$, and has minimum possible value — namely, zero — at each other point. For Gödel functions, G_n-regions play the rôle of Boolean points, although — as detailed below — the minimum possible value outside of such a region need not be zero.

In order to introduce the next results we need some more definitions. By a *prefix* of an ordered partition $Q_1 < \ldots < Q_k$ we mean a possibly empty (*i.e.*, with no blocks) ordered partition $Q_1 < \ldots < Q_h$ of $Q_1 \cup \ldots \cup Q_h$ for $h \in \{1, \ldots, k\}$. The *longest common prefix* of two distinct ordered partitions $\rho_1 = U_1 < \ldots < U_u$ and $\rho_2 = V_1 < \ldots < V_v$ is an ordered partition $W_1 < \ldots < W_w$ such that for every $i \in \{1, \ldots, w\}$, $U_i = V_i = W_i$ and $U_{w+1} \neq V_{w+1}$ (note that $0 \leq w < \min\{u, v\}$). If, on the other hand, $\rho_1 = \rho_2$, then their longest common prefix is ρ_1 itself.

LEMMA 4.1.3. *Let $W_1 < \ldots < W_w$ be the longest common prefix of $\rho_1 = U_1 < \ldots < U_u$ and $\rho_2 = V_1 < \ldots < V_v$. Then for any formula φ, if $\varphi^{\boldsymbol{F}^n_{\mathbb{G}}}$ coincides with a function $g \in W_i$ over $[\rho_1]$ then it coincides with $g \in W_i$ even over $[\rho_2]$.*

Proof. The proof proceeds by structural induction on φ. □

LEMMA 4.1.4. *If $\rho \preceq \tau$ and $\varphi^{\boldsymbol{F}^n_{\mathbb{G}}}$ is equal to 1 over $[\tau]$ then $\varphi^{\boldsymbol{F}^n_{\mathbb{G}}}$ is equal to 1 over $[\rho]$.*

Proof. Indeed displaying $\rho = Q_1 < \cdots < Q_{h-1} < Q_h$ and $\tau = Q_1 < \cdots < Q_{h-1} < R_h < \cdots < R_k$, if $\varphi^{\boldsymbol{F}^n_{\mathbb{G}}}$ were not equal to 1 over $[\rho]$ then $\varphi^{\boldsymbol{F}^n_{\mathbb{G}}} \in Q_i$ with $i < h$ hence even on $[\tau]$ it would be different from 1. □

LEMMA 4.1.5. *If φ and ψ are such that $\{\mathbf{x} \in [0,1]^n \mid \varphi^{\boldsymbol{F}^n_{\mathbb{G}}}(\mathbf{x}) = 1\} = \{\mathbf{x} \in [0,1]^n \mid \psi^{\boldsymbol{F}^n_{\mathbb{G}}}(\mathbf{x}) = 1\}$ then $\varphi^{\boldsymbol{F}^n_{\mathbb{G}}} = \psi^{\boldsymbol{F}^n_{\mathbb{G}}}$, i.e., functions associated with Gödel formulas are determined by their 1-set.*

Proof. First we prove two auxiliary facts.

1) If there is a Gödel partition ρ such that $\varphi^{\boldsymbol{F}^n_{\mathbb{G}}} = 0$ over $[\rho]$ then, by Lemma 4.1.3, $\varphi^{\boldsymbol{F}^n_{\mathbb{G}}} = 0$ over the uniquely determined minimal partition $\rho_m \preceq \rho$.

2) Let $[\rho] \subseteq \{\mathbf{x} \in [0,1]^n \mid \varphi^{\boldsymbol{F}^n_{\mathbb{G}}}(\mathbf{x}) = 1\}$ and display $\rho = U_1 < \cdots < U_u$. Let $\rho \prec \tau$ with $\tau = U_1 < \cdots < U_{n-1} < V < U_n \setminus V$ where $V \subseteq U_n \setminus \{\top\}$ (hence τ covers ρ with respect to the order $\preceq$). If $\varphi^{\boldsymbol{F}^n_{\mathbb{G}}} < 1$ over $[\tau]$ then, by Lemma 4.1.3, $\varphi^{\boldsymbol{F}^n_{\mathbb{G}}} \in V$ over every τ' with $\tau \preceq \tau'$.

Now, if $\varphi^{\boldsymbol{F}^n_{\mathbb{G}}} = 0$ over $[\rho]$, by 1) we can assume ρ is minimal, and then, by Lemma 4.1.4, $\psi^{\boldsymbol{F}^n_{\mathbb{G}}} \neq 1$ over $[\rho]$, and by minimality of ρ, $\psi^{\boldsymbol{F}^n_{\mathbb{G}}} = 0$ over $[\rho]$, too.

There remains to consider those partitions τ such that $\varphi^{\boldsymbol{F}^n_{\mathbb{G}}}$ is not constant over $[\tau]$. Then there exists a partition ρ covered by τ such that $\varphi^{\boldsymbol{F}^n_{\mathbb{G}}}$ is constantly 1 over $[\rho]$. By our current assumptions $\psi^{\boldsymbol{F}^n_{\mathbb{G}}}$ is constantly 1 over $[\rho]$, too. By 2) $\varphi^{\boldsymbol{F}^n_{\mathbb{G}}}$ and $\psi^{\boldsymbol{F}^n_{\mathbb{G}}}$ coincide over $[\pi]$ for each $\pi \succ \rho$.

Hence $\varphi^{\boldsymbol{F}^n_{\mathbb{G}}} = \psi^{\boldsymbol{F}^n_{\mathbb{G}}}$ over every $\pi \in P_n$, that is, they are the same function over $[0,1]^n$. □

Let us define the following derived connective:

$$x \lhd y = (y \to x) \to y\,.$$

Note that when interpreted in $[0,1]$ we have $x \lhd y = 1$ if and only if $x < y$ or $x = y = 1$ and $x \lhd y = y$ otherwise.

LEMMA 4.1.6. *Let ρ be a Gödel n-partition such that $[\rho] = \{\mathbf{u} \mid \bot \prec_0 u_{\sigma(1)} \prec_1 \cdots \prec_{n-1} u_{\sigma(n)} \prec_n \top\}$. For the sake of conciseness, let $x_{\sigma(0)} = \bot$ and $x_{\sigma(n+1)} = \top$. Set*

$$\chi_\rho = \bigwedge_{i=0}^{n} \delta_i,$$

where

$$\delta_i = \begin{cases} x_{\sigma(i)} \leftrightarrow x_{\sigma(i+1)} & \textit{iff } \prec_i \textit{ is } =, \\ x_{\sigma(i)} \lhd x_{\sigma(i+1)} & \textit{iff } \prec_i \textit{ is } <, \end{cases}$$

for every $i \in \{0, \ldots, n\}$.

Then $\chi_\rho^{\boldsymbol{F}^n_{\mathbb{G}}}$ is the smallest term function which coincides with 1 over $[\rho]$ and further it is the unique function such that $\chi_\rho^{\boldsymbol{F}^n_{\mathbb{G}}}(\mathbf{x}) = 1$ if and only if $\mathbf{x} \in C_\rho$.

Proof. From the definition of $\leftrightarrow$ and $\lhd$ it is immediate to check that $\chi_\rho^{\boldsymbol{F}^n_{\mathbb{G}}}$ is constantly 1 over $[\rho]$. By Lemma 4.1.4 it is constantly 1 over C_ρ, too. In order to prove that $\chi_\rho^{\boldsymbol{F}^n_{\mathbb{G}}}$ is the smallest function with such property, pick any $\tau \in P_n$ such that $\tau \not\preceq \rho$. Display ρ as $Q_1 < Q_2 < \cdots < Q_h$. Let π be the longest common prefix of τ and ρ. Notice that if $\rho \prec \tau$ then $\pi = Q_1 < Q_2 < \cdots < Q_{h-1}$, as the hth block of ρ is the union of all blocks of τ of index greater than $h-1$. In this case one can check that one of the conjuncts of χ_ρ evaluates to a function $x_{\sigma(i)} \neq \top$ belonging to the hth block of τ. By Lemma 4.1.3, this is the smallest value a Gödel term function which is 1 over $[\rho]$ can take over $[\tau]$. If ρ and τ are incomparable, then there is $w < h-1$ such that the longest common prefix of ρ and τ has the form $Q_1 < Q_2 < \cdots < Q_w$. In this case one of the conjuncts of χ_ρ evaluates to a function $x_{\sigma(i)} \neq \top$ belonging to the wth block of τ. Again by Lemma 4.1.3, this is the smallest value a Gödel term function which is 1 over $[\rho]$ can take over $[\tau]$. The uniqueness requirement follows at once by Lemma 4.1.5. □

The formula χ_π is called the (*Gödel*) *minterm* associated with π. Just like Boolean minterms afford a disjunctive normal form theorem, one can canonically express each Gödel function as a disjunction of Gödel minterms. Namely,

LEMMA 4.1.7. *Let φ be a term in n variables. Then for every $\pi \in M_n$ there exists a unique $\rho(\varphi, \pi) \in P_n$ with $\rho(\varphi, \pi) \preceq \pi$ such that $\varphi^{\boldsymbol{F}^n_{\mathbb{G}}}(\mathbf{x}) = \chi_{\rho(\varphi,\pi)}^{\boldsymbol{F}^n_{\mathbb{G}}}(\mathbf{x})$, for all $\mathbf{x} \in C_\pi$. Moreover, φ can be equivalently written as*

$$\varphi \equiv_{\mathbb{G}} \bigvee_{\pi \in M_n} \chi_{\rho(\varphi,\pi)}\,.$$

Proof. Immediate, from Lemma 4.1.6. □

DEFINITION 4.1.8. *A function* $f : [0,1]^n \to [0,1]$ *is a* Gödel function *(of* n *variables) if for every* G_n*-region* C_ρ *there exists a function* $g \in \{\bot, x_1, \ldots, x_n, \top\}$ *such that* $f(\mathbf{t}) = g(\mathbf{t})$ *for each* $\mathbf{t} = \langle t_1, \ldots, t_n \rangle \in C_\rho$*. Let* $\boldsymbol{G}_n$ *be the Gödel algebra of Gödel functions equipped with pointwise operations.*

THEOREM 4.1.9. $\boldsymbol{F}^n_{\mathbb{G}} \cong \boldsymbol{G}_n$.

Proof. Immediate, from Lemma 4.1.7. □

In the next section we shall see that the poset P_n of all Gödel n-partitions can be seen as the object corresponding to the free n-generated Gödel algebra in a category that is dually equivalent to the category of finite Gödel algebras and their homomorphisms. As a matter of fact P_n is order-isomorphic to the *prime spectrum* of $\boldsymbol{F}^n_{\mathbb{G}}$, that is, the set of all the prime filters of this free algebra, ordered by reverse inclusion. The prime spectra of finite Gödel algebras, together with suitable defined maps between them, will constitute the dual category we shall deal with. It is easy to see that a Gödel n-partition encodes both a prime filter of $\boldsymbol{F}^n_{\mathbb{G}}$, and the chain obtained by taking the quotient of $\boldsymbol{F}^n_{\mathbb{G}}$ with respect to the congruence canonically associated with the filter. As a matter of fact, using Theorem 4.1.9, an element ρ of M_n corresponds to the minimal prime filter constituted by all and only those functions constantly 1 over C_ρ. The chain $\boldsymbol{C}_\rho$ obtained from $\boldsymbol{F}^n_{\mathbb{G}}$ by restricting the domain of the Gödel functions to C_ρ is isomorphic to the quotient of $\boldsymbol{F}^n_{\mathbb{G}}$ by the described minimal prime filter. An element $\tau \prec \rho$ of P_n corresponds to the filter constituted by all and only those functions which coincides over C_ρ with one of the functions in the maximum block of τ. Roots of P_n then correspond to maximal filters, and quotienting the free algebra by each one of them gives the two-element Boolean chain.

4.2 Finite Spectral Duality for Gödel Algebras

We recall that two categories $\mathscr{C}$ and $\mathscr{D}$ are *equivalent* if there exist two functors $F\colon \mathscr{C} \to \mathscr{D}$ and $G\colon \mathscr{D} \to \mathscr{C}$ whose compositions FG and GF are naturally isomorphic to the identity functors.

That is, for each object C in $\mathscr{C}$ and D in $\mathscr{D}$ there are isomorphisms $\iota_C\colon C \to GF(C)$ and $\kappa_D\colon D \to FG(D)$ such that the following diagrams

$$\begin{array}{ccc} C_1 & \xrightarrow{f} & C_2 \\ \downarrow{\scriptstyle \iota_{C_1}} & & \downarrow{\scriptstyle \iota_{C_2}} \\ GF(C_1) & \xrightarrow{GF(f)} & GF(C_2) \end{array} \qquad \begin{array}{ccc} D_1 & \xrightarrow{g} & D_2 \\ \downarrow{\scriptstyle \kappa_{D_1}} & & \downarrow{\scriptstyle \kappa_{D_2}} \\ FG(D_1) & \xrightarrow{FG(g)} & FG(D_2) \end{array}$$

commute for each $f\colon C_1 \to C_2$ in $\mathscr{C}$ and $g\colon D_1 \to D_2$ in $\mathscr{D}$.

Equivalently, $\mathscr{C}$ and $\mathscr{D}$ are equivalent if there exists a functor

$$F\colon \mathscr{C} \to \mathscr{D}$$

such that F is *full and faithful*, that is

$$\mathrm{Hom}(C_1, C_2) \cong \mathrm{Hom}(F(C_1), F(C_2))$$

for all objects C_1, C_2 in $\mathscr{C}$, and F is *essentially surjective*, that is, for each object D in $\mathscr{D}$ there exists an object C in $\mathscr{C}$ such that

$$D \cong F(C)\,.$$

The *opposite* of a category $\mathscr{D}$ is the category $\mathscr{D}^{\mathrm{op}}$ obtained by reversing all arrows in $\mathscr{D}$. Two categories $\mathscr{C}$ and $\mathscr{D}$ are *dually equivalent* if $\mathscr{C} \equiv \mathscr{D}^{\mathrm{op}}$. In this case, the full and faithful functor $F\colon \mathscr{C} \to \mathscr{D}$ implementing the duality is obviously contravariant, that is

$$h \in \mathrm{Hom}(C_1, C_2) \mapsto F(h) \in \mathrm{Hom}(F(C_2), F(C_1))\,.$$

In this subsection we shall introduce a category that is dual to the category G of finite Gödel algebras and their homomorphisms. Since $\mathbb{G}$ is a locally finite variety, G includes all finitely generated free Gödel algebras.

We shall then show how to compute products in the dual category, and characterise the object dual to the Gödel algebra $\boldsymbol{F}^1_{\mathbb{G}}$. We shall then use Lemma 1.0.13 to obtain a representation of finitely generated free Gödel algebras via finite duality.

Throughout, *poset* is short for partially ordered set. If $\langle P, \leq\rangle$ is a poset and $S \subseteq P$, the *downset* generated by S is

$$\downarrow S = \{p \in P \mid p \leq s \text{ for some } s \in S\}\,.$$

(When S is a singleton $\{s\}$, we shall write $\downarrow s$ for $\downarrow \{s\}$.) A subposet $S \subseteq P$ is a *downset* if $\downarrow S = S$. *Upsets* and $\uparrow S$ are defined analogously. The one-element poset is denoted $\mathbf{1}$. A *forest* is a poset F such that $\downarrow x$ is totally ordered for any $x \in F$. If P is a poset, by $P_\perp$ we denote the poset obtained by adding a new bottom element $\perp$ to P. A *tree* is a forest with a minimum element, hence for each forest F, $F_\perp$ is a tree.

Gödel algebras are Heyting algebras satisfying the prelinearity condition. From any finite poset P one reconstructs a Heyting algebra, as follows. Let $\mathbf{Sub}P$ be the family of all downsets of P. When partially ordered by inclusion, $\mathbf{Sub}P$ is a finite distributive lattice, and thus carries a unique Heyting implication adjoint to the lattice meet operation via *residuation*. Explicitly, if $\boldsymbol{L}$ is a finite distributive lattice, then its Heyting implication is given by

$$x \to y = \bigvee\{z \in \boldsymbol{L} \mid z \wedge x \leq y\}$$

for all $x, y \in L$. Accordingly, we regard $\mathbf{Sub}P$ as a Heyting algebra.

Conversely, one can obtain a finite poset from any finite Heyting algebra $\boldsymbol{H}$. Let $\mathrm{Spec}\boldsymbol{H}$ denote the poset of prime filters of $\boldsymbol{H}$, ordered by reverse inclusion. Equivalently, one can think of $\mathrm{Spec}\boldsymbol{H}$ as the poset of minterms (i.e. join-irreducible elements) of $\boldsymbol{H}$, with the order they inherit from $\boldsymbol{H}$. (We recall that a *filter* of $\boldsymbol{H}$ is an upset of $\boldsymbol{H}$ closed under meets; it is *prime* if it does not contain the bottom element of $\boldsymbol{H}$, and contains either y or z whenever it contains $y \vee z$. We further recall that $x \in \boldsymbol{H}$ is *join-irreducible* if it is not the bottom element of $\boldsymbol{H}$, and whenever $x = y \vee z$ for $y, z \in \boldsymbol{H}$, then either $x = y$ or $x = z$.)

The constructions of the two preceding paragraphs are inverse to each other, in the sense that for any finite Heyting algebra $\boldsymbol{H}$ one has an isomorphism of Heyting algebras

$$\mathbf{Sub}\mathrm{Spec}\boldsymbol{H} \cong \boldsymbol{H}\ . \tag{6}$$

Let us now restrict attention to finite *Gödel* algebras. Horn proved [50, 2.4] that a Heyting algebra $\boldsymbol{H}$ is a Gödel algebra if and only if its prime filters are a forest under reverse inclusion, *i.e.* if $\mathrm{Spec}\boldsymbol{H}$ is a forest. Note that a downset of a forest F is itself a forest, and we shall call it a *subforest* of F. Summing up, a combinatorial representation of finite Gödel algebras is obtained as follows. Let $\boldsymbol{G}$ be a finite Gödel algebra. Then $\boldsymbol{G}$ is canonically isomorphic to the Gödel algebra of all subforests of a finite forest $\mathrm{Spec}\boldsymbol{G}$ as in (6), where $\mathrm{Spec}\boldsymbol{G}$ is unique to within a poset isomorphism. Under the representation (6), each element $g \in \boldsymbol{G}$ corresponds to a uniquely determined subforest of $\mathrm{Spec}\boldsymbol{G}$ (and conversely). Such a correspondence can be extended to a duality of categories.

We now give a detailed construction of this dual equivalence. Given two forests F and G, an order preserving map $f\colon F \to G$ is *open* if $x' \leq f(x)$ in G implies that there exists $y \leq x$ in F such that $f(y) = x'$. Open maps carry downsets to downsets.

Let G be the category of finite Gödel algebras where arrows are Gödel homomorphisms and let F be the category whose objects are forests and whose arrows are open maps.

We shall define the functors $\mathbf{Sub}\colon \mathsf{F} \to \mathsf{G}$ and $\mathrm{Spec}\colon \mathsf{G} \to \mathsf{F}$ implementing the dual equivalence.

Let $\mathrm{Sub}\, F$ be the set of all subforests (*i.e.*, downsets) of a forest $F \in \mathsf{F}$. We shall equip $\mathrm{Sub}\, F$ with the structure of a Gödel algebra as follows. For all subforests H, K of F, let

$$H \to K = F \setminus \uparrow (H \setminus K)\,. \tag{7}$$

Then, the structure

$$\mathbf{Sub}F = \langle \mathrm{Sub}\, F, \cap, \to, \emptyset \rangle$$

is a Gödel algebra. To prove this, we only need to show that (7) defines the residuum of subforest intersection, that is:

$$H \to K = \bigvee \{L \in \mathrm{Sub}\, F \mid L \cap H \leq K\}\,. \tag{8}$$

Pick $x \in \uparrow (H \setminus K)$. Then there exists $y \leq x$ such that $y \in H \setminus K$. Clearly, no such x belongs to a subforest L such that $L \cap H \subseteq K$. An easy check now shows that $F \setminus \uparrow (H \setminus K)$ is the greatest subforest L of F such that $L \cap H \subseteq K$, thus settling (8).

To turn $\mathbf{Sub}$ into a functor from F to G we specify how it carries morphisms to the other side. For each order-preserving open map $f\colon F \to G$ we let $\mathbf{Sub}f\colon \mathbf{Sub}G \to \mathbf{Sub}F$ be defined by taking preimages. For any $a \in \mathbf{Sub}G$:

$$(\mathbf{Sub}f)(a) = f^{-1}(a)\,.$$

Let now $\mathrm{Spec}\boldsymbol{A}$ be the set of all prime filters of the Gödel algebra $\boldsymbol{A} \in \mathsf{G}$, ordered by reverse inclusion. For each homomorphism $h\colon \boldsymbol{A} \to \boldsymbol{B}$ in G, we let $\mathrm{Spec}h\colon \mathrm{Spec}\boldsymbol{B} \to$

$\mathrm{Spec}\boldsymbol{A}$ be defined again by taking preimages. For any $\mathfrak{p} \in \mathrm{Spec}\boldsymbol{B}$:

$$(\mathrm{Spec}h)(\mathfrak{p}) = h^{-1}(\mathfrak{p})\,.$$

Then, $\mathrm{Spec}\colon \mathsf{G} \to \mathsf{F}$ is a functor from the category of finite Gödel algebras to the category of finite forests.

Note that $\mathrm{Spec}\boldsymbol{A}$ is order-isomorphic to the poset of join-irreducible elements of $\boldsymbol{A}$, with the inherited order. As a matter of fact, each prime filter $\mathfrak{p}$ of $\boldsymbol{A}$ is principal and is generated by a unique join-irreducible element, which we denote $j_{\mathfrak{p}}$.

Once we have defined the functors $\mathbf{Sub}$ and Spec, we proceed in showing they realize the desired dual equivalence. We first show that for each Gödel algebra $\boldsymbol{A} \in \mathsf{G}$ the following holds.

$$\boldsymbol{A} \cong \mathbf{Sub}\,\mathrm{Spec}\,\boldsymbol{A}\,. \tag{9}$$

Let $\varphi_{\boldsymbol{A}}$ be the map $x \in \boldsymbol{A} \mapsto \{\mathfrak{p} \mid \mathfrak{p} \in \mathrm{Spec}\,\boldsymbol{A},\, j_{\mathfrak{p}} \leq x\}$. Clearly, $\{\mathfrak{p} \mid \mathfrak{p} \in \mathrm{Spec}\,\boldsymbol{A},\, j_{\mathfrak{p}} \leq x\}$ is a downset of $\mathrm{Spec}\,\boldsymbol{A}$, hence $\varphi_{\boldsymbol{A}}$ maps to $\mathbf{Sub}\,\mathrm{Spec}\,\boldsymbol{A}$. Observe that if x, y are two distinct elements of $\boldsymbol{A}$ then there exists a join irreducible element $j \in \boldsymbol{A}$ such that, either $j \leq x$ and $j \not\leq y$, or $j \not\leq x$ and $j \leq y$; hence $\varphi_{\boldsymbol{A}}(x) \neq \varphi_{\boldsymbol{A}}(y)$, and the injectivity of $\varphi_{\boldsymbol{A}}$ is proved. Surjectivity of $\varphi_{\boldsymbol{A}}$ follows by noting that $\varphi_{\boldsymbol{A}}(\bigvee_{\mathfrak{p}\in S} j_{\mathfrak{p}}) = S$ for each downset S of $\mathrm{Spec}\,\boldsymbol{A}$. It remains to show that $\varphi_{\boldsymbol{A}}$ is a homomorphism of Gödel algebras. Trivially, $\varphi_{\boldsymbol{A}}(\bot_{\boldsymbol{A}}) = \emptyset$. For what concerns conjunction, $\varphi_{\boldsymbol{A}}(x \wedge y) = \{\mathfrak{p} \mid \mathfrak{p} \in \mathrm{Spec}\,\boldsymbol{A},\, j_{\mathfrak{p}} \leq x \wedge y\} = \{\mathfrak{p} \mid \mathfrak{p} \in \mathrm{Spec}\,\boldsymbol{A},\, j_{\mathfrak{p}} \leq x\} \cap \{\mathfrak{p} \mid \mathfrak{p} \in \mathrm{Spec}\,\boldsymbol{A},\, j_{\mathfrak{p}} \leq y\} = \varphi_{\boldsymbol{A}}(x) \cap \varphi_{\boldsymbol{A}}(y)$. To show that $\varphi_{\boldsymbol{A}}(x \to y) = \mathrm{Spec}\,\boldsymbol{A} \setminus \uparrow (\varphi_{\boldsymbol{A}}(x) \setminus \varphi_{\boldsymbol{A}}(y))$ we first note that $\varphi_{\boldsymbol{A}}(x \to y) = \{\mathfrak{p} \mid j_{\mathfrak{p}} \leq x \to y\} = \{\mathfrak{p} \mid x \wedge j_{\mathfrak{p}} \leq y\}$, by residuation. Pick $\mathfrak{q} \in \mathrm{Spec}\,\boldsymbol{A}$. Then $x \wedge j_{\mathfrak{q}} \leq y$ iff $x \wedge j_{\mathfrak{q}} \leq y \wedge j_{\mathfrak{q}}$. Observe that for each $z \in \boldsymbol{A}$ and each $\mathfrak{r} \in \mathrm{Spec}\,\boldsymbol{A}$, either $z \wedge j_{\mathfrak{r}} = \bot_{\boldsymbol{A}}$, or $z \wedge j_{\mathfrak{r}} = j_{\mathfrak{s}}$ for some $\mathfrak{s} \in \mathrm{Spec}\,\boldsymbol{A}$. A direct computation now shows $x \wedge j_{\mathfrak{q}} > y \wedge j_{\mathfrak{q}}$ iff there is $\mathfrak{t} \in \mathrm{Spec}\,\boldsymbol{A}$ such that $j_{\mathfrak{t}} \leq j_{\mathfrak{q}}$ and $\mathfrak{t} \in \varphi_{\boldsymbol{A}}(x)$ but $\mathfrak{t}_{\boldsymbol{A}} \notin \varphi(y)$, that is, $x \wedge j_{\mathfrak{q}} > y \wedge j_{\mathfrak{q}}$ iff $j_{\mathfrak{q}} \in \uparrow (\varphi_{\boldsymbol{A}}(x) \setminus \varphi_{\boldsymbol{A}}(y))$, and we are done. Thus, $\varphi_{\boldsymbol{A}}\colon \boldsymbol{A} \to \mathbf{Sub}\,\mathrm{Spec}\,\boldsymbol{A}$ is the desired isomorphism.

We next show that for each finite forest $F \in \mathsf{F}$ the following holds.

$$F \cong \mathrm{Spec}\,\mathbf{Sub}\,F\,. \tag{10}$$

Notice that for each $x \in F$, its downset $\downarrow x$ is a join irreducible element of $\mathbf{Sub}\,F$. Let then ψ_F be the map $x \in F \mapsto \langle \downarrow x \rangle$, where $\langle \downarrow x \rangle$ is the prime filter of $\mathbf{Sub}\,F$ generated by $\downarrow x$. Trivially, ψ_F is an order-preserving bijection. It remains to show that ψ_F is open. Pick $\mathfrak{p} \leq \langle \downarrow x \rangle$ in $\mathrm{Spec}\,\mathbf{Sub}\,F$. Let $y = \psi_F^{-1}(\mathfrak{p})$. Now, $\langle \downarrow y \rangle \supseteq \langle \downarrow x \rangle$ iff $\downarrow y \subseteq \downarrow x$ iff $y \leq x$, as it has to be shown. We conclude that ψ_F is an isomorphism in F.

To complete the picture we have to show that the functors $\mathbf{Sub}$ and Spec well behave on morphisms. We first show that for each $h\colon \boldsymbol{A} \to \boldsymbol{B}$ it holds that

$$\varphi_{\boldsymbol{B}} \circ h = \mathbf{Sub}\mathrm{Spec}h \circ \varphi_{\boldsymbol{A}}\,.$$

Pick $x \in \boldsymbol{A}$, then $\varphi_{\boldsymbol{B}}(h(x)) = \{\mathfrak{q} \mid \mathfrak{q} \in \mathrm{Spec}\boldsymbol{B}, j_{\mathfrak{q}} \leq h(x)\}$. Now, $\varphi_{\boldsymbol{A}}(x) = \{\mathfrak{p} \mid \mathfrak{p} \in \mathrm{Spec}\boldsymbol{A}, j_{\mathfrak{p}} \leq x\}$. Hence, we have to show

$$(\mathbf{Sub}\mathrm{Spec}h)(\{\mathfrak{p} \mid \mathfrak{p} \in \mathrm{Spec}\boldsymbol{A}, j_{\mathfrak{p}} \leq x\}) = \{\mathfrak{q} \mid \mathfrak{q} \in \mathrm{Spec}\boldsymbol{B}, j_{\mathfrak{q}} \leq h(x)\}\,. \tag{11}$$

By definition of the functors $\mathbf{Sub}$ and Spec, the left hand side of (11) reduces to $\{h(\mathfrak{p}) \mid \mathfrak{p} \in \mathrm{Spec}\boldsymbol{A}, j_{\mathfrak{p}} \leq x\}$. Since $x = \bigvee_{j_{\mathfrak{p}} \leq x} j_{\mathfrak{p}}$ and $h(x) = \bigvee_{j_{\mathfrak{q}} \leq h(x)} j_{\mathfrak{q}}$, the identity (11) holds.

Lastly, we have to show that for each $f\colon F \to G$ it holds that

$$\psi_G \circ f = \mathrm{Spec}\mathbf{Sub}f \circ \psi_F \,.$$

Pick $x \in F$, then $\psi_G(f(x)) = \langle\downarrow f(x)\rangle$, while $\psi_F(x) = \langle\downarrow x\rangle$. It must hold that

$$(\mathrm{Spec}\mathbf{Sub}f)(\langle\downarrow x\rangle) = \langle\downarrow f(x)\rangle \,, \tag{12}$$

which is obvious from the definitions of Spec and $\mathbf{Sub}$.

We have thus shown the following.

THEOREM 4.2.1. *The categories* G *and* F *are dually equivalent through the functors* $\mathbf{Sub}\colon \mathsf{F} \to \mathsf{G}$ *and* $\mathrm{Spec}\colon \mathsf{G} \to \mathsf{F}$.

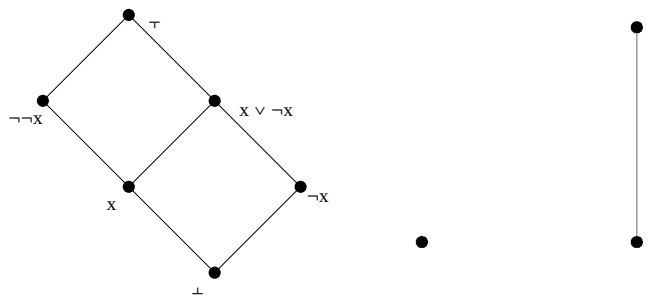
As an application of Theorem 4.2.1, for each integer $n \geq 0$ the free Gödel algebra $\boldsymbol{F}^n_{\mathbb{G}}$ can be characterised through a combinatorial description of the forest $\mathrm{Spec}\boldsymbol{F}^n_{\mathbb{G}}$. Throughout, we write

$$\mathscr{F}_n = \mathrm{Spec}\boldsymbol{F}^n_{\mathbb{G}} \,.$$

Let us denote $\boldsymbol{G}_i$ the unique (up to isomorphism) Gödel chain of cardinality $i+1$.

LEMMA 4.2.2. $\mathscr{F}_1 = \{a, b, c\}$ *where the order relation is the reflexive closure of* $b < c$.

Proof. By direct inspection, or by applying the functional representation theorem 4.1.9, the singly generated free Gödel algebra is $\boldsymbol{F}^1_{\mathbb{G}} \cong \boldsymbol{G}_1 \times \boldsymbol{G}_2$. Clearly, $\boldsymbol{G}_1 \times \boldsymbol{G}_2$ has exactly three join irreducible elements: denoted x the generator, these elements are $x, \neg x, \neg\neg x$, with $x < \neg\neg x$ as the only pair of elements in covering relation. □



(a) $\boldsymbol{F}^1_{\mathbb{G}}$. (b) $\mathrm{Spec}\,\boldsymbol{F}^1_{\mathbb{G}}$.

Figure 8. The Hasse diagram of $\boldsymbol{F}^1_{\mathbb{G}}$ and its prime spectrum.

By Lemma 1.0.13 we can hence describe the free Gödel algebra through its dual forest. Since coproducts in G correspond to products in F, by Lemma 1.0.13 we have

Figure 9. The subforests $F(g)$ of $\mathscr{F}_1$ corresponding to elements $g \in \boldsymbol{F}^1_{\mathbb{G}}$.

THEOREM 4.2.3.

$$\mathrm{Spec}\boldsymbol{F}^n_{\mathbb{G}} = \prod_{i=1}^{n} \mathscr{F}_1 \,.$$

We hence need to describe how to compute finite products in the category F. Given two forests $\langle F, \leq_F \rangle$ and $\langle G, \leq_G \rangle$, we denote by $F + G$ the coproduct of F and G, that is the forest $\langle F \uplus G, \leq_{F+G} \rangle$ where $F \uplus G$ is the disjoint set union of F and G and $x \leq_{F+G} y$ if and only if either $x, y \in F$ and $x \leq_F y$ or $x, y \in G$ and $x \leq_G y$. It is easy to check that $\langle F \uplus G, \leq_{F+G} \rangle$ is indeed the coproduct of F and G.

LEMMA 4.2.4. *Let F, G and H be three forests. Then:*

1. *If $|F| = 1$ then $F \times G \cong G$;*
2. *$(F + G) \times H \cong (F \times H) + (G \times H)$;*
3. *$F_\perp \times G_\perp \cong ((F \times G_\perp) + (F \times G) + (F_\perp \times G))_\perp$.*

Proof. 1. It immediately follows from the fact that the singleton is the terminal object of F.

2. It follows by a standard argument, since coproducts in F are disjoint unions.

3. Let us display $F_\perp$ isomorphically as $(F_1 + F_2 + \cdots + F_u)_\perp$ for a uniquely determined family $\{F_i\}_{i=1}^u$ of trees. Analogously $G_\perp \cong (G_1 + G_2 + \cdots + G_v)_\perp$, for a uniquely determined family $\{G_j\}_{j=1}^v$ of trees. By applying 2., we can rewrite the forest $(F \times G_\perp) + (F \times G) + (F_\perp \times G)$ isomorphically as the following coproducts of trees

$$\sum_{i=1}^{u} (F_i \times G_\perp) \;+\; \sum_{i=1}^{u} \sum_{j=1}^{v} (F_i \times G_j) \;+\; \sum_{j=1}^{v} (F_\perp \times G_j) \,.$$

We denote r_0 and s_0 the roots of $F_\perp$ and $G_\perp$, respectively. Further, we denote t_0 the root of the tree $((F \times G_\perp) + (F \times G) + (F_\perp \times G))_\perp$.

We next define the maps $\pi_{F_\perp} \colon ((F \times G_\perp) + (F \times G) + (F_\perp \times G))_\perp \to F_\perp$ and $\pi_{G_\perp} \colon ((F \times G_\perp) + (F \times G) + (F_\perp \times G))_\perp \to G_\perp$ as follows. First we set $\pi_{F_\perp}(t_0) = r_0$ and $\pi_{G_\perp}(t_0) = s_0$. Now, each $x \in (((F \times G_\perp) + (F \times G) + (F_\perp \times G))_\perp) \setminus \{t_0\}$, must belong to a unique tree either of the form $F_i \times G_\perp$ or $F_i \times G_j$ or $F_\perp \times G_j$. Writing F_0 and G_0 for $F_\perp$ and $G_\perp$, respectively, then $x \in F_i \times G_j$, for a uniquely determined pair $\langle i, j \rangle$ with $i + j > 0$. We define $\pi_{F_\perp}(x) = \iota_{F_i}(\pi_{F_i}(x))$, where $\pi_{F_i} \colon F_i \times G_j \to F_i$ is the projection function, while $\iota_{F_i} \colon F_i \to F_\perp$ is the set-theoretic inclusion of the support of F_i into $F_\perp$. Similarly, we define $\pi_{G_\perp}(x) = \iota_{G_j}(\pi_{G_j}(x))$. It is clear that both $\pi_{F_\perp}$ and $\pi_{G_\perp}$ are well-defined morphisms of finite forests.

Now, let us take a forest H and two morphisms $f\colon H \to F_\perp$ and $g\colon H \to G_\perp$. We shall construct a map $\langle f, g\rangle\colon H \to ((F \times G_\perp) + (F \times G) + (F_\perp \times G))_\perp$ such that $\pi_{F_\perp} \circ \langle f, g\rangle = f$ and $\pi_{G_\perp} \circ \langle f, g\rangle = g$. To accomplish this task we partition the set H as follows. Let $R_0 = f^{-1}(r_0)$ and $R_1 = H \setminus R_0$. Analogously, let $S_0 = g^{-1}(s_0)$ and $S_1 = H \setminus S_0$. Then the following is a partition of the set H:

$$\{R_0 \cap S_0, R_0 \cap S_1, R_1 \cap S_0, R_1 \cap S_1\}\,.$$

We refine this partition by further subdividing $R_1 \cap S_1$, as follows. Let R_2 be the set of all $x \in R_1 \cap S_1$ such that there is $y < x$ in H with $f(y) = r_0$ and $g(y) \neq s_0$. Notice that, since g is an order-preserving open map, if G_j is the unique tree in $\{G_h\}_{h=1}^{v}$ such that $g(x) \in G_j$, then also $g(y) \in G_j$. Similarly, let S_2 be the set of all $x \in R_1 \cap S_1$ such that there is $y < x$ in H with $f(y) \neq r_0$ and $g(y) = s_0$. Finally let $T_2 = (R_1 \cap S_1) \setminus (R_2 \cup S_2)$. Then

$$\{R_0 \cap S_0, R_0 \cap S_1, R_1 \cap S_0, R_2, S_2, T_2\}$$

is a partition of the set H. For each $x \in R_0 \cap S_0$ we let $\langle f, g\rangle(x) = t_0$; for each $x \in R_0 \cap S_1$ we note that there is a unique tree G_j such that $g(x) \in G_j$: we then let $\langle f, g\rangle(x)$ be the uniquely determined element t of $F_\perp \times G_j$ such that $\pi_{F_\perp}(t) = f(x) = r_0$ and $\pi_{G_j}(t) = g(x)$, where these two maps are the projections of the product $F_\perp \times G_j$; for each $x \in R_1 \cap S_0$ we reason analogously, letting $\langle f, g\rangle(x)$ be the uniquely determined element t of $F_i \times G_\perp$ such that $\pi_{F_i}(t) = f(x)$ and $\pi_{G_\perp}(t) = g(x) = s_0$. Some additional care is needed to deal with the remaining cases. For each $x \in R_2$ we note that there are uniquely determined trees F_i and G_j such that $f(x) \in F_i$ and $g(x) \in G_j$. Since $x \in R_2$, there is $y < x$ in H such that $f(y) = r_0$ and $g(y) \neq s_0$. As morphisms of finite forests must carry downsets to downsets, $\langle f, g\rangle(x)$ must belong to $F_\perp \times G_j$, but now, reasoning as in the preceding cases we let $\langle f, g\rangle(x)$ be the uniquely determined element t of $F_\perp \times G_j$ such that $\pi_{F_\perp}(t) = f(x)$ and $\pi_{G_j}(t) = g(x)$. For each $x \in S_2$ we reason analogously, letting $\langle f, g\rangle(x)$ be the uniquely determined element t of $F_i \times G_\perp$ such that $\pi_{F_i}(t) = f(x)$ and $\pi_{G_\perp}(t) = g(x)$. Finally, the last case $x \in T_2$ is again similarly dealt with, as we let $\langle f, g\rangle(x)$ be the uniquely determined element t of $F_i \times G_j$ such that $\pi_{F_i}(t) = f(x)$ and $\pi_{G_j}(t) = g(x)$. A simple check now shows $\pi_{F_\perp} \circ \langle f, g\rangle = f$ and $\pi_{G_\perp} \circ \langle f, g\rangle = g$ as desired.

There remains to show that $\langle f, g\rangle$ is the only map with this property. Let $h\colon H \to ((F \times G_\perp) + (F \times G) + (F_\perp \times G))_\perp$ be a morphism in F such that $\pi_{F_\perp} \circ h = f$ and $\pi_{G_\perp} \circ h = g$. Clearly h must coincide with $\langle f, g\rangle$ over $R_0 \cap S_0$. Notice that if $x \in R_0 \cap S_1$ then $f(x) = r_0$ and $g(x) \neq s_0$, whence $h(x)$ must belong to $F_\perp \times G_j$ for a uniquely determined G_j. But then $h(x) = \langle f, g\rangle(x)$ as otherwise $F_\perp \times G_j$, together with its projections, would not be the product in F of $F_\perp$ and G_j. A completely analogous argument shows that h must coincide with $\langle f, g\rangle$ over $R_1 \cap S_0$. If $x \in R_1 \cap S_1$ then there are uniquely determined F_i and G_j, $i \neq 0 \neq j$, such that $\pi_{F_i}(h(x)) = f(x)$ and $\pi_{G_j}(h(x)) = g(x)$, where projections are those of the product $F_i \times G_j$. If $x \in R_2 \subseteq R_1 \cap S_1$ then there is $y < x$ in H such that $\pi_{F_\perp}(h(y)) = r_0$ while $\pi_{G_\perp}(h(y)) \neq s_0$, that is $y \in R_0 \cap S_1$, and hence $h(y) \in F_\perp \times G_j$. Since h is order-preserving and open, x must belong to the isomorphic copy of F_i included as a set in $F_\perp$, but then $h(x) = \langle f, g\rangle(x)$

as otherwise $F_\perp \times G_j$, together with its projections, would not be the product in F of $F_\perp$ and G_j. Similar arguments hold for $x \in S_2$ or $x \in T_2$. Hence $h = \langle f, g\rangle$ and the proof is complete.

□

Note that, writing $\mathbf{1}$ for the one-element poset, we have $\mathscr{F}_1 \cong \mathbf{1} + \mathbf{1}_\perp$. Then $\mathscr{F}_2 \cong (\mathbf{1}+\mathbf{1}_\perp)\times(\mathbf{1}+\mathbf{1}_\perp) \cong \mathbf{1}+\mathbf{1}_\perp+\mathbf{1}_\perp+(\mathbf{1}_\perp\times\mathbf{1}_\perp) \cong \mathbf{1}+\mathbf{1}_\perp+\mathbf{1}_\perp+(\mathbf{1}_\perp+\mathbf{1}+\mathbf{1}_\perp)_\perp$.

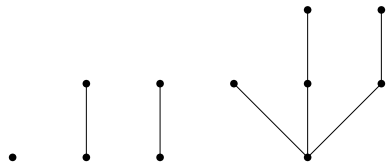


Figure 10. $\mathscr{F}_1 \times \mathscr{F}_1 = \mathscr{F}_2$.

EXAMPLE 4.2.5. *As shown in Figure 10, product of forests obviously fails to be the cartesian product of posets, and its underlying set generally has more points than the cartesian product of the underlying sets.*

Lemma 4.2.4 allows us to compute all finite products. In particular we have the following.

THEOREM 4.2.6.

$$\mathscr{F}_n = H_n + (H_n)_\perp$$

where $H_0 = \emptyset$ *and*

$$H_n = \sum_{i=0}^{n-1} \binom{n}{i} (H_i)_\perp \,. \tag{13}$$

Proof. Clearly $\mathscr{F}_0 = H_0 + (H_0)_\perp \cong \mathbf{1} \cong \mathrm{Spec}\boldsymbol{F}^0_{\mathbb{G}}$. By induction one shows that $H_n + (H_n)_\perp = (H_{n-1} + (H_{n-1})_\perp) \times (H_1 + (H_1)_\perp)$. □

Let $\boldsymbol{G}_\perp$ be the uniquely determined Gödel algebra whose lattice reduct is obtained appending a new bottom element to the lattice reduct of a Gödel algebra $\boldsymbol{G}$. Notice that $\mathbf{Sub}\, H_0$ is the trivial, one-element, algebra.

THEOREM 4.2.7. *For each integer* $n > 0$*,*

$$\boldsymbol{F}^n_{\mathbb{G}} \cong \prod_{i=0}^{n-1} ((\mathbf{Sub}\, H_i)_\perp)^{\binom{n}{i}} \times \left(\prod_{i=0}^{n-1} ((\mathbf{Sub}\, H_i)_\perp)^{\binom{n}{i}} \right)_\perp .$$

Proof. Immediate, from Theorem 4.2.6. □

It is now easy to give a recurrence formula for the cardinality of $\boldsymbol{F}^n_{\mathbb{G}}$.

COROLLARY 4.2.8.

$$|\boldsymbol{F}^n_{\mathbb{G}}| = c_n^2 + c_n \,,$$

for $c_0 = 1$ *and*

$$c_n = \prod_{i=0}^{n-1} (c_i + 1)^{\binom{n}{i}} \,.$$

4.3 Gödel Hoops

We recall the reader that Gödel hoops are precisely the $\perp$-free subreducts of Gödel algebras. Clearly, they constitute a variety, denoted $\mathbb{GH}$.

For each finite Gödel hoop G, Let $\mathrm{Spec}^*(G) = \mathrm{Spec}(G) \cup \{G\}$ be the map assigning to each finite Gödel hoop G the tree obtained by adding G to the prime spectrum of G, ordered by reverse inclusion. Similarly, let $\mathbf{Sub}^*$ be the map assigning each finite tree T to the Gödel hoop of *nonempty* subforests of T. Define Spec^* and $\mathbf{Sub}^*$ on morphisms by taking preimages, as in Section 4.2. Then $\mathbf{Sub}^*$ and Spec^* realizes a dual equivalence between the categoy T of finite trees and their morphisms, and the category GH of finite Gödel hoops and their homomorphisms.

T is a subcategory of F which inherits products, while the coproduct $T_1 + T_2 + \cdots + T_k$ is obtained collapsing into one element all the roots of the trees in the forest $T_1 \uplus T_2 \uplus \cdots \uplus T_k$. Note also $\mathbf{Sub}^*\, F_\perp = \mathbf{Sub}\, F$.

THEOREM 4.3.1.

$$\boldsymbol{F}^n_{\mathbb{GH}} \cong \mathbf{Sub}^*((H_n)_\perp) \cong \mathbf{Sub}(H_n) \cong \prod_{i=0}^{n-1} (\mathbf{Sub}\,(H_i)_\perp)^{\binom{n}{i}} \,.$$

Moreover,

$$|\boldsymbol{F}^n_{\mathbb{GH}}| = c_n \,,$$

where the sequence of integers $\{c_i\}_{i \in \mathsf{N}}$ *is defined as in Corollary 4.2.8.*

Proof. Immediate, from Theorems 4.2.6, 4.2.7 and Corollary 4.2.8. □

Turning to functional representation, Theorem 4.3.1 is equivalently formulated as follows. Call a function $f\colon S \to [0,1]$ *positive on* S if $f(s) > 0$ for all $s \in S$.

THEOREM 4.3.2. $\boldsymbol{F}^n_{\mathbb{GH}}$ *is the algebra of restrictions to* $(0,1]^n$ *of Gödel functions positive on* $(0,1]^n$*, equipped with pointwise defined operations.*

4.4 Finite-valued Gödel Algebras

Let $\mathbb{G}_k$ be the subvariety of $\mathbb{G}$ generated by the $k+1$ element Gödel chain $\boldsymbol{G}_k$.

The *height* of an element in a forest its the cardinality of its downnset. The height of a forest is the maximum height of its elements. Let F_k be the category formed by all finite forests with height at most k.

The following results are easily derived corollaries of Theorem 4.2.1 and Theorem 4.1.9.

THEOREM 4.4.1. *The category* G_k *of all finite Gödel algebras generated by the* $\boldsymbol{G}_k$ *is dually equivalent to* F_k*. Moreover, the dual object to* $\boldsymbol{F}^n_{\mathbb{G}_k}$ *is obtained from* $\mathscr{F}_n$ *by deletion of all elements of height greater than* k*.*

THEOREM 4.4.2. $\boldsymbol{F}^n_{\mathbb{G}_k}$ *is isomorphic to the algebra of restrictions of functions in* $\boldsymbol{F}^n_{\mathbb{G}}$ *to the set of all rational points in* $[0,1]^n$ *whose denominator divides* k*.*

4.5 Nilpotent Minimum Algebras

In this section we shall adapt the spectral duality representation given for Gödel algebras in the previous subsection to the case of Nilpotent Minimum algebras. We obtain representations of finitely generated free NM-algebras as the dual objects to suitably defined finite $\{0,1\}$-labelled forests. We then obtain the functional representation theorem for free NM-algebras in the form of algebras of certain $[0,1]$-valued functions over $[0,1]^n$.

Recall that negation in NM-algebras is involutive, that is $\neg\neg x = x$ for any element x. Moreover, choice of the negation equips every chain C with a unique structure of NM-algebra:

$$\boldsymbol{C} = \langle C, \odot, \rightarrow_\odot, 0\rangle\,,$$

where

$$x \odot y = \begin{cases} \min\{x,y\} & \text{if } x > \neg y\,, \\ 0 & \text{otherwise.} \end{cases}$$

and the residual implication $\rightarrow_\odot$ is given by

$$x \rightarrow_\odot y = \begin{cases} 1 & \text{if } x \leq y\,, \\ \max\{\neg x, y\} & \text{otherwise.} \end{cases}$$

The variety $\mathbb{NM}$ of Nilpotent Minimum algebras is generated by the standard algebra $[0,1]$ determined by the standard involutive negation $\neg x = 1 - x$.

Given an NM-chain $\boldsymbol{C}$, one can define its *positive* and *negative parts*, respectively by

$$C^+ = \{x \in C \mid x > \neg x\} \qquad \text{and} \qquad C^- = \{x \in C \mid x < \neg x\}\,.$$

An NM-chain $\boldsymbol{C}$ has at most one fixpoint of negation, that is, an element e such that $\neg e = e$. If $\boldsymbol{C}$ has the negation fixpoint, then the domain of $\boldsymbol{C}$ is $C^- \cup \{e\} \cup C^+$; if, on the other hand, $\boldsymbol{C}$ lacks the negation fixpoint then its domain is $C^- \cup C^+$.

For any NM-algebra $\boldsymbol{A}$, we denote $\mathbf{B}(\boldsymbol{A})$ the *Boolean skeleton* of $\boldsymbol{A}$, that is, the largest subalgebra of $\boldsymbol{A}$ that is a Boolean algebra; analogously we denote $\mathbf{MV}(\boldsymbol{A})$ the largest subalgebra of $\boldsymbol{A}$ that is an MV-algebra, or, the MV-*skeleton* of $\boldsymbol{A}$. It can be shown that both $\mathbf{B}(\boldsymbol{A})$ and $\mathbf{MV}(\boldsymbol{A})$ always exist.

One can check that $\mathbf{B}(\boldsymbol{C}) \cong \{0,1\}$ for any NM-chain $\boldsymbol{C}$, and that either $\mathbf{MV}(\boldsymbol{C}) \cong \text{Ł}_1 (\cong \{0,1\})$, or $\mathbf{MV}(\boldsymbol{C}) \cong \text{Ł}_2 \cong \{0, 1/2, 1\}$: the former case obtains if $\boldsymbol{C}$ lacks the negation fixpoint, the latter if $\boldsymbol{C}$ has it. Moreover $\boldsymbol{C}$ actually retracts onto its MV-skeleton. Due to the subdirect representation theorem, these properties extend from chains to the directly indecomposable NM-algebras, as follows.

PROPOSITION 4.5.1. *An NM-algebra $\boldsymbol{A}$ is directly indecomposable if and only if $\mathbf{B}(\boldsymbol{A}) \cong \{0,1\}$. Moreover, $\boldsymbol{A}$ is directly indecomposable if and only if either $\mathbf{MV}(\boldsymbol{C}) \cong \mathbf{Ł}_1$ or $\mathbf{MV}(\boldsymbol{C}) \cong \mathbf{Ł}_2$. Moreover, $\boldsymbol{A}$ has the negation fixpoint if and only if $\mathbf{MV}(\boldsymbol{C}) \cong \mathbf{Ł}_2$.*

As another consequence of the subdirect representation by chains, one notes that given a directly indecomposable NM-algebra $\boldsymbol{A}$, the restriction of its operations to its positive part equips A^+ with a unique structure of Gödel hoop. Vice versa, any Gödel hoop can be seen both as the positive part of a uniquely determined directly indecomposable NM-algebra with negation fixpoint, and as the positive part of a uniquely determined directly indecomposable NM-algebra without negation fixpoint. This allows to strengthen Proposition 4.5.1, as follows.

DEFINITION 4.5.2. *Let $\boldsymbol{D} = \langle D, \&, \to, \top\rangle$ be a Gödel hoop. The* disconnected rotation *of $\boldsymbol{D}$ is the algebra*

$$\mathbf{DR}(\boldsymbol{D}) = \langle D \times \{1\} \cup D \times \{0\}, \otimes, \Rightarrow, \sqcap, \bot\rangle$$

with operations defined as follows:

$$\begin{aligned}
(x,i) \sqcap (y,j) &= (y,j) \sqcap (x,i) = \begin{cases} (x \wedge y, 1) & \text{if } i = j = 1, \\ (x \vee y, 0) & \text{if } i = j = 0, \\ (x, 0) & \text{if } i < j. \end{cases} \\
(x,i) \otimes (y,j) &= (y,j) \otimes (x,i) = \begin{cases} (x \,\&\, y, 1) & \text{if } i = j = 1, \\ (\top, 0) & \text{if } i = j = 0, \\ (y \to x, 0) & \text{if } i < j. \end{cases} \\
(x,i) \Rightarrow (y,j) &= \begin{cases} (x \to y, 1) & \text{if } i = j = 1, \\ (y \to x, 0) & \text{if } i = j = 0, \\ (x \,\&\, y, 0) & \text{if } i > j, \\ (\top, 1) & \text{if } i < j. \end{cases} \\
\bot &= (\top, 0)
\end{aligned}$$

The connected rotation of $\boldsymbol{D}$ is

$$\mathbf{CR}(\boldsymbol{D}) = \left\langle D \times \{1\} \cup \left\{\left(\frac{1}{2}, \frac{1}{2}\right)\right\} \cup D \times \{0\}, \otimes, \Rightarrow, \sqcap, \bot \right\rangle$$

where the operations $\otimes$, $\Rightarrow$, $\sqcap$ and $\bot$ are given as in the definition of the disconnected rotation extended by

$$\begin{aligned}
(x,i) \sqcap \left(\frac{1}{2}, \frac{1}{2}\right) = \left(\frac{1}{2}, \frac{1}{2}\right) \sqcap (x,i) &= \begin{cases} \left(\frac{1}{2}, \frac{1}{2}\right) & \text{if } i = 1, \\ (x,i) & \text{otherwise,} \end{cases} \\
(x,i) \otimes \left(\frac{1}{2}, \frac{1}{2}\right) = \left(\frac{1}{2}, \frac{1}{2}\right) \otimes (x,i) &= \begin{cases} (x,i) & \text{if } i = 1, \\ (\top, 0) & \text{otherwise,} \end{cases} \\
(x,i) \Rightarrow \left(\frac{1}{2}, \frac{1}{2}\right) &= \begin{cases} \left(\frac{1}{2}, \frac{1}{2}\right) & \text{if } i = 1, \\ (\top, 1) & \text{otherwise,} \end{cases} \\
\left(\frac{1}{2}, \frac{1}{2}\right) \Rightarrow (x,i) &= \begin{cases} \left(\frac{1}{2}, \frac{1}{2}\right) & \text{if } i = 0, \\ (\top, 1) & \text{otherwise,} \end{cases}
\end{aligned}$$

A routine verification shows that the disconnected rotation of a Gödel hoop is an NM-algebra without negation fixpoint, while its connected rotation is an NM-algebra with fixpoint.

PROPOSITION 4.5.3. *An* NM*-algebra* $\boldsymbol{A}$ *is directly indecomposable if and only if it is isomorphic either to the connected or to the disconnected rotation of a Gödel hoop* $\boldsymbol{G}(\boldsymbol{A})$.

We recall that the prime spectrum $\mathrm{Spec}\boldsymbol{A}$ of an NM-algebra $\boldsymbol{A}$ is the poset of prime filters of $\boldsymbol{A}$ ordered by reverse inclusion.

PROPOSITION 4.5.4. *If* $\boldsymbol{A}$ *is a directly indecomposable* NM*-algebra, then* $\mathrm{Spec}\boldsymbol{A}$ *is isomorphic to the poset of prime filters of the Gödel algebra* $\boldsymbol{G}(\boldsymbol{A})_{\perp}$.

Proof. An easy check shows that the set of prime filters of $\boldsymbol{A}$ is the union of the set of prime filters of the Gödel hoop A^{+} together with A^{+} itself. □

Note that for each Gödel hoop $\boldsymbol{H}$, the rotations $\mathbf{CR}(\boldsymbol{H})$ and $\mathbf{DR}(\boldsymbol{H})$ have the same prime spectrum:

$$\mathrm{Spec}\mathbf{CR}(\boldsymbol{H}) \cong \mathrm{Spec}\boldsymbol{H}_{\perp} \cong \mathrm{Spec}\mathbf{DR}(\boldsymbol{H})\,.$$

We need an additional bit of information to distinguish these two cases. We start defining a category of trees with a bit attached. Each tree will correspond to a directly indecomposable finite NM-algebra: the bit will specify whether this algebra has the negation fixpoint. Then our duality is extended to a category of forests of labelled trees: each such forest will correspond to a direct product of directly indecomposable finite NM-algebras, and viceversa.

DEFINITION 4.5.5. *A* labelled tree *is a pair* (T, i) *where* T *is a tree and* $i \in \{0, 1\}$. *A* labelled forest *is a finite set of labelled trees. A morphism of labelled trees* (T, i) *and* (T', i') *is an open order-preserving map from* T *to* T' *such that* $i \leq i'$. *A morphism of labelled forests* $\phi : F_1 \to F_2$ *is specified by morphisms of labelled trees* $\phi_j : (T_j, i_j) \to (T'_j, i'_j)$ *where* $(T'_j, i'_j) \in F_2$ *for every* $(T_j, i_j) \in F_1$.

It is trivial to check that labelled forests and their morphisms form a category that we denote by LF. Let LT be the subcategory of labelled trees and DNM the category of directly indecomposable NM-algebras and their homomorphisms.

For any directly indecomposable NM-algebra A, let

$$\mathrm{Spec}^{+}\boldsymbol{A} = (\mathrm{Spec}\boldsymbol{A}, i)$$

where $i = 0$ if A has a negation fixpoint and $i = 1$ otherwise. For each homomorphism $h\colon \boldsymbol{A} \to \boldsymbol{B}$ in DNM, let $\mathrm{Spec}^{+}h : \mathrm{Spec}^{+}\boldsymbol{B} \to \mathrm{Spec}^{+}\boldsymbol{A}$ be given, as usual, by taking counterimages

$$(\mathrm{Spec}^{+}h)(\mathfrak{p}) = h^{-1}(\mathfrak{p}) = \{x \in \boldsymbol{A} \mid h(x) \in \mathfrak{p}\}\,.$$

Note that since there are no homomorphisms from NM-algebras with negation fixpoint into algebras without fixpoint, the duals of the morphisms in DNM are well defined morphisms in LT.

THEOREM 4.5.6. *The category* DNM *and* LT *are dually equivalent via* Spec^+.

Proof. A tedious check shows that Spec^+ is full, faithful, and essentially surjective. □

Let NM be the category of finite NM-algebras.

THEOREM 4.5.7. *The category* NM *and* LF *are dually equivalent via* Spec^+.

Proof. Immediate, from Theorem 4.5.6. □

We can hence give a description of the free NM-algebra in terms of its dual using the same approach we have applied to Gödel algebras in the preceding subsection. In order to do this, we need to describe the product in the category of labelled trees. Coproduct is just disjoint union.

PROPOSITION 4.5.8. *Let* (S, i) *and* (T, j) *be labelled trees, and let* $S \times T$*, together with its projections* π_S, π_T *be the product of* S *and* T *in the category* F *of forests. The labelled tree*

$$(S \times T, ij)\,,$$

with the projection maps

$$\pi_{(S,i)} : (S \times T, ij) \to (S, i) \qquad \pi_{(T,j)} : (S \times T, ij) \to (T, j)$$

induced by π_S *and* π_T*, is the product of* (S, i) *and* (T, j) *in the category* LT.

To compute products in LF *it is sufficient to apply the distribution law: Let* (R, h) *be a labelled tree in* LT*, then the following holds in* LF*:*

$$(R, h) \times ((S, i) + (T, j)) = ((R, h) \times (S, i)) + ((R, h) \times (T, j))\,.$$

Proof. Clearly, $\pi_{(S,i)}$ and $\pi_{(T,j)}$ are morphisms in LT (and in LF) as $ij \leq i, j$. Take two morphisms $f_S\colon (R, k) \to (S, i)$ and $f_T\colon (R, k) \to (T, j)$ in LT. By definition, $k \leq i, j$, and hence $k \leq ij$. Then, by the properties of product in the category F of finite forests, there exists a unique morphism $g\colon (R, k) \to (S \times T, ij)$ such that $\pi_{(S,i)} \circ g = f_S$ and $\pi_{(T,j)} \circ g = f_T$.

The distribution law follows immediately from Lemma 4.2.4.2. □

Let $\boldsymbol{F}^n_{\mathbb{NM}}$ be the free NM-algebra over n generators. Let $\mathscr{LF}_n$ be the labelled forest such that

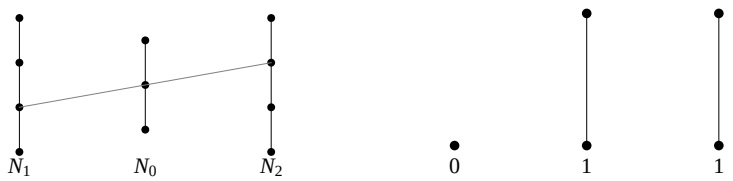
$$\mathrm{Spec}^+ \boldsymbol{F}^n_{\mathbb{NM}} = \mathscr{LF}_n\,.$$

Let C_n denote a chain of $n + 1$ elements.

PROPOSITION 4.5.9. $\mathscr{LF}_1$ *is the forest consisting of the labelled tree* $(C_0, 0)$ *and two copies of the labelled tree* $(C_1, 1)$.

Proof. All singly generated NM-chains are isomorphic either to the three element chain ordered by $\bot < x < \top$ (where x is the generating element); or to the four element chain $\{\bot, x, \neg x, \top\}$, that can be generated in two ways: $\bot < \neg x < x < \top$ and $\bot < x < \neg x < \top$; or they have less than three elements, and in this case they are

homomorphic images of the preceding ones. Let us denote the three element chain as N_0 and the two four element chains by N_1 and N_2. Then $\boldsymbol{F}^1_{\mathbb{NM}}$ embeds in the product $N_0 \times N_1 \times N_2$. By direct computation one finds NM-terms in one variable evaluating to all 48 elements of $N_0 \times N_1 \times N_2$. Hence, $\boldsymbol{F}^1_{\mathbb{NM}} \cong N_0 \times N_1 \times N_2$. The spectrum of N_0 is C_0 and it has the negation fixpoint. The spectrum of N_1 and N_2 is C_1, and they lack the negation fixpoint. □



(a) $F^1_{\mathbb{NM}} \cong N_0 \times N_1 \times N_2$ with generator.

(b) $\mathscr{LF}_1$.

Figure 11. The free singly generated NM-algebra and its labelled prime spectrum.

Recall that, by Lemma 1.0.13, $\mathrm{Spec}^+ \boldsymbol{F}^n_{\mathbb{NM}} = \prod_{i=1}^n \mathscr{LF}_1$.

THEOREM 4.5.10.

$$\mathscr{LF}_n \cong 2^n((H_n)_\perp, 1) + \sum_{i=0}^{n-1} 2^i \binom{n}{i} ((H_i)_\perp, 0)\,,$$

where the poset H_i, for $i = 0, \ldots, n$, is defined by (13).

Proof. Immediate, from Proposition 4.5.8 and Proposition 4.5.9. □

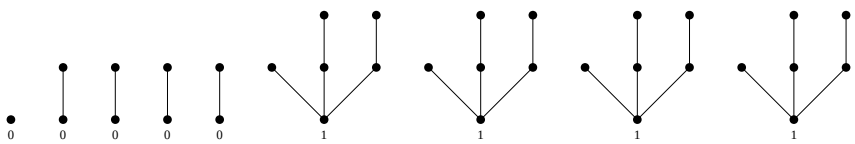


Figure 12. $\mathscr{LF}_2 = \mathscr{LF}_1 \times \mathscr{LF}_1$.

COROLLARY 4.5.11.

$$\boldsymbol{F}^n_{\mathbb{NM}} \cong (\mathbf{DR}(\mathbf{Sub}H_n))^{2^n} \times \prod_{i=0}^{n-1} (\mathbf{CR}(\mathbf{Sub}H_i))^{2^i\binom{n}{i}}\,.$$

$$|\boldsymbol{F}^n_{\mathbb{NM}}| = (2c_n)^{2^n} \cdot \prod_{i=0}^{n-1} (2(c_i+1))^{2^i\binom{n}{i}}\,,$$

where the sequence of integers $\{c_i\}_{i\in\mathsf{N}}$ is defined as in Corollary 4.2.8.

4.6 Functional Representation of Free NM-algebras

By a *signed ordered partition* of $\{x_1, \ldots, x_n\}$ we mean a pair (π, λ), where π is an ordered partition and λ is a function $\lambda\colon \{1, \ldots, n\} \to \{-1, 0, 1\}$ called the *sign function*, satisfying the following

1. $\pi = Q_1 <_\pi \cdots <_\pi Q_h$ is an ordered partition of the set

$$\{x_i \mid i \in \{1, \ldots, n\},\ \lambda(i) \neq 0\} \cup \{\top\}\,,$$

2. $\top \in Q_h$.

We impose a partial order on signed ordered partitions of $\{x_1, \ldots, x_n\}$ by the following stipulation. We set $(\pi_1, \lambda_1) \prec (\pi_2, \lambda_2)$ if the following hold:

- $\pi_1 = Q_1 <_{\pi_1} \cdots <_{\pi_1} Q_{h-1} <_{\pi_1} Q_h$ and $\pi_2 = Q_1 <_{\pi_2} \cdots <_{\pi_2} Q_{h-1} <_{\pi_2} R_h <_{\pi_2} \cdots <_{\pi_2} R_k$ with $R_h \neq Q_h$.
- $\lambda_1 = \lambda_2$.

The relation $\prec$ is clearly irreflexive and transitive. The partial order $\preceq$ induced by $\prec$ is the desired partial order on the set of all signed ordered partitions of $\{x_1, \ldots, x_n\}$.

It is easy to see that the poset $(F_n, \preceq)$ of all signed ordered partitions of $\{x_1, \ldots, x_n\}$ is a finite forest. In particular $(\pi, \lambda_1) \vee_\preceq (\rho, \lambda_2)$ exists if and only if (π, λ_1) and (ρ, λ_2) are $\preceq$-comparable, and hence it must be the case $\lambda_1 = \lambda = \lambda_2$, and then obviously $(\pi, \lambda_1) \vee_\preceq (\rho, \lambda_2) = \max_\preceq\{(\pi, \lambda), (\rho, \lambda)\}$; for what regards $\preceq$-meets, $(\pi, \lambda_1) \wedge_\preceq (\rho, \lambda_2)$ exists if and only if $\lambda_1 = \lambda = \lambda_2$, and can be described as follows. To avoid trivialities, assume $\pi \neq \rho$. Display $\pi = B_1 <_\pi \cdots <_\pi B_h <_\pi R_{h+1} \cdots <_\pi R_u$ and $\rho = B_1 <_\rho \cdots <_\rho B_h <_\rho S_{h+1} \cdots <_\rho S_v$, with $h \in \{0, 1, \ldots, \min(u, v) - 1\}$, and $R_{h+1} \neq S_{h+1}$ (clearly, if $h = 0$ already the first blocks of π and ρ are distinct). Then $(\pi, \lambda_1) \wedge_\preceq (\rho, \lambda_2) = (\tau, \lambda)$ for

$$\tau = B_1 <_\tau \cdots <_\tau B_h <_\tau \bigcup_{i=h+1}^{u} R_i\,.$$

Note that $\bigcup_{i=h+1}^{u} R_i = \bigcup_{i=h+1}^{v} S_i$. The sequence of blocks $B_1 <_\tau \cdots <_\tau B_h$ is the *longest common prefix* of (π, λ) and (ρ, λ).

The set of trees of F_n is in bijection with sign functions: as a matter of fact, for each sign function $\lambda\colon \{1, \ldots, n\} \to \{-1, 0, 1\}$, the signed ordered partition (π_λ, λ) where π_λ is composed by the single block $\{x_i \mid \lambda(i) \neq 0\} \cup \{\top\}$, is $\preceq$ than every $(\rho, \lambda) \in F_n$, while it is incomparable to every (ρ, λ') for $\lambda \neq \lambda'$. We denote T_λ the tree with root (π_λ, λ).

For each $t \in [0, 1]$, let $||t|| = \max\{t, 1 - t\}$.

The *basic region* $[\pi, \lambda]$ associated with a signed ordered partition (π, λ), with $\pi = Q_1 <_\pi \cdots <_\pi Q_h$, is the set of all points $\langle t_1, \ldots, t_n\rangle \in [0, 1]^n$ such that for all $i \in \{1, \ldots, n\}$,

1. $$\lambda(i) = \begin{cases} 1 & \text{if } t_i > \frac{1}{2}\,, \\ 0 & \text{if } t_i = \frac{1}{2}\,, \\ -1 & \text{if } t_i < \frac{1}{2}\,; \end{cases}$$

2. $x_i \in Q_h$ if and only if $||t_i|| = 1$;

3. x_i belongs to a block of π that is $<_\pi$ than the block x_j belongs to, if and only if $||t_i|| < ||t_j||$.

A straightforward verifications shows the following.

LEMMA 4.6.1. *The set of all basic regions associated with signed ordered partitions of* $\{x_1, \ldots, x_n\}$ *is a partition of* $[0,1]^n$.

We define the region associated with a signed ordered partition (π, λ) as

$$N(\pi, \lambda) = \bigcup_{(\rho,\lambda) \preceq (\pi,\lambda)} [\rho, \lambda]\,.$$

We denote $\mathcal{N}_n$ the set of all regions associated with signed ordered partitions of the set $\{x_1, \ldots, x_n\}$. We further denote $N(\lambda) = \bigcup_{(\rho,\lambda)\in T_\lambda} [\rho, \lambda]$.

LEMMA 4.6.2. $N(\lambda) = \bigcup_{(\rho,\lambda)\in T_\lambda} N(\rho, \lambda)$ *for each* $\lambda \in \{-1, 0, 1\}^n$. *Moreover, the set of all regions* $N(\lambda)$*, for* λ *ranging over all the sign functions (there are* 3^n *of them), is a partition of* $[0,1]^n$. *Further,* $(\mathcal{N}_n, \subseteq)$ *is order-isomorphic to* $(F_n, \preceq)$ *via the map* $N(\pi, \lambda) \mapsto (\pi, \lambda)$.

Proof. It follows immediately from Lemma 4.6.1 and the definitions of basic region and of $N(\pi, \lambda)$. □

We now introduce normal forms for NM-terms, based on a suitable definition of minterms. First we introduce the following derived connectives:

$$\begin{array}{lcl} \circ x & = & (x \leftrightarrow \neg x) \,\&\, (x \leftrightarrow \neg x)\,, \\ +x & = & \neg((\neg(x \,\&\, x)) \,\&\, (\neg(x \,\&\, x)))\,, \\ -x & = & +(\neg x)\,. \end{array}$$

The above defined operations have the following properties in the standard algebra $[0,1]$. $\circ x = 1$ if and only if $x = \neg x$, $\circ x = 0$ otherwise; $+x = 1$ iff $x > \neg x$, $+x = 0$ otherwise; $-x = 1$ iff $x < \neg x$, $-x = 0$ otherwise.

Let $\lambda\colon \{1, \ldots, n\} \to \{-1, 0, 1\}$ be a sign function. For each $j \in \{-1, 0, 1\}$, let $T_j = \{i \mid \lambda(i) = j\} \subseteq \{1, \ldots, n\}$. We define

$$\sigma_\lambda = \bigwedge_{i\in T_{-1}} -x_i \wedge \bigwedge_{i\in T_0} \circ x_i \wedge \bigwedge_{i\in T_1} +x_i\,.$$

One checks that $\sigma_\lambda^{\boldsymbol{F}^n_{\mathbb{NM}}}(t_1, \ldots, t_n) = 1$ if and only if $\langle t_1, \ldots, t_n\rangle \in N(\lambda)$, and that $\sigma_\lambda^{\boldsymbol{F}^n_{\mathbb{NM}}}(t_1, \ldots, t_n) = 0$ otherwise.

DEFINITION 4.6.3. *Consider a signed ordered partition* (π, λ). *Display* $\pi = Q_1 < \cdots < Q_h$. *Denote* $X_\lambda = \{i \mid \lambda(i) \neq 0\}$. *Let* $\sigma\colon X_\lambda \to X_\lambda$ *be a permutation and* $\prec_i\colon \{1, \ldots, n\} \to \{<, =\}$ *be a map such that* $[\pi, \lambda]$ *is the set of all points* $\langle t_1, \ldots, t_n\rangle \in [0, 1]^n$ *satisfying*

- $t_i = 1/2$ *if and only if* $\lambda(i) = 0$,
- $t_{\sigma(1)}^{(\epsilon(1))} \prec_1 t_{\sigma(2)}^{(\epsilon(2))} \prec_2 \cdots \prec_{n-1} t_{\sigma(n)}^{(\epsilon(n))} \prec_n 1$, *where* $\epsilon(i) = \lambda(\sigma(i))$, *and* $t_i^{(1)} = t_i$ *while* $t_i^{(-1)} = 1 - t_i$.

Set $x_{\sigma(|X_\lambda|+1)} = \top$ *and define*

$$\chi^+_{\pi,\lambda} = \sigma_\lambda \wedge \bigwedge_{i=1}^{|X_\lambda|} \delta_i\,,$$

where

$$\delta_i = \begin{cases} x_{\sigma(i)}^{\epsilon(i)} \leftrightarrow x_{\sigma(i+1)}^{\epsilon(i+1)} & \textit{iff } \prec_i \textit{ is } =, \\ x_{\sigma(i)}^{\epsilon(i)} \lhd x_{\sigma(i+1)}^{\epsilon(i+1)} & \textit{iff } \prec_i \textit{ is } <, \end{cases}$$

for every $i \in \{1, \ldots, |X_\lambda|\}$ *and where* $x_i^{(1)} = x_i$ *while* $x_i^{(-1)} = \neg x_i$. *The formula* $\chi^+_{(\pi,\lambda)}$ *is called the* positive minterm *at* (π, λ).

LEMMA 4.6.4. *The function* $(\chi^+_{\pi,\lambda})^{\boldsymbol{F}^n_{\mathbb{NM}}}$ *is the smallest term function constantly* 1 *over* $[\pi, \lambda]$; *further, it is constantly* 0 *over* $[0, 1]^n \setminus N(\lambda)$ *and is the smallest term function constantly* 1 *over* $N(\pi, \lambda)$.

Proof. Recalling that the restriction of the operations to the positive part $(1/2, 1]$ of the standard NM-chain $[0, 1]$ makes it into a Gödel hoop, the proof follows from Lemma 4.1.6 and the properties of σ_λ. □

We shall introduce *negative minterms*, too.

LEMMA 4.6.5. *For any* (π_λ, λ) *set* $\chi^-_{\pi_\lambda,\lambda} = \bot$. *For any* (π, λ) *such that* $(\pi, \lambda) \neq (\pi_\lambda, \lambda)$, *let* (ρ, λ) *denote the unique* $\preceq$*-predecessor of* (π, λ). *Let*

$$\chi^-_{\pi,\lambda} = \sigma_\lambda \wedge \neg \bigvee_{(\tau,\lambda)\in\max_{\preceq}(T_\lambda\setminus\uparrow(\pi,\lambda))} \chi^+_{\tau,\lambda}\,.$$

Then, $(\chi^-_{\pi,\lambda})^{\boldsymbol{F}^n_{\mathbb{NM}}}(t_1, \ldots, t_n) =$

$$= \begin{cases} 1 - (\chi^+_{\rho,\lambda})^{\boldsymbol{F}^n_{\mathbb{NM}}}(t_1, \ldots, t_n) & \textit{if } (t_1, \ldots, t_n) \in \bigcup_{(\tau,\lambda)\in\uparrow(\pi,\lambda)} N(\tau, \lambda) \\ 0 & \textit{otherwise.} \end{cases}$$

Proof. We note that $(\bigvee_{(\tau,\lambda)\in\max_{\preceq}(T_\lambda\setminus\uparrow(\pi,\lambda))} \chi^+_{\tau,\lambda})^{\boldsymbol{F}^n_{\mathbb{NM}}}(t_1, \ldots, t_n) = 1$ if and only if $\langle t_1, \ldots, t_n\rangle \in \bigcup_{(\tau,\lambda)\in T_\lambda\setminus\uparrow(\pi,\lambda)} N(\tau, \lambda)$ and it coincides with $(\chi^+_{\rho,\lambda})^{\boldsymbol{F}^n_{\mathbb{NM}}}(t_1, \ldots, t_n)$ for all $\langle t_1, \ldots, t_n\rangle \in \bigcup_{(\tau,\lambda)\in\uparrow(\pi,\lambda)} N(\tau, \lambda)$. □

We also introduce *fixpoint minterms*.

LEMMA 4.6.6. *Assume there is $i \in \{1, \ldots, n\}$ such that $\lambda(i) = 0$. Let*

$$\chi^{\circ}_{\pi_\lambda,\lambda} = \sigma_\lambda \wedge x_i \,.$$

Then

$$(\chi^{\circ}_{\pi_\lambda,\lambda})^{\boldsymbol{F}^n_{\mathbb{NM}}}(t_1, \ldots, t_n) = \begin{cases} \frac{1}{2} & \text{if } \langle t_1, \ldots, t_n\rangle \in N(\lambda) \\ 0 & \text{otherwise .} \end{cases}$$

Proof. Immediate, from the properties of σ_λ. □

LEMMA 4.6.7. *Let φ be a term in n variables and (π, λ) be a maximal signed ordered partition of $\{x_1, \ldots, x_n\}$. Then there exists a unique signed ordered partition $(\rho, \lambda) \preceq (\pi, \lambda)$, which we denote $\rho(f, \pi, \lambda)$, and a unique $*(f, \lambda) \in \{-, \circ, +\}$ such that*

$$\varphi^{\boldsymbol{F}^n_{\mathbb{NM}}}(t_1, \ldots, t_n) = (\chi^{*(f,\lambda)}_{\rho(f,\pi,\lambda)})^{\boldsymbol{F}^n_{\mathbb{NM}}}(t_1, \ldots, t_n)$$

for all $\langle t_1, \ldots, t_n\rangle \in N(\pi, \lambda)$.Moreover, φ can be equivalently written as

$$\varphi \equiv_{\mathbb{NM}} \bigvee_{\lambda\in\{-1,0,1\}^n} \bigvee_{(\pi,\lambda)\in\max_{\preceq} T_\lambda} \chi^{*(f,\lambda)}_{\rho(f,\pi,\lambda)} \,.$$

Proof. From Lemma 4.6.4, Lemma 4.6.5, and Lemma 4.6.6. □

DEFINITION 4.6.8. *A function $f\colon [0, 1]^n \to [0, 1]$ is an* NM*-function over n variables if for each region $N(\pi, \lambda)$ there is a function $g \in \{\bot, 1-x_1, \ldots, 1-x_n, x_1, \ldots, x_n, \top\}$ such that $f(t_1, \ldots, t_n) = g(t_1, \ldots, t_n)$ for all $\langle t_1, \ldots, t_n\rangle \in N(\pi, \lambda)$. The* NM*-algebra of all* NM*-functions equipped with pointwise operations will be denoted by $\mathbf{NM}_n$.*

THEOREM 4.6.9. $\boldsymbol{F}^n_{\mathbb{NM}} \cong \mathbf{NM}_n$.

Proof. Immediate, from Lemma 4.6.7. □

An interesting property of NM-functions, analogous to Lemma 4.1.6, is the following.

PROPOSITION 4.6.10. *Let $f, g \in \mathbf{NM}_n$ be such that $f^{-1}(1) = g^{-1}(1)$ and $f^{-1}(0) = g^{-1}(0)$. Then $f = g$.*

Finally, notice that the posets $(F_n, \preceq)$ and $\mathrm{Spec}\boldsymbol{F}^n_{\mathbb{NM}}$ are order-isomorphic via the map $(\pi, \lambda) \mapsto \langle \chi^+_{\pi,\lambda}\rangle$, as positive minterms are exactly those join-irreducible elements of $\boldsymbol{F}^n_{\mathbb{NM}}$ that singly generate its prime filters.

5 BL-algebras and Related Structures

In this section we shall introduce the functional representation theorem for free n-generated BL-algebras. We shall then show how to adapt this construction to obtain the representation of free n-generated SBL-algebras. We shall then deal with locally finite subvarieties of $\mathbb{BL}$. However, we first discuss the case of $\mathrm{BL}_\triangle$-algebras, as it is studied in [59].

5.1 Free $\mathrm{BL}_\triangle$-algebras

For each integer $n \geq 0$, the subvariety of $\mathbb{BL}$ generated by the class of n-generated BL-algebras is in turns generated by the algebra

$$(n+1)[0,1]_{Ł}\,,$$

that is, the ordinal sum of $n+1$ many copies of the standard MV-algebra. By Lemma 1.0.12 it is thus sufficient, in order to characterise the free n-generated BL-algebra $\boldsymbol{F}^n_{\mathbb{BL}}$, to identify the subalgebra of the algebra

$$(n+1)[0,1]_{Ł}^{((n+1)[0,1]_{Ł})^n}$$

generated by the projections.

This has proved not to be an easy task to tackle. The problem lies in the complex interaction of the behaviour of functions in $\boldsymbol{F}^n_{\mathbb{BL}}$ over different regions of the domain $[0,n+1]^n$. The intricacies of this interaction will be dealt with in Section 5.2: as a first step towards the description of elements of $\boldsymbol{F}^n_{\mathbb{BL}}$, we focus on free algebras on a related variety, obtained by expanding the language of BL with Baaz's $\triangle$ unary connective. The resulting variety $\mathbb{BL}_\triangle$ is much easier to deal with, as the operator $\triangle$ allows for definition by cases, killing off most of the interacting effects between regions.

DEFINITION 5.1.1. *A* $\mathrm{BL}_\triangle$*-algebra* $\boldsymbol{A}$ *is a* BL*-algebra with an additional unary operator* $\triangle$ *satisfying, for each* $x, y \in \boldsymbol{A}$*:*

1. $\triangle x \vee \neg\triangle x = 1$,
2. $\triangle 1 = 1$,
3. $\triangle(x \to y) \leq \triangle x \to \triangle y$,
4. $\triangle x \leq x$,
5. $\triangle\triangle x = \triangle x$.

Let $\mathbb{BL}_\triangle$ *denote the variety of all* $\mathrm{BL}_\triangle$*-algebras.*

LEMMA 5.1.2. *For every* $\mathrm{BL}_\triangle$ *chain* $\boldsymbol{C}$ *and for each* $x \in \boldsymbol{C}$*,*

$$\triangle x = \begin{cases} 1 & \textit{if } x = 1 \\ 0 & \textit{otherwise}\,. \end{cases}$$

Further, $\mathbb{BL}_\triangle$ *enjoys standard completeness, and in particular, for each integer* $n \geq 0$*, the subvariety of* $\mathbb{BL}_\triangle$ *generated by the* n*-generated* $\mathrm{BL}_\triangle$*-algebras is generated by the* $\mathrm{BL}_\triangle$*-algebra* $(n+1)[0,1]_{Ł}$*.*

Note that we can identify the universe of $(n+1)[0,1]_{Ł}$ with the interval $[0,n+1]$ in the obvious way, and the top element of the algebra is then the integer $n+1$.

Hence again, by Lemma 1.0.12, to characterise $\boldsymbol{F}^n_{\mathbb{BL}_\triangle}$, one has to identify the functions $f\colon ((n+1)[0,1]_{Ł})^n \to (n+1)[0,1]_{Ł}$ generated by the projections. The structure of ordinal sum of $(n+1)[0,1]_{Ł}$ suggests to partition the domain $[0,n+1]^n$ into *cells* as follows.

DEFINITION 5.1.3. *For each integer $n \geq 0$, a* cell *is a subset C of $[0, n+1]^n$ of the form $C = C_1 \times C_2 \times \cdots \times C_n$, where each C_i is either the singleton $I_{n+1} = \{n+1\}$ or is of the form $I_j = [j, j+1)$ for some integer $j \in \{0, 1, \ldots, n\}$.*

It turns out that the behaviour of any element $f \in \boldsymbol{F}^n_{\mathbb{BL}_\triangle}$ is redundant over some cells, that is, the restriction of f to a one cell may be completely determined by the restriction of f to another, distinct, cell. The complement of the set of redundant cells are exactly those cells $I_{k_1} \times \cdots \times I_{k_n}$ such that $(\bigcup_{i=1}^n \{k_i\} \cup \{0\}) \setminus \{n+1\}$ is an initial segment of N, that is, it is of the form $\{0, 1, \ldots, h\}$ for some h. Let $I(n)$ be the set-theoretic union of all non-redundant cells. Then the algebra of restrictions of functions in $\boldsymbol{F}^n_{\mathbb{BL}_\triangle}$ to the domain $I(n)$ is a $\mathrm{BL}_\triangle$-algebra isomorphic to $\boldsymbol{F}^n_{\mathbb{BL}_\triangle}$.

The restriction of a function $f \in \boldsymbol{F}^n_{\mathbb{BL}_\triangle}$ to a cell is a sort of "generalised" McNaughton function.

DEFINITION 5.1.4. *Let $\{y_1, \ldots, y_k\}$ be a subset of the set of variables $\{x_1, \ldots, x_n\}$, and let $f \in \boldsymbol{F}^k_{\mathbb{MV}}$ be a McNaughton function. For each $i \in \{0, , \ldots, n+1\}$ we define a function $f_{i,n} \colon I_i^k \to I_i \cup \{0, n+1\}$ as follows:*

1. *If $i = 0$ and $f(y_1, \ldots, y_k) \neq 1$ then $f_{i,n}(y_1, \ldots, y_k) = f(y_1, \ldots, y_k)$;*
2. *If $i = 0$ and $f(y_1, \ldots, y_k) = 1$ then $f_{i,n}(y_1, \ldots, y_k) = n+1$;*
3. *If $i \in \{1, \ldots, n\}$ and $f(1, 1, \ldots, 1) = 1$ and $f(y_1 - i, \ldots, y_k - i) \neq 1$ then $f_{i,n}(y_1, \ldots, y_k) = f(y_1 - i, \ldots, y_k - i) + i$;*
4. *If $i \in \{1, \ldots, n\}$ and $f(1, 1, \ldots, 1) = 1$ and $f(y_1 - i, \ldots, y_k - i) = 1$ then $f_{i,n}(y_1, \ldots, y_k) = n+1$;*
5. *If $i = n+1$ and $f(1, 1, \ldots, 1) = 1$ then $f_{i,n}(y_1, \ldots, y_k) = n+1$;*
6. *If $i > 0$ and $f(1, 1, \ldots, 1) = 0$ then $f_{i,n}(y_1, \ldots, y_k) = 0$.*

We are now ready to subdivide cells into regions whose characteristic function can be expressed with a suitable $\mathrm{BL}_\triangle$-term. These regions, namely, *linear semialgebraic sets*, play roughly the same role as the polyhedra in the complex Σ of Lemma 2.1.4 play for McNaughton functions.

DEFINITION 5.1.5. *For any set S and any subset K of $\{1, \ldots, n\}$, a function $f \colon S^n \to S$ is* essentially $|K|$-ary *over S^K if, for any two points $\langle x_1, \ldots, x_n \rangle, \langle y_1, \ldots, y_n \rangle$ of S^n such that $x_i = y_i$ for all $i \in K$, it holds that $f(x_1, \ldots, x_n) = f(y_1, \ldots, y_n)$. If $i \in K$ then f* depends on x_i. *If K is the smallest set such that f is essentially $|K|$-ary over S^K then f* only depends on $\{x_i \mid i \in K\}$.

The following definition introduces a slight variant of the Q-semialgebraic sets introduced in Definition 2.4.1.

DEFINITION 5.1.6. *Let C be a cell. A* linear semialgebraic subset *of C is the set of solutions in C of a finite set of linear inequalities E of the form*

$$f(x_1, \ldots, x_n) \lhd g(x_1, \ldots, x_n)\,,$$

for $\lhd \in \{\leq, <\}$, satisfying the following stipulations.

1. *For each inequality $e \in E$ of the form $f(x_1, \ldots, x_n) \lhd g(x_1, \ldots, x_n)$, there is an integer $i(e) \in \{0, \ldots, n+1\}$ such that for all $j \in \{1, \ldots, n\}$, if either f or g depends on the variable x_j, then for all $\langle a_1, \ldots, a_n \rangle \in C$, $\lfloor a_j \rfloor = i(e)$.*

2. *For each inequality $e \in E$ of the form $f(x_1, \ldots, x_n) \lhd g(x_1, \ldots, x_n)$, there is a subset $\{y_1, \ldots, y_k\}$ of variables, and $h, k \in \boldsymbol{F}^k_{\mathbb{MV}}$ such that f and g only depend on $\{y_1, \ldots, y_k\}$, and for all $\langle x_1, \ldots, x_n \rangle \in C$ it holds that*

$$f(x_1, \ldots, x_n) = h_{i(e),n}(y_1, \ldots, y_k)$$

and

$$g(x_1, \ldots, x_n) = k_{i(e),n}(y_1, \ldots, y_k)\,.$$

3. *If $i(e) > 0$ then $h(1, 1, \ldots, 1) = 1 = k(1, 1, \ldots, 1)$.*

The following Lemma shows the usefulness of the operator Δ in the task of isolating the behaviour of a $\mathrm{BL}_\triangle$-function f over a linear semialgebraic subset, that is, for any linear semialgebraic subset Y it is possible to express with a term the characteristic function of Y. When dealing with BL-algebras (without $\triangle$), it is in general not possible to express such characteristic functions.

LEMMA 5.1.7. *For every linear semialgebraic subset Y of a cell C, there is a $\mathrm{BL}_\triangle$-term φ_Y such that for all $\langle t_1, \ldots, t_n \rangle \in [0, n{+}1]^n$ it holds that $\varphi_Y^{(n+1)[0,1]_L}(t_1, \ldots, t_n) \in \{0, n+1\}$ and*

$$\langle t_1, \ldots, t_n \rangle \in Y \quad \textit{if and only if} \quad \varphi_Y^{(n+1)[0,1]_L}(t_1, \ldots, t_n) = n+1\,.$$

Lemma 5.1.7 allows to prove the functional representation theorem for $\boldsymbol{F}^n_{\mathbb{BL}_\triangle}$.

DEFINITION 5.1.8. *A* $\mathrm{BL}_\triangle$-partition *of $I(n)$ is a partition of $I(n)$ into linear semialgebraic subsets of cells. An* elementary $\mathrm{BL}_\triangle$-function *is a function f with domain $\mathrm{dom}(f) = Y$, for Y a linear semialgebraic subset of some cell C, and is such that either f is constant on Y, and then this constant value is either 0 or $n+1$, or there exist $i(f) \in \{0, \ldots, n+1\}$ and $g \in \boldsymbol{F}^k_{\mathbb{MV}}$ with $f = g_{i(f),n}$ and for each $j \in \{1, \ldots, n\}$, if g depends on the variable x_j, then $\lfloor a_j \rfloor = i(f)$ for all $\langle a_1, \ldots, a_n \rangle \in C$.*

THEOREM 5.1.9. *For each integer $n \geq 0$, the free n-generated $\mathrm{BL}_\triangle$-algebra $\boldsymbol{F}^n_{\mathbb{BL}_\triangle}$ is the algebra whose universe is the set of all functions $f\colon I(n) \to [0, n+1]$ such that there exist a $\mathrm{BL}_\triangle$-partition $\{P_1, \ldots, P_m\}$ and elementary $BL_\triangle$-functions $f_1, \ldots, f_m$ satisfying, for each index $i \in \{1, \ldots, m\}$:*

1. $\mathrm{dom}(f_i) = P_i$;

2. *f coincides with f_i over P_i.*

The operations are defined pointwise.

5.2 Free BL-algebras

In this section we state the functional representation theorem for the n-generated free BL-algebra $\boldsymbol{F}^n_{\mathbb{BL}}$. Recall that the subvariety generated by the class of all n-generated BL-algebras is singly generated by the algebra $(n+1)[0,1]_{\text{Ł}}$, hence $\boldsymbol{F}^n_{\mathbb{BL}}$ is (isomorphic to) the subalgebra of the algebra of all the functions $f\colon ((n+1)[0,1]_{\text{Ł}})^n \to (n+1)[0,1]_{\text{Ł}}$ generated by the projections. By Lemma 1.0.12, it is thus sufficient to characterise this set.

Roughly speaking, we characterize elements of $\boldsymbol{F}^n_{\mathbb{BL}}$, which we call *n-ary BL-functions*, in terms of a finite collection of suitable elements of the free n-generated Wajsberg hoop $\boldsymbol{F}^n_{\mathbb{WH}}$, suitably arranged over a finite, partially ordered, combinatorial structure, which we call the *Fubini tree* of the set $\{1,\dots,n\}$ (see Definition 5.2.1). Each node of a Fubini tree corresponds bijectively to a non-redundant cell $\prod_{i=1}^n I_{i_k}$, with each $i_k \neq n+1$, of Definition 5.1.3; the additional partially ordered structure imposed on the set of cells will record the interdependencies between cells. We call these arrangements *(n-ary) encodings*. Each encoding uniquely determines an n-ary real valued function over $[0,n+1]^n$, in fact, a BL-function. The set of n-ary BL-functions, equipped with operations defined pointwise by the fundamental operations of $(n+1)[0,1]_{\text{Ł}}$, is the free n-generated BL-algebra.

Throughout the section let $[n]$ denote the set $\{1,2\dots,n\}$. We start with some basic definitions. Let $K \subseteq [n]$ have cardinality $m = |K|$, and let S be a set. We call any function $\mathbf{x}\colon K \to S$ an *m-tuple indexed by K*, and we safely regard $\mathbf{x}$ as a point in the m-dimensional space S^m whose coordinates are indexed by K. For each point $\mathbf{x} = (x_1,\dots,x_n) \in S^n$ its *projection over S^K*, denoted $\pi_K(\mathbf{x})$, is the m-tuple indexed by K defined as $k \in K \mapsto x_k = (\mathbf{x})_k$. If $R \subseteq S^n$, the *projection of R over S^K* is $\pi_K(R) = \{\pi_K(\mathbf{x}) \mid \mathbf{x} \in R\}$. Hence, $S^{[n]} = \pi_{[n]}(S^n) = S^n$. Even though $S^{[n]} = S^n$ we consistently write $S^{[n]}$ throughout the rest of this section, since we shall very often be dealing with points in S^K for subsets K of $[n]$.

Assume $\mathbf{x}_1 \in S^{K_1}$ and $\mathbf{x}_2 = S^{K_2}$ for $K_1, K_2 \subseteq [n]$ and $K_1 \cap K_2 = \emptyset$. Then we define the point

$$\mathbf{x}_1 \dot{\times} \mathbf{x}_2$$

as the point $\mathbf{x} \in S^{K_1 \cup K_2}$ such that $\pi_{K_1}(\mathbf{x}) = \mathbf{x}_1$ and $\pi_{K_2}(\mathbf{x}) = \mathbf{x}_2$, that is, $\mathbf{x}$ is the point whose coordinate of index $j \in K_1 \cup K_2$ equals coordinate $(\mathbf{x}_1)_j$ if j is in K_1 and coordinate $(\mathbf{x}_2)_j$ if j is in K_2. For subsets R_1, R_2 of S we denote $R_1^{K_1} \dot{\times} R_2^{K_2}$ the set $\{\pi_{K_1}(\mathbf{x}) \dot{\times} \pi_{K_2}(\mathbf{y}) \mid \mathbf{x} \in R_1^{[n]},\ \mathbf{y} \in R_2^{[n]}\}$.

DEFINITION 5.2.1. *A* Fubini partition *R of $[n]$ is an ordered partition of $[n]$ into nonempty subsets, that is, $R = (\{B_1,\dots,B_l\},\leq)$ where the set $\{B_1,\dots,B_l\}$ is a partition of $[n]$ and $B_i \leq B_j$ if $i \leq j$. In short, we write $R = (B_1,\dots,B_l)$. We denote $\mathscr{F}_{[n]}$ the set of Fubini partitions of $[n]$.*

Let $R, S \in \mathscr{F}_{[n]}$. We say that R precedes *S, in symbols $R \leq_{\mathscr{F}} S$, if R has the form $(B_1,\dots,B_l,B_{l+1})$ and $S = (B_1,\dots,B_l,C_1,\dots,C_m)$ for $m \geq 1$. Say that S* covers *R, or that R is* the predecessor *of S, in symbols $R \prec_{\mathscr{F}} S$, if R has the form $(B_1,\dots,B_l,B_{l+1})$ and $S = (B_1,\dots,B_l,C_1,C_2)$.*

Note $(\mathscr{F}_{[n]}, \leq_{\mathscr{F}})$ is a tree; by a slight abuse of notation we shall denote $\min(\mathscr{F}_{[n]}) = ([n])$ simply by $[n]$.

EXAMPLE 5.2.2. *Let $n = 2$. Then, $(\mathscr{F}_{[2]}, \leq_{\mathscr{F}})$ is a tree with root $[2]$, covered by the leaves $(\{1\}, \{2\})$ and $(\{2\}, \{1\})$. Let $n = 3$. Then, $(\mathscr{F}_{[3]}, \leq_{\mathscr{F}})$ is a tree with root $[3]$, covered by the leaves $(\{1, 2\}, \{3\})$, $(\{1, 3\}, \{2\})$, $(\{2, 3\}, \{1\})$, and the nodes $(\{1\}, \{2, 3\})$, $(\{2\}, \{1, 3\})$, and $(\{3\}, \{1, 2\})$; the node $(\{1\}, \{2, 3\})$ is covered by the leaves $(\{1\}, \{2\}, \{3\})$ and $(\{1\}, \{3\}, \{2\})$; the node $(\{2\}, \{1, 3\})$ is covered by the leaves $(\{2\}, \{1\}, \{3\})$ and $(\{2\}, \{3\}, \{1\})$; the node $(\{3\}, \{1, 2\})$ is covered by the leaves $(\{3\}, \{1\}, \{2\})$ and $(\{3\}, \{2\}, \{1\})$.*

On the basis of $F_{[n]}$, we define a partition of the domain $[0, n + 1]^{[n]}$ of n-ary BL-functions.

We let $\{P_0, P_1\}$ be the following partition of $[0, n + 1]^{[n]}$: $P_1 = [1, n + 1]^{[n]}$ and $P_0 = [0, n + 1]^{[n]} \setminus P_1$.

For $b \in \{0, 1\}$, each point $\mathbf{x} \in P_b$ is associated with a Fubini partition $R_{\mathbf{x}}$, that reflects the linear order among the integer parts of the components of $\mathbf{x}$, in the following terms.

DEFINITION 5.2.3. *For each $b \in \{0, 1\}$ and every point $\mathbf{x} \in P_b$, we let $R_{\mathbf{x}}$ denote the Fubini partition* of $\mathbf{x}$, *defined as follows. Let $J = \{\lfloor x_j \rfloor \mid j \in [n]\}$. For each $j \in J$ let $B_j = \{k \in [n] \mid \lfloor x_k \rfloor = j\}$. If $\{n, n + 1\} \subseteq J$ then let $J' = J \setminus \{n + 1\}$, and $C_n = B_n \cup B_{n+1}$; let further $C_j = B_j$ for each $j \in J \setminus \{n, n + 1\}$. If $\{n, n + 1\} \not\subseteq J$ then let $J' = J$ and $C_j = B_j$ for each $j \in J$. Set $R_{\mathbf{x}} = \{C_j \mid j \in J'\}$ with $C_i \leq C_j$ iff $i \leq j$. For every $b \in \{0, 1\}$ and every Fubini partition R of $[n]$, we let*

$$P_{b,R} = \{\mathbf{x} \in P_b \mid R_{\mathbf{x}} = R\}.$$

EXAMPLE 5.2.4. *Let $n = 3$, and let $\mathbf{x} = (x_1, x_2, x_3) = (1.2, 4, 0.2) \in [0, 4]^{[3]}$. Then $J = \{0, 1, 4\}$, $B_0 = \{3\}$, $B_1 = \{1\}$, and $B_4 = \{2\}$. Here, $J' = J$, $C_0 = B_0$, $C_1 = B_1$, $C_4 = B_4$, and $R_{\mathbf{x}} = (C_0, C_1, C_4)$. Also, $P_{0,(\{3\},\{1\},\{2\})} = \{(x_1, x_2, x_3) \in [0, 4]^{[3]} \mid 0 = \lfloor x_3 \rfloor < \lfloor x_1 \rfloor < \lfloor x_2 \rfloor\}$, and $P_{1,(\{3\},\{1\},\{2\})} = \{(x_1, x_2, x_3) \in [0, 4]^{[3]} \mid 1 = \lfloor x_3 \rfloor < \lfloor x_1 \rfloor < \lfloor x_2 \rfloor\}$. The point $\mathbf{x}$ above is in $P_{0,(\{3\},\{1\},\{2\})}$.*

Note that $\{P_{b,R} \mid b \in \{0, 1\},\ R \in \mathscr{F}_{[n]}\}$ is a partition of $[0, n + 1]^n$. As a matter of fact each set $P_{b,R}$ is the union of a finite number of cells as defined in Definition 5.1.3. In particular, $P_{b,R}$ contains exactly one non-redundant cell: the behaviour of a BL-function f over other cells in $P_{b,R}$ is determined by the behaviour of f over the non-redundant one.

EXAMPLE 5.2.5. *Let $n = 2$. Then $\{P_{b,[2]}, P_{b,(\{1\},\{2\})}, P_{b,(\{2\},\{1\})} \mid b = 0, 1\}$ forms a partition of $[0, 3]^{[2]}$. In particular, $P_{0,[2]} = \{(x_1, x_2) \mid 0 \leq x_1, x_2 < 1\}$, $P_{0,(\{1\},\{2\})} = \{(x_1, x_2) \mid 0 \leq x_1 < 1 \leq x_2 \leq 3\}$, $P_{0,(\{2\},\{1\})} = \{(x_1, x_2) \mid 0 \leq x_2 < 1 \leq x_1 \leq 3\}$, $P_{1,[2]} = \{(x_1, x_2) \mid 1 \leq x_1, x_2 < 2$ or $2 \leq x_1, x_2 \leq 3\}$, $P_{1,(\{1\},\{2\})} = \{(x_1, x_2) \mid 1 \leq x_1 < 2 \leq x_2 \leq 3\}$, $P_{1,(\{2\},\{1\})} = \{(x_1, x_2) \mid 1 \leq x_2 < 2 \leq x_1 \leq 3\}$.*

We now need to adapt the definition of polyhedral complex given in Section 2.1 to the BL-functions setting.

DEFINITION 5.2.6. *An* open polyhedron *is the relative interior of a polyhedron. An* open polyhedral complex *is the set of relative interiors of the polyhedra of a polyhedral complex.*

Note that linear semialgebraic subsets are finite unions of open polyhedra, or, equivalently, supports of open polyhedral complexes. It will turn out that for every BL-function f there exists an open polyhedral complex Σ whose support is $[0, n+1]^n$ such that f is linear over each open polyhedron in Σ.

We shall associate with each Fubini partition $R \in \mathscr{F}_{[n]}$ a partially defined function f over $[0,1]^{[n]}$, which we call a *prismwise* Wajsberg function. The domain of f will be an open polyhedral complex, and over each element of this complex, f will coincide with a Wajsberg function in $\boldsymbol{F}^n_{\mathbb{WH}}$.

Recall Definition 5.1.5 for the notion of a function essentially $|K|$-ary over $[0,1]^K$.

DEFINITION 5.2.7 (Prismwise Wajsberg Function). *Let K be a nonempty subset of $[n]$. Let $\{\Delta_i \mid i \in [l]\}$ be an open polyhedral complex in $[0,1]^{[n]\setminus K}$ such that $\Delta_i \subseteq [0,1)^{[n]\setminus K}$ for every $i \in [l]$. An* n-ary prismwise Wajsberg function *is a function f from*

$$\mathrm{dom}(f) = \bigcup_{i\in[l]} \Delta_i \dot{\times} [0,1]^K$$

to $[0,1]$ such that for each Δ_i, there exists a Wajsberg function g in $\boldsymbol{F}^n_{\mathbb{WH}}$, essentially $|K|$-ary over $[0,1]^K$, such that the restriction of f to the prism $\Delta_i \dot{\times} [0,1]^K$ *coincides with g, namely,*

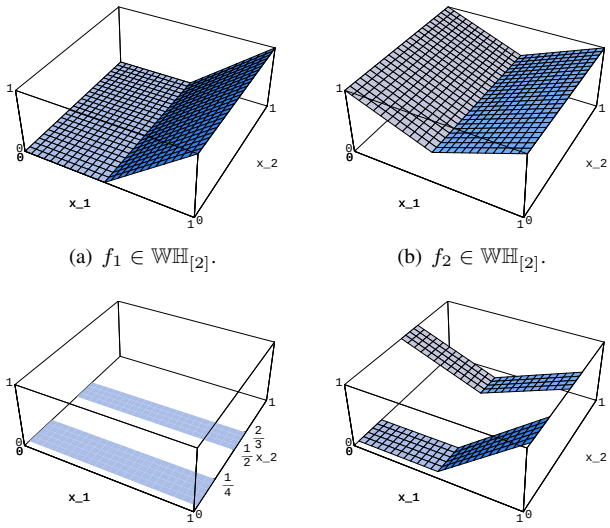
$$f \restriction \Delta_i \dot{\times} [0,1]^K = g.$$

We call $\{\Delta_i\}_{i\in[l]}$ the base *of f and g the* realization *of f over Δ_i. If $\mathbf{x} \in \mathrm{dom}(f)$, then there exists a unique Δ_i in the base of f such that $\mathbf{x} \in \Delta_i \dot{\times} [0,1]^K$. Therefore, we also say that g is the realization of f over $\mathbf{x}$. We let $\mathbb{PW}_{[n]}$ denote the set of n-ary prismwise Wajsberg functions.*

We let $\emptyset$ denote the unique n-ary prismwise Wajsberg function with empty domain. Note that in particular, the Wajsberg function $g \in \boldsymbol{F}^n_{\mathbb{WH}}$ is the n-ary prismwise Wajsberg function coinciding with g over the domain $[0,1]^{[n]}$.

Observe that both $\mathrm{dom}(f)$ and $[0,1]^{[n]} \setminus \mathrm{dom}(f)$ are supports of open polyhedral complexes whose polyhedra are prisms of the form $\Delta \dot{\times} [0,1]^K$ for suitable open polyhedra Δ. We remark that each prism $\Delta \dot{\times} [0,1]^K$ is a linear semialgebraic subset of $[0,1]^n$. On the other hand, each linear semialgebraic subset of a cell $C = \prod_{i=1}^n I_{k_i}$, with each $k_i \leq n$, has the form $\{\langle k_1, \ldots, k_n\rangle + \mathbf{x} \mid \mathbf{x} \in \bigcup_{i\in I} \Delta_i \dot{\times} [0,1)^K\}$ for some open polyhedral complex $\{\Delta_i\}_{i\in I}$ and $K \subseteq [n]$.

We are now ready to formally introduce the notion of *encodings* of BL-functions.



(a) $f_1 \in \mathbb{WH}_{[2]}$. (b) $f_2 \in \mathbb{WH}_{[2]}$.

(c) $\mathrm{dom}(f)$. (d) $f \in \mathbb{PW}_{[2]}$.

Figure 13. Sampling Definition 5.2.7 with $n = 2$. Let $K = \{1\}$. (a) and (b) sketch Wajsberg functions f_1 and f_2 in $\mathbb{WH}_{[2]}$. Both f_1 and f_2 are essentially unary over $[0,1]^{\{1\}}$. Let $\Delta_1 = \{(x_2) \mid 0 < x_2 < 1/4\} \subseteq [0,1)^{\{2\}}$, and let $\Delta_2 = \{(x_2) \mid 1/2 < x_2 < 2/3\} \subseteq [0,1)^{\{2\}}$. The base of f is $\{(x_2) \mid 0 < x_2 < 1/4 \text{ or } 1/2 < x_2 < 2/3\} \subseteq [0,1)^{\{2\}}$. (c) and (d) describe the prismwise Wajsberg function f in $\mathbb{PW}_{[2]}$ whose realizations over Δ_1 and Δ_2 are f_1 and f_2 respectively. Namely, f coincides with f_1 over $\Delta_1 \times [0,1]^{\{1\}}$, and with f_2 over $\Delta_2 \times [0,1]^{\{1\}}$.

DEFINITION 5.2.8 (Encoding). *The set $A_{[n]}$ of n-ary* encodings *is the set of all pairs of functions*

$$(L\colon \mathscr{F}_{[n]} \to \mathbb{PW}_{[n]}, H\colon \mathscr{F}_{[n]} \to \mathbb{PW}_{[n]})$$

satisfying the following conditions.

(i) $\mathrm{dom}(L([n])) = [0,1]^{[n]}$ *and* $L([n]) \in \boldsymbol{F}^n_{\mathbb{WH}}$.

(ii) $\mathrm{dom}(H([n])) = [0,1]^{[n]}$ *and* $H([n]) \in \boldsymbol{F}^n_{\mathbb{WH}}$, *or* $\mathrm{dom}(H([n])) = \emptyset$.

(iii) *Let $I \in \{L, H\}$, $S \prec_{\mathscr{F}} R \in \mathscr{F}_{[n]}$ and let K be the maximum block of R. Let further F_K be the $([n] \setminus K)$-dimensional* semi-open face

$$F_K = [0,1)^{[n]\setminus K} \dot{\times} \{1\}^K ,$$

and

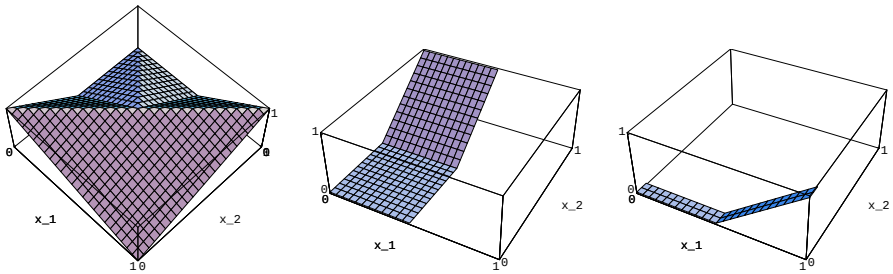
$$\Delta_{K,I,S} = \pi_{[n]\setminus K}\left(I(S)^{-1}(b) \cap F_K\right) ,$$

where $b \in \{0,1\}$ is equal to 0 if and only if: $I = L$, $S = [n]$, and $\mathrm{dom}(H(S)) = \emptyset$. Then, for $I \in \{L, H\}$, the function I is essentially $|K|$-ary over K and its

domain is:

$$\mathrm{dom}(I(R)) = \Delta_{K,I,S} \dot{\times} [0,1]^K.$$

The role of the bit b in Definition 5.2.8 can be understood by comparison with Theorem 2.2.4.



(a) h_1. (b) h_2. (c) h_3.

Figure 14. Sampling Definition 5.2.8 with $b = 0$. $E_0 = (L_0, H_0) \in A_{[2]}$. $L_0 \colon F_{[2]} \to \mathbb{PW}_{[2]}$ is the map $[2] \mapsto h_1$, $(\{1\},\{2\}) \mapsto h_2$, $(\{2\},\{1\}) \mapsto h_3$. Here, $\mathrm{dom}(h_1) = [0,1]^{[2]}$. As $b = 0$, we have $H_0([2]) = \mathrm{dom}(H_0([2])) = \emptyset$, and since then $H_0([2])^{-1}(1) = \emptyset$, we have that the map $H_0 \colon F_{[2]} \to \mathbb{PW}_{[2]}$ is identically $\emptyset$. By inspection, $\mathrm{dom}(h_2) = \{(x_1) \in [0,1)^{\{1\}} \mid 0 \le x_1 \le 1/2\} \times [0,1]^{\{2\}}$, since $\pi_{\{1\}}\left(h_1^{-1}(0) \cap \{(x_1,x_2) \mid 0 \le x_1 < 1, x_2 = 1\}\right) = \{(x_1) \in [0,1)^{\{1\}} \mid 0 \le x_1 \le 1/2\}$; $\mathrm{dom}(h_3) = \{(x_2) \in [0,1)^{\{2\}} \mid 0 = x_2\} \times [0,1]^{\{1\}}$, since $\pi_{\{2\}}\left(h_1^{-1}(0) \cap \{(x_1,x_2) \mid 0 \le x_2 < 1, x_1 = 1\}\right) = \{(x_2) \in [0,1)^{\{1\}} \mid 0 = x_2\}$.

In most cases, we can get easily rid of the necessity of explicitly dealing with the value of b by using the following simplification.

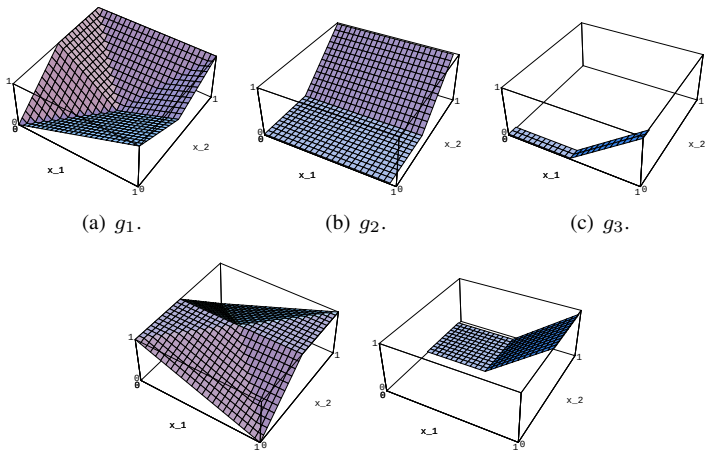
Let $(L,H) \in A_{[n]}$ and $I \in \{L,H\}$. Then we denote by $I_+(R)$ the function

$$I_+(R)(\mathbf{x}) \quad = \begin{cases} 1 - I(R)(\mathbf{x}) & I = L,\ R = [n] \text{ and } H([n]) = \emptyset, \\ I(R)(\mathbf{x}) & \text{otherwise.} \end{cases}$$

Analogously, if $g \in \boldsymbol{F}^n_{\mathbb{WH}}$ is the function realizing $I(R)$ over Δ_i (for $\{\Delta_i\}_{i \in I}$ being the base of $I(R)$) then we let $g_+ = 1 - g$ if $I = L$, $R = [n]$ and $H([n]) = \emptyset$, and $g_+ = g$, otherwise.

As the terminology suggests, the purpose of encodings is that of *specifying* certain real valued functions from $[0,n+1]^{[n]}$ to $[0,n+1]$, in fact, BL-functions. The task now in order is the following. Given an encoding $A \in A_{[n]}$, describe a uniquely determined map from $[0,n+1]^{[n]}$ to $[0,n+1]$ in terms of the information encapsulated in A: the BL-function encoded by A.

For each $x \in [0,n+1]$, we let $\lfloor x \rceil = 1$ if $x = n+1$, and $\lfloor x \rceil = x - \lfloor x \rfloor$ otherwise. For each $\mathbf{x} = (x_1,\ldots,x_n) \in [0,n+1]^{[n]}$, we let $\lfloor \mathbf{x} \rceil = (\lfloor x_1 \rceil,\ldots,\lfloor x_n \rceil) \in [0,1]^{[n]}$.



(a) g_1. (b) g_2. (c) g_3.

(d) g_4. (e) g_5.

Figure 15. Sampling Definition 5.2.8 with $b = 1$. $E_1 = (L_1, H_1) \in A_{[2]}$. $L_1 \colon F_{[2]} \to \mathbb{PW}_{[2]}$ is the map $[2] \mapsto g_1$, $(\{1\},\{2\}) \mapsto g_2$, $(\{2\},\{1\}) \mapsto g_3$. $H_1 \colon F_{[2]} \to \mathbb{PW}_{[2]}$ is the map $[2] \mapsto g_4$, $(\{1\},\{2\}) \mapsto \emptyset$, $(\{2\},\{1\}) \mapsto g_5$. g_1 and g_4 are in $\mathbb{WH}_{[2]}$, with $\mathrm{dom}(g_1) = \mathrm{dom}(g_4) = [0,1]^{[2]}$. By inspection, $\mathrm{dom}(g_2) = \{(x_1) \in [0,1)^{\{1\}} \mid 0 \le x_1 < 1\} \times [0,1]^{\{2\}}$, since $\pi_{\{1\}}\left(g_1^{-1}(1) \cap \{(x_1,x_2) \mid 0 \le x_1 < 1, x_2 = 1\}\right) = \{(x_1) \in [0,1)^{\{1\}} \mid 0 \le x_1 < 1\}$; $\mathrm{dom}(g_3) = \{(x_2) \in [0,1)^{\{2\}} \mid 0 = x_2\} \times [0,1]^{\{1\}}$, since $\pi_{\{2\}}\left(g_1^{-1}(1) \cap \{(x_1,x_2) \mid 0 \le x_2 < 1, x_1 = 1\}\right) = \{(x_2) \in [0,1)^{\{1\}} \mid 0 = x_2\}$; $\mathrm{dom}(g_5) = \{(x_2) \in [0,1)^{\{2\}} \mid 1/2 \le x_2 < 1\} \times [0,1]^{\{1\}}$, since $\pi_{\{2\}}\left(g_4^{-1}(1) \cap \{(x_1,x_2) \mid 0 \le x_2 < 1, x_1 = 1\}\right) = \{(x_2) \in [0,1)^{\{1\}} \mid 1/2 \le x_2 < 1\}$. Notice that $H((\{1\},\{2\})) = \emptyset$, since $\pi_{\{1\}}\left(g_4^{-1}(1) \cap \{(x_1,x_2) \mid 0 \le x_1 < 1, x_2 = 1\}\right) = \emptyset$.

To evaluate the encoding $A = (L, H)$ at $\mathbf{x}$ we shall visit the branch constituted by the partitions $\le_{\mathscr{F}} R_{\mathbf{x}}$ in the tree $\mathrm{dom}(L)$ if $\mathbf{x} \in P_0$, and in the tree $\mathrm{dom}(H)$ if $\mathbf{x} \in P_1$. As a matter of fact, if $\lfloor \mathbf{x} \rceil$ does not belong to the domain of $I(R_{\mathbf{x}})$, then the value of the BL-function encoded by A at $\mathbf{x}$ is given by a function $I(S)$ for some $S <_{\mathscr{F}} R_{\mathbf{x}}$ at a point $\mathbf{x}'$ which precedes $\lfloor \mathbf{x} \rceil$ in the so-called *ascending sequence* of $\mathbf{x}$. The ascending sequence of a point starts from the zeroth step $(1, 1, \ldots, 1) = \{1\}^{[n]}$, and then fixes at the ith step only the (fractional part of the) components of $\mathbf{x}$ that appear in the i-th block of $R_{\mathbf{x}}$.

DEFINITION 5.2.9. *Consider $A = (I_0, I_1) \in A_{[n]}$ and let $b \in \{0,1\}$. For each $R = (B_1, \ldots, B_k) \in \mathscr{F}_{[n]}$ let $\mathbf{x} \in P_{b,R}$. Display the totally ordered subset $\{S \in \mathscr{F}_{[n]} \mid S \le_{\mathscr{F}} R\}$ as*

$$[n] = R_1 \prec_{\mathscr{F}} R_2 \prec_{\mathscr{F}} \cdots \prec_{\mathscr{F}} R_k = R\,.$$

We define the ascending sequence *of* $\mathbf{x}$ *as*

$$\{1\}^{[n]} = \mathbf{x}_0, \mathbf{x}_1, \ldots, \mathbf{x}_k = \lfloor \mathbf{x} \rceil\,,$$

where for each $i \in \{0, 1, \ldots, k-1\}$ *and* $j \in [n]$, $(\mathbf{x}_i)_j = 1$ *if* j *belongs to the maximum block of* R_{i+1} *and* $(\mathbf{x}_i)_j = x_j - \lfloor x_j \rfloor$ *otherwise. Equivalently,* $\mathbf{x}_i = \pi_{[n]\setminus\bigcup_{j=i+1}^{k} B_j}(\mathbf{x}_k - \lfloor \mathbf{x}_k \rfloor) \dot{\times} \{1\}^{\bigcup_{j=i+1}^{k} B_j}$.

The evaluation depth $d^A_{\mathbf{x}}$ *of* $\mathbf{x}$ *in* A *is*

$$d^A_{\mathbf{x}} = \max\{i \in [k] \mid \mathbf{x}_i \in \mathrm{dom}(I_b(R_i))\},$$

if this maximum exists, $d^A_{\mathbf{x}} = 0$ *otherwise. Notice that* $d^A_{\mathbf{x}} = 0$ *if and only if* $b = 1$ *and* $\mathrm{dom}(I_1([n])) = \emptyset$. *In this case* $d^A_{\mathbf{y}} = 0$ *for all* $\mathbf{y} \in P_1$.

We call

$$\mathbf{f}^A_{\mathbf{x}} = \mathbf{x}_{d^A_{\mathbf{x}}}$$

the fractional part *of* $\mathbf{x}$ *in* A. *If* $d^A_{\mathbf{x}} > 0$ *we call the integer* $i^A_{\mathbf{x}}$ *such that*

$$i^A_{\mathbf{x}} = \min(\{\lfloor x_j \rfloor \mid j \in B_{d^A_{\mathbf{x}}}\} \cup \{n\}),$$

the integer part *of* $\mathbf{x}$ *in* A. *Further, we call the Wajsberg function* g *realizing* $I_b(R_{d^A_{\mathbf{x}}})$ *over* $\mathbf{x}_{d^A_{\mathbf{x}}}$ *the* responsible *for* $\mathbf{x}$ *in* A. *Finally, if* $d^A_{\mathbf{x}} = 0$ *then the* responsible *of* $\mathbf{x}$ *in* A *is the function constantly* 0 *and the integer part* $i^A_{\mathbf{x}}$ *is set to* 0.

EXAMPLE 5.2.10. *Let* $\mathbf{x} = (.2, 2.7, 4)$, $\mathbf{y} = (.2, 2.7, 3.5)$, $\mathbf{z} = (.6, 2.4, 3.8)$ *be points in* $[0, 4]^{[3]}$. *In particular note that* $\mathbf{x}, \mathbf{y}, \mathbf{z} \in P_{0,(\{1\},\{2\},\{3\})}$. *Then, the ascending sequence of* $\mathbf{x}$ *is* $\mathbf{x}_0 = (1, 1, 1)$, $\mathbf{x}_1 = (.2, 1, 1)$, $\mathbf{x}_2 = (.2, .7, 1)$, $\mathbf{x}_3 = (.2, .7, 1)$*; analogously, the ascending sequence of* $\mathbf{y}$ *is* $\mathbf{y}_0 = (1, 1, 1)$, $\mathbf{y}_1 = (.2, 1, 1)$, $\mathbf{y}_2 = (.2, .7, 1)$, $\mathbf{y}_3 = (.2, .7, .5)$*; finally, the ascending sequence of* $\mathbf{z}$ *is* $\mathbf{z}_0 = (1, 1, 1)$, $\mathbf{z}_1 = (.6, 1, 1)$, $\mathbf{z}_2 = (.6, .4, 1)$, $\mathbf{z}_3 = (.6, .4, .8)$. *Assume now that* $A = (L, H) \in A_{[3]}$ *is such that* $\mathrm{dom}(L(\{1, 2, 3\}))) = [0, 1]^{[3]}$, $\mathrm{dom}(L((\{1\}, \{2, 3\}))) = [.3, .6]^{\{1\}} \times [0, 1]^{\{2,3\}}$, *and* $\mathrm{dom}(L((\{1\}, \{2\}, \{3\}))) = [.3, .6]^{\{1\}} \times [.5, .7]^{\{2\}} \times [0, 1]^{\{3\}}$. *Then* $d^A_{\mathbf{x}} = d^A_{\mathbf{y}} = 1$ *and* $d^A_{\mathbf{z}} = 2$*;* $\mathbf{f}^A_{\mathbf{x}} = \mathbf{x}_1 = (.2, 1, 1) = \mathbf{y}_1 = \mathbf{f}^A_{\mathbf{y}}$ *and* $\mathbf{f}^A_{\mathbf{z}} = \mathbf{z}_2 = (.6, .4, 1)$*; finally,* $i^A_{\mathbf{x}} = 0 = i^A_{\mathbf{y}}$ *and* $i^A_{\mathbf{z}} = 2$.

Finally, we define the map F, that sends the encodings in $A_{[n]}$ to functions from $[0, n+1]^{[n]}$ to $[0, n+1]$.

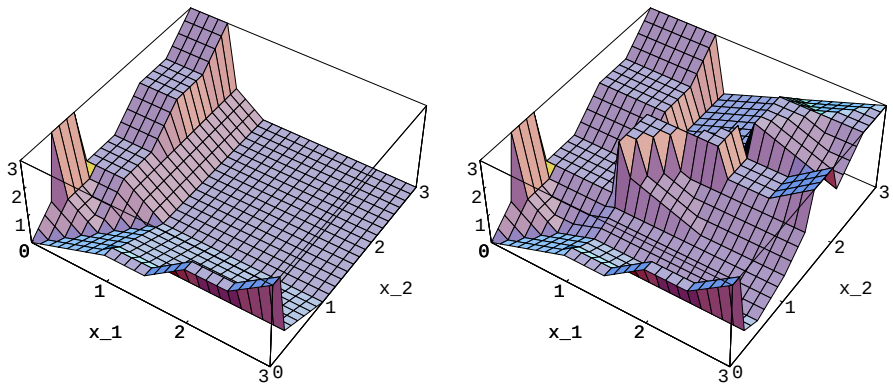
DEFINITION 5.2.11 (BL-functions). *We let* F *be the map*

$$A = (I_0, I_1) \in A_n \mapsto f_A \in [0, n+1]^{[0,n+1]^{[n]}}$$

defined as follows. Let $\mathbf{x} = (x_1, \ldots, x_n) \in [0, n+1]^{[n]}$, *let* $b \in \{0, 1\}$ *be such that* $\mathbf{x} \in P_b$, *and let* $R = R_{d^A_{\mathbf{x}}} \in \mathscr{F}_{[n]}$. *Then,*

$$f_A(\mathbf{x}) \quad = \begin{cases} I_{b+}(R)(\mathbf{f}^A_{\mathbf{x}}) + i^A_{\mathbf{x}} & \text{if } I_{b+}(R)(\mathbf{f}^A_{\mathbf{x}}) < 1, \\ n+1 & \text{otherwise.} \end{cases}$$

We let $F(A_{[n]}) = \{F(A) \mid A \in A_{[n]}\}$ *denote the set of* n*-ary BL-functions.*



(a) $F(E_0)\colon [0,3]^{[2]} \to [0,3]$. (b) $F(E_1)\colon [0,3]^{[2]} \to [0,3]$.

Figure 16. Sampling Definition 5.2.11, with the encodings E_0 and E_1 given in Figure 14 and Figure 15 respectively.

THEOREM 5.2.12 (Functional Representation). *The algebra*

$$\langle F(A_{[n]}), \odot, \to, \bot \rangle,$$

where the operations are pointwise defined as the corresponding operations of $(n+1)[0,1]_{\mathbf{L}}$*, is the free* n*-generated BL-algebra.*

It is possible to directly equip the set $A_{[n]}$ with a structure of an BL-algebra, turning it into the free n-generated BL-algebra. Details are given in [5].

We remark that the n-generated free $\mathbb{BL}_\Delta$-algebra can be represented as a pair of functions $(L\colon \mathscr{F}_{[n]} \to \mathbb{PW}_{[n]}, H\colon \mathscr{F}_{[n]} \to \mathbb{PW}_{[n]})$ satisfying conditions (i),(ii) of Definition 5.2.8, while in condition (iii) the requirements on $\Delta_{K,I,S}$ are relaxed to $\Delta_{K,I,S} = \pi_{[n]\setminus K}(|\Sigma|)$ for some open polyhedral complex Σ such that $|\Sigma| \subseteq F_K$.

The proof of Theorem 5.2.12 splits in two parts. The proof that each element of $\boldsymbol{F}^n_{\mathbb{BL}}$ is the image under the map F of some encoding amounts to a lengthy verification. The proof that for each encoding $(I_0, I_1) \in A_{[n]}$ it is possible to effectively construct a term φ such that

$$\varphi^{\boldsymbol{F}^n_{\mathbb{BL}}} = F((I_0, I_1))$$

goes through a series of preparatory lemmas, as follows.

Until the end of this Subsection, not to burden notiation, we shall write $\varphi(\mathbf{t})$ instead of $\varphi^{\boldsymbol{F}^n_{\mathbb{BL}}}(\mathbf{t})$ for the interpretation of a term φ in $\boldsymbol{F}^n_{\mathbb{BL}}$.

We start introducing the derived connectives $\triangleleft$ and $\diamond$:

$$\begin{aligned} x \triangleleft y &= (x \to y) \odot ((y \to x) \to x), \\ x \diamond y &= ((x \triangleleft y) \to y) \wedge ((y \triangleleft x) \to x). \end{aligned}$$

LEMMA 5.2.13. *For each* $i, j \in [n]$ *and* $\mathbf{t} = \langle t_1, \ldots, t_n \rangle \in [0, n+1]^{[n]}$,

$$
\begin{aligned}
(x_i \triangleleft x_j)(\mathbf{t}) &= \begin{cases} n+1 & \textit{if } \lfloor t_i \rfloor < \lfloor t_j \rfloor, \\ t_j & \textit{otherwise,} \end{cases} \\
(x_i \diamond x_j)(\mathbf{t}) &= \begin{cases} n+1 & \textit{if } \lfloor t_i \rfloor = \lfloor t_j \rfloor, \\ \max\{t_i, t_j\} & \textit{otherwise,} \end{cases} \\
(\bot \triangleleft x_i)(\mathbf{t}) = \neg\neg x_i(\mathbf{t}) &= \begin{cases} n+1 & \textit{if } 0 < \lfloor t_i \rfloor, \\ t_i & \textit{otherwise,} \end{cases} \\
(\bot \diamond x_i)(\mathbf{t}) = (\neg\neg x_i \to x_i)(\mathbf{t}) &= \begin{cases} n+1 & \textit{if } 0 = \lfloor t_i \rfloor, \\ t_i & \textit{otherwise.} \end{cases}
\end{aligned}
$$

Let $R = (B_1, \ldots, B_k) \in \mathscr{F}_{[n]}$ be such that $i \in B_k$. Let $C = \{(j, j') \in B_m \times B_m \mid m \in [k]\}$, and let $D = \{(j, j') \in B_m \times B_{m+1} \mid m \in [k-1]\}$. We define terms $x_{i,0,R}$ and $x_{i,1,R}$, as follows:

$x_{i,0,R} =$

$$= \Big(\Big(\bigwedge_{j \in B_1} \bot \diamond x_j\Big) \wedge \Big(\bigwedge_{(j,j') \in C} x_j \diamond x_{j'}\Big) \wedge \Big(\bigwedge_{(j,j') \in D} x_j \triangleleft x_{j'}\Big)\Big) \to x_i$$

and $x_{i,1,R} =$

$$= \Big(\Big(\bigwedge_{j \in B_1} \bot \triangleleft x_j\Big) \wedge \Big(\bigwedge_{(j,j') \in C} x_j \diamond x_{j'}\Big) \wedge \Big(\bigwedge_{(j,j') \in D} x_j \triangleleft x_{j'}\Big)\Big) \to x_i$$

LEMMA 5.2.14 (Cuboidwise Variable Isolation). *Let* $i \in [n]$, *let* $b \in \{0, 1\}$, *and let* $R = (B_1, \ldots, B_k) \in \mathscr{F}_{[n]}$ *be such that* $i \in B_k$. *For every* $\mathbf{t} = \langle t_1, \ldots, t_n \rangle \in [0, n+1]^{[n]}$,

$$
x_{i,b,R}(\mathbf{t}) = \begin{cases} t_i & \textit{if } \mathbf{t} \in P_{b,R}, \\ t_i & \textit{if } \mathbf{t} \in P_{b,R'},\ R \prec_{\mathscr{F}} (B_1, \ldots, B'_k, B''_k) \leq_{\mathscr{F}} R',\ i \in B'_k, \\ n+1 & \textit{otherwise.} \end{cases}
$$

Let $b \in \{0, 1\}$, let $R = (B_1, \ldots, B_k) \in \mathscr{F}_{[n]}$, and let $g \in \mathbb{WH}_{[n]}$ be essentially $|B_k|$-ary over $[0, 1]^{B_k}$. Let ϕ_g be a Wajsberg hoop term such that $\phi_g^{\boldsymbol{F}_{\mathbb{WH}}^{|B_k|}} = g$. For every $i \in B_k$, we let $\phi_{g,b,R}$ denote the term obtained by substituting in ϕ_g, for every $i \in B_k$, the variable x_i by the term $x_{i,b,R}$. In symbols:

$$\phi_{g,b,R} = \phi_g[x_i \leftarrow x_{i,b,R} \mid i \in B_k].$$

Let $b \in \{0, 1\}$, let $S, R \in \mathscr{F}_{[n]}$ such that

$$S = (B_1, \ldots, B_{k-1} \cup B_k) \prec_{\mathscr{F}} (B_1, \ldots, B_{k-1}, B_k) = R,$$

let g be a realization of $I_b(R)$, so that

$$I_b(R)|_{\Delta_1 \times \cdots \times \Delta_{k-1} \times [0,1]^{B_k}} = g$$

where $\Delta_{k-1} \subseteq [0,1)^{B_{k-1}}$ is the relative interior of a polyhedron $\delta \subseteq [0,1]^{B_{k-1}}$. Let γ be any nonempty face of δ, and let $f_\gamma \in \boldsymbol{F}^n_{\mathbb{WH}}$ be a Wajsberg function essentially $|B_{k-1}|$-ary over $[0,1]^{B_{k-1}}$ such that

$$f_\gamma^{-1}(1) = (\gamma \cup \{1\}^{B_{k-1}}) \times [0,1]^{[n]\setminus B_{k-1}}.$$

LEMMA 5.2.15 (Prismwise Masking). *For every $i \in B_k$, and every point $\mathbf{t}$ in $[0, n+1]^{[n]}$ such that $x_{i,b,R}(\mathbf{x}) = t_i$, if $\pi_{B_{k-1}}(\lfloor \mathbf{t} \rceil) \notin \gamma$, then $\phi_{f_\gamma,b,S}(\mathbf{t}) < t_i$, otherwise, $\phi_{f_\gamma,b,S}(\mathbf{t}) = n+1$.*

For every $b \in \{0,1\}$, every $R \in \mathscr{F}_{[n]}$, every realization g of $I_b(R)$, and every $i \in B_k$, we define terms $x_{i,b,R,g}$ as follows:

$$\begin{aligned} x_{i,b,[n],g} &= x_{i,b,[n]}, \\ x_{i,b,R,g} &= \left(\bigwedge_{\gamma \neq \delta} (\phi_{f_\gamma,b,S} \to x_{i,b,R}) \right) \to (\phi_{f_\delta,b,S} \to x_{i,b,R}), \end{aligned}$$

where γ ranges over any proper nonempty face of δ. This construction satisfies the following statement.

LEMMA 5.2.16 (Prismwise Variable Isolation). *Let $b \in \{0,1\}$, let $R = (B_1, \ldots, B_k) \in \mathscr{F}_{[n]}$, let g be a realization of $I_b(R)$, and let $i \in B_k$. Let $\mathbf{t} = \langle t_1, \ldots, t_n \rangle \in [0, n+1]^{[n]}$. Then,*

$$x_{i,b,R,g}(\mathbf{t}) = \begin{cases} t_i & \text{if } \mathbf{t} \in P_{b,R,g}, \\ n+1 & \text{otherwise.} \end{cases}$$

where $P_{b,R,g} = \{\mathbf{t} \in P_b \mid g \text{ realization of } I_b(R) \text{ responsible for } \mathbf{t} \text{ in } (I_0, I_1)\}$.

Let $b \in \{0,1\}$, $R = (B_1, \ldots, B_k) \in \mathscr{F}_{[n]}$ with $k \geq 1$, let g be any realization of $I_b(R)$, and let ϕ_g be a Wajsberg hoop term such that $\phi_g^{\boldsymbol{F}^n_{\mathbb{WH}}} = g$. Let $\theta_{b,R,g} = \phi_g[x_i \leftarrow x_{i,b,R,g} \mid i \in B_k]$.

LEMMA 5.2.17 (Prismwise Term Isolation). *Let $\mathbf{t} = \langle t_1, \ldots, t_n \rangle \in [0, n+1]^{[n]}$. Then,*

$$\theta_{b,R,g}(\mathbf{t}) = \begin{cases} g(\mathbf{f}_{\mathbf{t}}^{(I_0,I_1)}) + i_{\mathbf{t}}^{(I_0,I_1)} & \text{if } \mathbf{t} \in P_{b,R,g} \text{ and } g(\mathbf{f}_{\mathbf{t}}^{(I_0,I_1)}) < 1, \\ n+1 & \text{otherwise.} \end{cases}$$

Finally we let the term φ be defined as follows. If $I_1([n]) = \emptyset$, then

$$\varphi = \neg\theta_{0,[n],g} \wedge \bigwedge_{R \neq [n]} \bigwedge_h \theta_{0,R,h},$$

where g is the realization of $I_0([n])$ and h ranges over the realizations of $I_0(R)$ for $R \in \mathscr{F}_{[n]}$; otherwise, if $I_1([n]) \neq \emptyset$,

$$\varphi = \bigwedge_{b \in \{1,0\}} \bigwedge_{R \in \mathscr{F}_{[n]}} \bigwedge_h \theta_{b,R,h},$$

where $h \in I_b(R)$ ranges over the realizations of $I_b(R)$ for $b \in \{0,1\}$ and $R \in \mathscr{F}_{[n]}$.

LEMMA 5.2.18 (Normal Form). *There is an algorithm that for every* $(I_0, I_1) \in A_{[n]}$ *outputs a term* φ *such that*

$$F((I_0, I_1)) = \varphi^{\boldsymbol{F}^n_{\mathrm{BL}}}.$$

5.3 SBL-algebras, and Some Revisitations

In this section we shall describe the free n-generated SBL-algebra as an algebra of suitable restrictions of the functions in the set $F(A_{[n]})$ of n-ary BL-functions. Analogously, we describe free algebras for some other subvarietes of $\mathbb{BL}$ as algebras of restrictions of $F(A_{[n]})$.

The SBL-algebra given by the ordinal sum $\{0,1\} \oplus n[0,1]_{\text{Ł}}$ is generic for the variety generated by all n-generated SBL-algebras. Since $\{0,1\} \oplus n[0,1]_{\text{Ł}}$ is a BL-subalgebra of $(n+1)[0,1]_{\text{Ł}}$, we have the following.

THEOREM 5.3.1. *For any integer* $n > 0$*, the free* n*-generated* SBL*-algebra* $\boldsymbol{F}^n_{\mathbb{SBL}}$ *is the algebra of restrictions of functions in* $F(A_{[n]})$ *to the set*

$$(\{0\} \cup [1, n+1])^n .$$

Analogous results hold for all subvarieties $\mathbb{V}$ of $\mathbb{BL}$ such that the variety generated by all n-generated $\mathbb{V}$-algebras is singly generated by a subalgebra of $(n+1)[0,1]_{\text{Ł}}$. As an example we give isomorphic representations of $\boldsymbol{F}^n_{\mathbb{MV}}$ and $\boldsymbol{F}^n_{\mathbb{G}}$. Note that the ordinal sum $\omega\{0,1\}$ of denumerably many copies of the two-element Boolean algebra is generic for $\mathbb{G}$.

THEOREM 5.3.2. *For any integer* $n > 0$*:*

- $\boldsymbol{F}^n_{\mathbb{MV}}$ *is isomorphic to the algebra of restrictions of functions in* $F(A_{[n]})$ *to the set*

$$([0,1) \cup \{n+1\})^n .$$

- $\boldsymbol{F}^n_{\mathbb{G}}$ *is isomorphic to the algebra of restrictions of functions in* $F(A_{[n]})$ *to the set*

$$\{0, 1, \ldots, n+1\}^n .$$

Section 5.4 combines this technique with the combinatorial spectral dualities to offer a framework where to deal with locally finite subvarieties of $\mathbb{BL}$.

It must be stressed that there are varieties of BL-algebras, corresponding to prominent logics, which are not amenable to the same treatment of Theorem 5.3.1 and Theorem 5.3.2. As a matter of fact, consider the variety of Product algebras $\mathbb{P}$. As is well known it does not exist any Product algebra that is generic for the variety generated by n-generated Product algebras and that embeds into $(n+1)[0,1]_{\text{Ł}}$. Hence we cannot represent $\boldsymbol{F}^n_{\mathbb{P}}$ as an algebra of restrictions of functions in $F(A_{[n]})$. Roughly speaking, to accomplish such a task we should replace each copy of $[0,1]$ with a non-standard unit interval $[0,1]^*$. In this case, for instance, $\boldsymbol{F}^1_{\mathbb{P}}$ would be the algebra of restrictions of BL-functions to the set $\{0\} \cup (2-\epsilon, 2]$, for ϵ being an infinitesimal.

5.4 Locally Finite Varieties of BL-algebras

In this section we devise a spectral duality for locally finite BL-algebras. This kind of duality affords a combinatorial representation of those algebras, and in particular of finitely generated free algebras. The combinatorial representation provides us a with a powerful tool to investigate these algebras: for instance we can effectively compute products and coproducts and determine the exact cardinalities of free algebras. Compare with the analogous constructions given for Gödel algebras and related structures in Section 4.

Let FBL denote the category of finite BL-algebras and their homomorphisms. Let $\boldsymbol{A}$ be a finite BL-algebra.

Recall that the *prime spectrum* of $\boldsymbol{A}$, in symbols $\mathrm{Spec}\boldsymbol{A}$, is the set of all prime filters of $\boldsymbol{A}$ ordered by reverse inclusion.

Given a finite BL-chain $\boldsymbol{C}$, let c denote its largest idempotent distinct from $\top^{\boldsymbol{C}}$. We define its *top part* as

$$T(\boldsymbol{C}) = \{x \in \boldsymbol{C} \mid x > c\}\,.$$

Let N^+ denote $\mathsf{N} \setminus \{0\}$. Further, for each prime filter $\mathfrak{p} \in \mathrm{Spec}\boldsymbol{A}$, we denote $\boldsymbol{A}/\mathfrak{p}$ the quotient of $\boldsymbol{A}$ modulo the congruence determined by $\mathfrak{p}$.

DEFINITION 5.4.1. *The* weighted spectrum *of $\boldsymbol{A}$ is the function*

$$\mathrm{wSpec}\,\boldsymbol{A}\colon \mathrm{Spec}\,\boldsymbol{A} \to \mathsf{N}^+$$

such that

$$(\mathrm{wSpec}\,\boldsymbol{A})(\mathfrak{p}) = |T(\boldsymbol{A}/\mathfrak{p})|\,,$$

for every prime filter $\mathfrak{p} \in \mathrm{Spec}\,\boldsymbol{A}$.

DEFINITION 5.4.2. *A* weighted forest *is a map $w\colon F \to \mathsf{N}^+$, where F is a forest.*

Given two weighted forests $w\colon F \to \mathsf{N}^+$ and $w'\colon F' \to \mathsf{N}^+$ a weighted forest morphism *$g\colon w \to w'$ is a map $g\colon F \to F'$ such that*

1. *g is* order-preserving*, that is, if $x \leq y \in F$, then $g(x) \leq g(y)$;*

2. *g is* open*, that is, whenever $x' \leq g(x) \in F'$, then there is $y \leq x \in F$ such that $g(y) = x'$;*

3. *g* respects weights*, that is, for each $x \in F$ there is $y \leq x$ such that $g(y) = g(x)$ and $w'(g(y))$ divides $w(y)$.*

Recall that an order-preserving open map carries downsets to downsets.

It is easy to check that weighted forests and their morphisms forms a category, denoted WF.

LEMMA 5.4.3. *Let the map* wSpec *be extended to* FBL*-morphisms as follows. For any pair of algebras* $\boldsymbol{A}, \boldsymbol{B}$ *in* FBL*, and for any homomorphism* $h\colon \boldsymbol{A} \to \boldsymbol{B}$*, let the map* $\mathrm{wSpec}\, h\colon \mathrm{wSpec}\, \boldsymbol{B} \to \mathrm{wSpec}\, \boldsymbol{A}$ *be defined by the following stipulation:*

$$\mathfrak{p} \in \mathrm{Spec}\, \boldsymbol{B} \mapsto h^{-1}(\mathfrak{p}) \in \mathrm{Spec}\, \boldsymbol{A}\,.$$

Then wSpec *is a contravariant functor from* FBL *to* WF.

wSpec realises a categorical equivalence between FBL and $\mathsf{WF}^{\mathsf{op}}$. We now describe the contravariant functor $\mathbf{wSub}\colon \mathsf{WF} \to \mathsf{FBL}$ implementing the duality in the opposite direction.

DEFINITION 5.4.4. *Given a weighted forest* $w\colon F \to \mathsf{N}^+$*, a* weighted subforest *of* w *is a map* $w'\colon F' \to \mathsf{N}^+$*, where* F' *is a subforest of* F *(that is, a downset of* F*) such that* $w'(x) \leq w(x)$ *for all* $x \in \max F'$*, and* $w'(x) = w(x)$ *otherwise.*

We let $\mathrm{wSub}\, w$ *denote the set of all weighted subforests of* w*.*

We shall equip $\mathrm{wSub}\, w$ with a structure of BL-algebra. To begin with, writing $\emptyset\colon \emptyset \to \mathsf{N}^+$ for the unique empty weighted forest, we set $\bot = \emptyset$, and $\top = w$. To define $\odot$, consider subforests $u\colon U \to \mathsf{N}^+$ and $v\colon V \to \mathsf{N}^+$ of w. Define a function $a\colon U \cap V \to \mathsf{N}^+ \cup \{0\}$ by

$$a(x) = \begin{cases} \max(0, u(x) + v(x) - w(x)) & \text{if } x \in \max U \cap \max V \\ u(x) & \text{if } x \in \max U \text{ and } x \notin \max V \\ v(x) & \text{if } x \notin \max U \text{ and } x \in \max V \\ w(x) & \text{otherwise,} \end{cases}$$

for each $x \in U \cap V$. Let $E = \{x \in U \cap V \mid a(x) > 0\}$, and define $u \odot v\colon E \to \mathsf{N}^+$ by the restriction $u \odot v = a \restriction E$.

Turning to implication, we define $u \to v$. First, we set

$$\begin{aligned} A &= F \setminus \uparrow (U \setminus V)\,, \\ B &= \{x \mid x \in \max U \cap \max V \text{ and } u(x) > v(x)\}\,. \end{aligned}$$

Then we set $E = (A \setminus \uparrow B) \cup B$. We define $(u \to v)\colon E \to \mathsf{N}^+$ by

$$(u \to v)(x) = \begin{cases} v(x) + w(x) - u(x) & \text{if } x \in B \\ v(x) & \text{if } x \in (U \cap V) \setminus (\max U \cap \max V) \\ w(x) - u(x) & \text{if } x \in \min U \setminus V \\ w(x) & \text{otherwise,} \end{cases}$$

for each $x \in E$.

It can now be proved that for any weighted forest $w\colon F \to \mathsf{N}^+$, the algebraic structure

$$\mathbf{wSub} w = \langle\, \mathrm{wSub}\, w, \odot, \to, \bot \,\rangle$$

is a (finite) BL-algebra. To turn $\mathbf{wSub}$ into a contravariant functor from WF to FBL, we take inverse images again. Namely, if $g\colon w \to w'$ is a morphism between the weighted forests $w\colon F \to \mathsf{N}^+$ and $w'\colon F' \to \mathsf{N}^+$, we define $\mathbf{wSub}g\colon \mathbf{wSub}w' \to \mathbf{wSub}w$ by

$$\mathbf{wSub}g\colon U \in \mathbf{Sub}F' \mapsto g^{-1}(U) \in \mathbf{Sub}F\,,$$

where $\mathbf{Sub}F'$ and $\mathbf{Sub}F$ are applications of the functor $\mathbf{Sub}\colon \mathsf{F} \to \mathsf{G}$ defined in Section 4 (see Theorem 4.2.1). One can prove that $\mathbf{wSub}g$ so defined is a homomorphism of BL-algebras. To sum up, $\mathbf{wSub}$ is a contravariant functor from WF to FBL.

THEOREM 5.4.5 (Finite Duality). *The category of finite BL-algebras and their homomorphisms is dually equivalent to the category of weighted forests and their morphisms. That is, the composite functors* $\mathrm{wSpec} \circ \mathbf{wSub}$ *and* $\mathbf{wSub} \circ \mathrm{wSpec}$ *are naturally isomorphic to the identity functors on* WF *and* FBL*, respectively.*

In particular, the functor wSpec is essentially surjective, and this yields the following representation theorem for finite BL-algebras.

COROLLARY 5.4.6. *Any finite BL-algebra is isomorphic to* $\mathbf{wSub}\,w$*, for a weighted forest* $w\colon F \to \mathsf{N}^+$ *that is unique to within an isomorphism of weighted forests.*

To manipulate weighted spectra of finite BL-algebras we have to introduce some further notions. In particular we shall define finite products and coproducts in WF.

Given a weighted forest $w\colon F \to \mathsf{N}^+$, we denote $w_\perp$ the weighted *tree* $w_\perp\colon F_\perp \to \mathsf{N}^+$, where $F_\perp = F \cup \{\perp\}$ with $\perp < x$ for all $x \in F$, and $w_\perp(x) = w(x)$ for all $x \in F$, while $w_\perp(\perp) = 1$. Each weighted tree arises from a weighted forest in this way. Further, for each $e \in \mathsf{N}^+$, we let w_e denote the weighted tree $w_e\colon \emptyset_\perp \to \mathsf{N}^+$ defined by $w_e(\perp) = e$.

LEMMA 5.4.7. *Finite products and coproducts in* WF *have the following properties.*

1. *The coproduct in* WF *of two finite weighted forests* $w\colon F \to \mathsf{N}^+$ *and* $w'\colon F' \to \mathsf{N}^+$ *is (isomorphic to) the weighted forest* $w + w'\colon F \uplus F' \to \mathsf{N}^+$*, where* $\uplus$ *is the disjoint union and* $(w + w')(x) = w(x)$ *if* $x \in F$*, while* $(w + w')(x) = w'(x)$ *if* $x \in F'$*.*

2. *Products distribute over coproducts, that is, for each* $u, v, w \in \mathsf{WF}$*,* $u \times (v + w) \cong (u + v) \times (u + w)$*.*

3. *For each pair of integers* $d, e \in \mathsf{N}^+$*,* $w_d \times w_e \cong w_{\mathrm{lcm}(d,e)}$*, where* lcm *denotes the least common multiple. Moreover, for each pair of weighted trees* $u_\perp$ *and* $v_\perp$*, it holds that* $(w_d \times u_\perp) \times (w_e \times v_\perp) \cong w_{\mathrm{lcm}(d,e)} \times (u_\perp \times v_\perp)$*, and* $w_1 \times u_\perp \cong u_\perp$*.*

4. *For each pair of weighted trees* $u_\perp$ *and* $v_\perp$ *in* WF*,*

$$u_\perp \times v_\perp \cong (u \times v_\perp + u \times v + u_\perp \times v)_\perp\,.$$

As an application of spectral duality we compute the exact structure of the weighted spectra of free algebras in the subvariety of $\mathbb{BL}$ satisfying Grigolia's axioms. Knowledge of this structure easily yields a large amount of information on algebras in this variety: for instance the computation of cardinalities of free algebra follows straightforwardly.

Let us define $x \oplus y$ as

$$((x \to (x \odot y)) \to y) \vee ((y \to (y \odot x)) \to x)\,.$$

We observe that in each MV-algebra this term coincides with $\neg(\neg x \odot \neg y)$, that is, with the Łukasiewicz strong disjunction.

For each integer $k > 0$, let $\mathbb{BL}_k$ be the subvariety of $\mathbb{BL}$ satisfying Grigolia's axioms given in Section 2.3. Let further $\boldsymbol{B}_k^l$ denote the ordinal sum of l copies of the $(k+1)$-element MV-chain $Ł_k$.

LEMMA 5.4.8. *For each integer $k > 0$ the following hold.*

1. *The variety $\mathbb{BL}_k$ is generated by $\{\boldsymbol{B}_k^l \mid l \in \mathsf{N}^+\}$.*
2. *The variety $\mathbb{BL}_k$ is locally finite.*
3. *For a finite BL-algebra $\boldsymbol{A}$, the following are equivalent.*
   (a) $\boldsymbol{A} \in \mathbb{BL}_k$.
   (b) $\mathrm{wSpec}\,\boldsymbol{A}$ *has range included in the set of divisors of k.*

Hence the weighted spectrum of a $\mathbb{BL}_k$-algebra is a finite forest the weights of whose elements are divisors of k. By Corollary 5.4.6, the viceversa holds, too: each such forest is the weighted spectrum of a $\mathbb{BL}_k$-algebra.

To compute the structure of $\mathrm{wSpec}\,\boldsymbol{F}^n_{\mathbb{BL}_k}$ we start by introducing a slight variant of the function $\alpha(n,d)$ of Lemma 2.3.2.

We let $\beta(0,d) = 0$ if $d > 1$ and 1 if $d = 1$, and, for $n \geq 1$,

$$\beta(n,d) = d^n + \sum_{\emptyset \neq X \subseteq \mathrm{CoAtDiv}(d)} (-1)^{|X|} (\gcd X)^n\,.$$

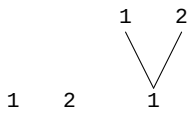
LEMMA 5.4.9. *For each integer $k > 0$ let $M_k^0 = w_1$, while for each integer $m > 0$, $M_k^m = \sum_{d|k} \beta(m,d) w_d$. Then for each integer $n > 0$:*

$$\mathrm{wSpec}\,\boldsymbol{F}^n_{\mathbb{BL}_k} \cong (M_k^1 + (M_k^1)_\perp)^n\,.$$

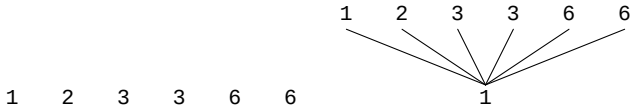
Computing the products in the right hand side of the equation for $\mathrm{wSpec}\,\boldsymbol{F}^n_{\mathbb{BL}_k}$ in Lemma 5.4.9 gives the following recursive formula.

THEOREM 5.4.10. *For each pair of positive integers k, d we define $T_{k,d}^0 = w_d$, while for each integer $m > 0$,*

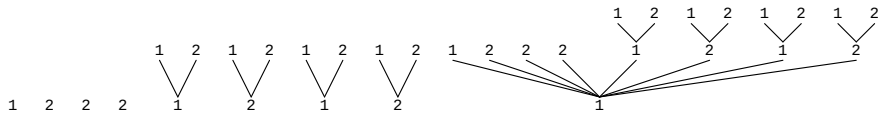
$$T_{k,d}^m = w_d \times \left(\sum_{i=1}^{n} \sum_{e|k} \binom{n}{i} \beta(i,e) T_{k,e}^{m-i} \right)_\perp\,.$$



(a) $\mathrm{wSpec}\,\boldsymbol{F}^1_{\mathbb{BL}_3}$.



(b) $\mathrm{wSpec}\,\boldsymbol{F}^1_{\mathbb{BL}_7}$.



(c) $\mathrm{wSpec}\,\boldsymbol{F}^2_{\mathbb{BL}_3}$.

Figure 17. Sampling Lemma 5.4.9.

Then, for each integer $n \geq 0$:

$$\mathrm{wSpec}\,\boldsymbol{F}^n_{\mathbb{BL}_k} \cong \sum_{i=0}^{n} \sum_{d|k} \binom{n}{i} \beta(i,d) T^{n-i}_{k,d}\,.$$

We write $t(k,n,d)$ for the cardinality of the $\mathbb{BL}_k$-algebra $\mathrm{Sub}\,T^n_{k,d}$.

COROLLARY 5.4.11. *For each integer $n \geq 0$ and $k, d \in \mathsf{N}^+$, it holds that $t(k,0,d) = d+1$ and $t(k,n,d) = d + \prod_{i=1}^{n} \prod_{e|k} t(k,n-i,e)^{\binom{n}{i}\beta(i,e)}$, and hence*

$$|\boldsymbol{F}^n_{\mathbb{BL}_k}| = \prod_{i=0}^{n} \prod_{d|k} t(k,n-i,d)^{\binom{n}{i}\beta(i,d)}\,.$$

We end this section with some observations and some further simple applications.

Clearly, $\mathbb{MV}_k$ is a subvariety of $\mathbb{BL}_k$, hence the weighted spectrum of any algebra $\boldsymbol{A} \in \mathbb{MV}_k$, is a forest of trees of the form w_d for d a divisor of k. Theorem 2.3.3 can then be reformulated in dual terms as $\mathrm{wSpec}\,\boldsymbol{F}^n_{\mathbb{MV}_k} \cong \sum_{d|k} \alpha(n,d) w_d$.

Analogously, the variety of Gödel algebras coincides with $\mathbb{BL}_1$, and hence the weighted spectrum of a Gödel algebra $\boldsymbol{A}$ is a forest of copies of w_1, and then, forgetting weights, the formula in theorem 5.4.10 specialises to the formula $\mathrm{wSpec}\,\boldsymbol{F}^n_{\mathbb{G}} \cong \sum_{i=0}^{n-1} \binom{n}{i} (H_i)_\perp + (\sum_{i=0}^{n-1} \binom{n}{i} (H_i)_\perp)_\perp$ of Theorem 4.2.6.

Given a locally finite variety $\mathbb{V}$ of BL-algebras, the subvariety constituted by all $\mathbb{V}$-algebras satisfying $x \wedge \neg x = 0$ is a locally finite variety of SBL-algebras. The corresponding subcategory of finite forests are precisely those weighted spectra of $\mathbb{V}$-algebras such that the weight of the root of every tree is 1.

6 Pierce Representation

A very useful construction for characterising free algebras over a set of generators of arbitrary cardinality is the weak Boolean product representation.

In this section we shall explore representation of free algebras in a variety $\mathbb{V}$ as weak Boolean products of the directly indecomposable members of $\mathbb{V}$.

In particular, for some subvarieties $\mathbb{V}$ of $\mathbb{MTL}$ it is possible to build all directly indecomposable algebras applying some construction involving algebras from an auxiliary variety $\mathbb{V}^*$. In this section we explain a construction of this kind and state the conditions of its applicability. As a matter of fact, we have already shown instances of this construction in previous sections of this chapter. Proposition 4.5.3 states that each directly indecomposable NM-algebra without negation fixpoint arises as the disconnected rotation of a Gödel hoop. The construction that yields a product algebra from a cancellative hoop is another example: actually Theorem 3.2.6 shows a decomposition of $\boldsymbol{F}^n_{\mathbb{P}}$ as a direct product of directly indecomposable product algebras.

When the set of directly indecomposable members needed to build an algebra is not finite, then direct products are no longer sufficient. Under some hypotheses that are always satisfied by the structures of our concern, one may consider to generalise direct products to weak Boolean products, as follows.

We recall the reader that a *Stone space* is a totally disconnected compact Hausdorff space.

DEFINITION 6.0.12. *Given a Stone space X, a* weak Boolean product *of a family $\{\boldsymbol{A}_x\}_{x\in X}$ of algebras in a variety $\mathbb{V}$ is a subdirect product $\boldsymbol{A} \hookrightarrow \prod_{x\in X} \boldsymbol{A}_x$ such that, for all $a, b \in \boldsymbol{A}$:*

1. *the set $\{x \in X \mid a(x) = b(x)\}$ is open in X;*
2. *for all clopen sets Z in X, the function $c\colon X \to \prod_{x\in X} \boldsymbol{A}_x$ defined as $c(x) = a(x)$ if $x \in Z$, while $c(x) = b(x)$ otherwise, is such that $c \in \boldsymbol{A}$.*

Clearly, $\boldsymbol{A} \in \mathbb{V}$. An algebra is representable as weak Boolean product *if it is isomorphic to a weak Boolean product.*

The variety $\mathbb{BRL}$ of bounded integral residuated lattices is arithmetical and has the Boolean Factor Congruence property. Hence every non-trivial bounded integral residuated lattice can be represented a weak Boolean product of directly indecomposable members of $\mathbb{BRL}$.

LEMMA 6.0.13. *The directly indecomposable members of $\mathbb{BRL}$ are precisely those lattices $\boldsymbol{A}$ such that their Boolean skeleton $\mathbf{B}(\boldsymbol{A})$ is isomorphic to $\{0, 1\}$.*

In the following, for any filter $\mathfrak{p}$ of a lattice $\boldsymbol{A} \in \mathbb{BRL}$, we write $\boldsymbol{A}/\mathfrak{p}$ for the quotient of $\boldsymbol{A}$ modulo the congruence whose top class is $\mathfrak{p}$. If $\mathfrak{q}$ is a filter of the Boolean skeleton $\mathbf{B}(\boldsymbol{A})$ of $\boldsymbol{A}$, we denote by $\langle \mathfrak{q} \rangle$ the filter of $\boldsymbol{A}$ generated by $\mathfrak{q}$.

THEOREM 6.0.14. *Let $\boldsymbol{A}$ be a lattice in $\mathbb{BRL}$. Let $\mathrm{BSpec}\,\boldsymbol{A}$ be the prime spectrum of the Boolean algebra $\mathbf{B}(\boldsymbol{A})$ equipped with the Stone topology.*

For any $\mathfrak{p} \in \mathrm{BSpec}\,\boldsymbol{A}$, the lattice $\boldsymbol{A}/\langle \mathfrak{p} \rangle$ is directly indecomposable, as $\mathbf{B}(\boldsymbol{A}/\langle \mathfrak{p} \rangle) \cong \{0, 1\}$.

The correspondence

$$a \mapsto (a/\langle \mathfrak{p} \rangle)_{\mathfrak{p} \in \mathrm{BSpec}\,\boldsymbol{A}}$$

gives a representation of $\boldsymbol{A}$ as weak Boolean product of the family of directly indecomposable lattices $\{\boldsymbol{A}/\langle \mathfrak{p} \rangle\}_{\mathfrak{p} \in \mathrm{BSpec}\,\boldsymbol{A}}$ over the space $\mathrm{BSpec}\,\boldsymbol{A}$. This representation is called the Pierce Representation *of $\boldsymbol{A}$.*

Obviously, $\mathbb{MTL} \subseteq \mathbb{BRL}$. We shall work with some subvarieties of $\mathbb{MTL}$. An MTL-algebra satisfying the equation

$$\neg\neg(\neg\neg x \to x) = \top \tag{14}$$

is called a *Glivenko* MTL-algebra. It is straightforward to check that $\mathbb{BL}$, $\mathbb{IMTL}$ and $\mathbb{SMTL}$ are all varieties of Glivenko MTL-algebras.

We introduce a condition and a construction, applicable when the condition holds, to produce directly indecomposable Glivenko MTL-algebras starting from basic semihoops (a.k.a. prelinear semihoops) in $\mathbb{MTLH}$.

Pick $\boldsymbol{A} = \langle A, \&, \to, \wedge, \vee, \top \rangle \in \mathbb{MTLH}$. A map $\delta\colon A \to A$ is *DL-admissible* if it satisfies the following, for any $a, b \in A$.

$$\begin{array}{ll} a \to \delta(a) = \top & \delta(a \wedge b) = \delta(a) \wedge \delta(b) \\ \delta(\delta(a)) = \delta(a) & \delta(a \vee b) = \delta(a) \vee \delta(b) \\ \delta(a \to b) = a \to \delta(b) & \delta(a \,\&\, b) = \delta(\delta(a) \,\&\, \delta(b)) \end{array}$$

DEFINITION 6.0.15. *Let δ be DL-admissible for $\boldsymbol{A} \in \mathbb{MTLH}$. The following operations are defined on $S(\boldsymbol{A}, \delta) = (A \times \{1\}) \cup (\delta(A) \times \{0\})$, for any $x, y \in \boldsymbol{A}$ and*

$i, j \in \{0, 1\}$:

$$
\begin{array}{rcl}
(x,i) \sqcap (y,j) & = & (y,j) \sqcap (x,i) = \begin{cases} (x \wedge y, 1) & \textit{if } i = j = 1, \\ (x \vee y, 0) & \textit{if } i = j = 0, \\ (x, 0) & \textit{if } i < j. \end{cases} \\
(x,i) \sqcup (y,j) & = & (y,j) \sqcup (x,i) = \begin{cases} (x \vee y, 1) & \textit{if } i = j = 1, \\ (x \wedge y, 0) & \textit{if } i = j = 0, \\ (y, 1) & \textit{if } i < j. \end{cases} \\
(x,i) \odot (y,j) & = & (y,j) \odot (x,i) = \begin{cases} (x \,\&\, y, 1) & \textit{if } i = j = 1, \\ (\top, 0) & \textit{if } i = j = 0, \\ (y \to x, 0) & \textit{if } i < j. \end{cases} \\
(x,i) \Rightarrow (y,j) & = & \begin{cases} (x \to y, 1) & \textit{if } i = j = 1, \\ (y \to x, 0) & \textit{if } i = j = 0, \\ \delta(x \,\&\, y, 0) & \textit{if } i > j, \\ (\top, 1) & \textit{if } i < j. \end{cases} \\
\top & = & (\top, 1) \\
\bot & = & (\top, 0)
\end{array}
$$

Let $\boldsymbol{S}(\boldsymbol{A}, \delta) = \langle S(\boldsymbol{A}, \delta), \odot, \Rightarrow, \sqcap, \sqcup, \bot, \top \rangle$.

Compare Definition 6.0.15 with Definition 4.5.2.

PROPOSITION 6.0.16. $\boldsymbol{S}(\boldsymbol{A}, \delta)$ *is a directly indecomposable Glivenko* MTL*-algebra. Moreover,* $\boldsymbol{S}(\boldsymbol{A}, \delta) \in \mathbb{IMTL}$ *if and only if* $\delta = \mathrm{id}_{\boldsymbol{A}}$*, while* $\boldsymbol{S}(\boldsymbol{A}, \delta) \in \mathbb{SMTL}$ *if and only if* δ *is constantly* $\top$.

Further, the equation

$$(\neg x \to \neg\neg x)^2 = \neg(x^2) \to \neg\neg(x^2) \tag{15}$$

holds in $\boldsymbol{S}(\boldsymbol{A}, \delta)$.

We denote $\mathbb{DL}$ the subvariety of Glivenko MTL-algebras satisfying the Di Nola-Lettieri Equation (15). Notice that $\mathbb{SMTL} \subseteq \mathbb{DL}$, and moreover, the variety generated by the Chang MV-algebra is $\mathbb{MV} \cap \mathbb{DL}$.

Let $\boldsymbol{A} \in \mathbb{DL}$. We denote $\nabla \colon \boldsymbol{A} \to \boldsymbol{A}$ the map $x \mapsto (\neg x \to \neg\neg x)^2$. Further we let $A^\top = \nabla^{-1}(\{\top\})$. It can be checked that the restriction of the hoop operations of $\boldsymbol{A}$ to $A^\top$ turns it into a basic semihoop $\boldsymbol{A}^\top$. An algebra $\boldsymbol{H} = \langle H, \&, \to, \wedge, \vee, \delta, \top \rangle$ is called a *kernel DL-algebra* if its hoop reduct $\boldsymbol{H}^-$ is a basic semihoop and $\delta \colon H \to H$ is DL-admissible. By definition of DL-admissible map, the class of kernel DL-algebras forms a variety $\mathbb{KDL}$. For each $\boldsymbol{A}^\top$ let $\delta_{\neg\neg} \colon A^\top \to A^\top$ be the DL-admissible map $x \mapsto \neg\neg x$. Note that $\delta_{\neg\neg}$ coincides with the identity map $\mathrm{id}_{\boldsymbol{A}^\top}$ if $\boldsymbol{A}$ is involutive, while it coincides with the constant function $\top^{\boldsymbol{A}^\top}$ if $\boldsymbol{A} \in \mathbb{SMTL}$. We let $\boldsymbol{P}(\boldsymbol{A})$ denote the KDL-algebra $\langle \boldsymbol{A}^\top, \delta_{\neg\neg} \rangle$.

We associate with each class $\mathbb{K}$ of DL-algebras the class $\mathbb{K}^*$ of KDL-algebras $\langle \boldsymbol{A}, \delta \rangle$ such that $\boldsymbol{S}(\boldsymbol{A}, \delta) \in \mathbb{K}$.

THEOREM 6.0.17. *$\mathbb{V}^*$ is a subvariety of $\mathbb{KDL}$ for each subvariety $\mathbb{V}$ of $\mathbb{DL}$.*

The operators $\boldsymbol{S}$ and $\boldsymbol{P}$ can be used, together with the Peirce representation, to obtain description of free algebras.

THEOREM 6.0.18. *For each subvariety $\mathbb{V}$ of DL-algebras, $\mathbf{B}(\boldsymbol{F}_{\mathbb{V}}^{|X|})$ is the free Boolean algebra over the set $\nabla X = \{\nabla(x) \mid x \in X\}$ of generators. Since X and ∇X have the same cardinality, $\mathbf{B}(\boldsymbol{F}_{\mathbb{V}}^{|X|})$ is the free Boolean algebra $\boldsymbol{F}_{\mathbb{B}}^{|X|}$ over $|X|$ generators.*

Recall that the Stone space of $\boldsymbol{F}_{\mathbb{B}}^{|X|}$ is the Cantor space 2^X, as the ultrafilters of $\boldsymbol{F}_{\mathbb{B}}^{|X|}$ are in bijection with the subsets of X.

Assume X is nonempty. Then the map $U \mapsto \{x \in X \mid \nabla(x) \in U\}$ is a bijection from the set of ultrafilters of $\mathbf{B}(\boldsymbol{F}_{\mathbb{V}}^{|X|})$ into 2^X. The inverse map is given by sending S to the ultrafilter U_S generated by $\nabla S \cup \{\neg\nabla(x) \mid x \in X \setminus S\}$.

THEOREM 6.0.19. *Let $\mathbb{V}$ be any subvariety of $\mathbb{DL}$ and let $X \neq \emptyset$ be any set of generating symbols. For each $S \in 2^X$, the algebra $\boldsymbol{F}_{\mathbb{V}}^{|X|}/\langle U_S\rangle$ is directly indecomposable. Further, the free algebra $\boldsymbol{F}_{\mathbb{V}}^{|X|}$ is isomorphic to the weak Boolean product of the family $\{\boldsymbol{S}(\boldsymbol{P}(\boldsymbol{F}_{\mathbb{V}}^{|X|}/\langle U_S\rangle))\}_{S\subseteq X}$ over the space 2^X.*

Usefulness of Theorem 6.0.19 shows itself when the structure of the KDL-algebras $\boldsymbol{P}(\boldsymbol{F}_{\mathbb{V}}^{|X|}/\langle U_S\rangle)$ is known. This has been accomplished for the case when $\mathbb{V}$ is $\mathbb{SMTL}$ or the variety of involutive DL-algebras.

In the following, for any $\boldsymbol{A} \in \mathbb{SMTL}$, let $\boldsymbol{\sigma}(\boldsymbol{A}) = \boldsymbol{S}(\boldsymbol{A}, \delta_\top)$, where $\delta_\top \colon A \to A$ is the constant function $\top$. Analogously, let $\boldsymbol{\iota}(\boldsymbol{A}) = \boldsymbol{S}(\boldsymbol{A}, \mathrm{id}_{\boldsymbol{A}})$ for any involutive DL-algebra $\boldsymbol{A}$.

THEOREM 6.0.20. *Let $\mathbb{V}$ be a nontrivial subvariety of $\mathbb{MTL}$.*

If $\mathbb{V}$ is a subvariety of $\mathbb{SMTL}$ then, for each set X the free algebra $\boldsymbol{F}_{\mathbb{V}}^{|X|}$ is isomorphic to the weak Boolean product of the family $\{\boldsymbol{\sigma}(\boldsymbol{F}_{\mathbb{V}^*}^{|S|})\}_{S\subseteq X}$ over the space 2^X.

If $\mathbb{V}$ is a subvariety of involutive DL-algebras then, for each set X the free algebra $\boldsymbol{F}_{\mathbb{V}}^{|X|}$ is isomorphic to the weak Boolean power of the family $\{\boldsymbol{\iota}(\boldsymbol{F}_{\mathbb{V}^*}^{|X|})\}_{S\subseteq X}$ over the space 2^X.

When X is finite the weak Boolean product reduces to a direct product.

THEOREM 6.0.21. *Let $k \leq 1$ be any integer.*

If $\mathbb{V}$ is a nontrivial subvariety of $\mathbb{SMTL}$ then $\boldsymbol{F}_{\mathbb{V}}^k \cong \prod_{i=0}^k \boldsymbol{\sigma}(\boldsymbol{F}_{\mathbb{V}^*}^i)^{\binom{k}{i}}$.

If $\mathbb{V}$ is a nontrivial subvariety of involutive DL-algebras then $\boldsymbol{F}_{\mathbb{V}}^k \cong \boldsymbol{\iota}(\boldsymbol{F}_{\mathbb{V}^*}^k)^{2^k}$.

We apply Theorem 6.0.20 to the varieties of Gödel algebras and of product algebras. Application of Theorem 6.0.21 to the same varieties gives back the results we already obtained.

First we note that $\mathbb{G}^* = \mathbb{GH}$. Hence, for each set X, the free Gödel algebra $\boldsymbol{F}_{\mathbb{G}}^{|X|}$ is isomorphic to the weak Boolean product of the family $\{\boldsymbol{\sigma}(\boldsymbol{F}_{\mathbb{GH}}^{|S|})\}_{S\subseteq X}$ over the space 2^X.

Next, consider the variety $\mathbb{CH}$ of cancellative hoops. Note that for $\boldsymbol{A} \in \mathbb{MTLH}$ it holds that

$$\boldsymbol{A} \in \mathbb{CH} \qquad \text{iff} \qquad \boldsymbol{\sigma}(\boldsymbol{A}) \in \mathbb{P} \qquad \text{iff} \qquad \boldsymbol{\iota}(\boldsymbol{A}) \in \mathbb{MV}\,.$$

It follows that if $\mathbb{K}$ is one of $\mathbb{P}$ or $\mathbb{MV}$ then $\mathbb{K}^* = \mathbb{CH}$.

Then, for each set X, the free product algebra $\boldsymbol{F}_{\mathbb{P}}^{|X|}$ is isomorphic to the weak Boolean product of the family $\{\boldsymbol{\sigma}(\boldsymbol{F}_{\mathbb{CH}}^{|S|})\}_{S\subseteq X}$ over the space 2^X. Moreover, for each set X, the free algebra in the variety $\mathbb{CMV} = \mathbb{MV} \cap \mathbb{DL}$ generated by the Chang's MV-algebra, is such that $\boldsymbol{F}_{\mathbb{CMV}}^{|X|}$ is isomorphic to the weak Boolean power of the family $\{\boldsymbol{\iota}(\boldsymbol{F}_{\mathbb{CH}}^{|X|})\}_{S\subseteq X}$ over the space 2^X.

For what concerns $\mathbb{NM}$ and its subvarieties, we note that the subvariety $\mathbb{NM}^-$ of NM-algebras without fixpoint, which is generated by $[0,1] \setminus \{1/2\}$, is such that $(\mathbb{NM}^-)^* = \mathbb{GH}$, and in particular $\boldsymbol{H}$ is a Gödel hoop if and only if $\boldsymbol{\iota}(\boldsymbol{H}) \in \mathbb{NM}^-$. It follows that for each set X, the free algebra $\boldsymbol{F}_{\mathbb{NM}^-}^{|X|}$ is isomorphic to the weak Boolean power of the family $\{\boldsymbol{\iota}(\boldsymbol{F}_{\mathbb{GH}}^{|X|})\}_{S\subseteq X}$ over the space 2^X.

To deal with $\mathbb{NM}$ itself, we first represent $\boldsymbol{F}_{\mathbb{NM}}^{|X|}$ isomorphically as the weak Boolean product of the family $\{\boldsymbol{F}_{\mathbb{NM}}^{|X|}/\langle U\rangle\}_{U\in \mathrm{BSpec}\,\boldsymbol{F}_{\mathbb{NM}}^{|X|}}$, and then for each $U \in \mathrm{BSpec}\boldsymbol{F}_{\mathbb{NM}}^{|X|}$, we have that the directly indecomposable NM-algebra $\boldsymbol{F}_{\mathbb{NM}}^{|X|}/\langle U\rangle$ is either isomorphic to the connected rotation – if it has the negation fixpoint – or the disconnected rotation – if it lacks the negation fixpoint – of the free Gödel hoop $\boldsymbol{F}_{\mathbb{GH}}^{|X_U|}$, where $X_U = \{x/\langle U\rangle \mid x \in X\,, x/\langle U\rangle > \neg x/\langle U\rangle\} \cup \{\neg x/\langle U\rangle \mid x \in X\,, x/\langle U\rangle < \neg x/\langle U\rangle\}$.

7 Open Problems

With the recent characterization of free BL-algebras (see Section 5.2) all the prominent schematic extensions of BL have their functional representation theorems. In stark contrast with BL, the picture is very incomplete for schematic extensions of MTL which are not extensions of BL. As a matter of fact, in the literature one finds concrete representation theorems of free algebras only for certain subvarieties of $\mathbb{WNM}$, in particular for the most prominent of them, Gödel and NM-algebras, and for some minor ones as NMG- and RDP-algebras. A representation of the free WNM-algebras is at present unavailable, but predicting that this gap is soon to be filled constitutes a safe bet. A common feature of the varieties of MTL having a functional representation is the knowledge of a generic chain for the variety. This knowledge is at present not available for the variety of MTL-algebras, and some of its prominent subvarieties, such as IMTL- and ΠMTL-algebras. The same varieties are still lacking a functional representation theorem for their free algebras. For what concerns extensions/expansions of MTL, the most interesting open problem is the characterization of free PMV-algebras. Such a result will solve the long-standing Birkhoff-Pierce conjecture either in the positive or in the negative.

8 Historical and Bibliographical Remarks

Standard references for Section 1 are the following Universal Algebra texts: [56], [23], [36].

For what concerns Section 2.1, McNaughton originally proved Theorem 2.1.2 in [57], where he used a version of Lemma 2.1.21. McNaughton's proof is not constructive, since at a certain stage it involves an unbounded search. The first constructive proof of Theorem 2.1.2 is found in [61]. See also [29]. This proof uses Lemma 2.1.21 to express Schauder hats formulas. The proof presented in the chapter is taken from Panti's paper [65] on the geometric proof of the completeness of Łukasiewicz calculus. The techniques used in the proof rely on convex geometry [44], on piecewise linear topology [66] and on De Concini-Procesi lemma on the elimination of points of indeterminacy in toric varieties [39] (Lemma 2.1.12 can be thought of as a non-homogeneous version of the De Concini-Procesi result). A different proof is given in [64]: the idea underlying this proof is to use an equivalent version of Lemma 2.1.21 to compute for each domain of linearity D of a given McNaughton function f an MV-term φ_D that in the free algebra evaluates to 1 exactly over D, together with an MV-term ψ_D whose evalutation coincides with f over D. It is then shown that there exists an integer $k(f, D)$ such that the evaluation of $\varphi_D^{k(f,D)}$ goes to 0 outside D with slopes that are sufficiently high to allow to patch together all terms $(\psi_D \& \varphi_D^{k(f,D)})$ with a $\bigvee$-combination which yields the desired function f. This proof can be extended to represent free algebras in the variety $\mathbb{MV}(\mathsf{Q})$ obtained expanding the MV signature with a denumerable set of constants $\{c_\delta\}_{\delta \in [0,1] \cap \mathsf{Q}}$ which in the standard algebra are forced to be interpreted in the corresponding rational numbers in $[0, 1]$ by way of suitable book-keeping axioms. Unsurprisingly, $\boldsymbol{F}^n_{\mathbb{MV}(\mathsf{Q})}$ is the class of continuous piecewise linear functions where each linear piece has the form $p(\mathbf{x}) = b_i + \sum_{i=1}^n a_i x_i$, where each a_i is an integer, while b_i may be rational. In this Chapter we have preferred to expound the proof of McNaugton's Theorem based on regular triangulations and Schauder bases because the notion of normal form as $\oplus$-sum of Schauder hats makes the latter concept the best known approximation to the notion of semantical minterm as an element of the free algebra that, as a function, takes on a prescribed value at a prescribed point, and is the smallest possible elsewhere. Moreover, the notion of Schauder bases constitutes a first step towards the development of the theory of the MV-partitions of unity and the related theory of abstract bases for MV-algebras, which are now lines of research very actively pursued, as witnessed by several chapters of the recent book [63]. See also [54], [53], and [15], [62] for some applications. In particular, the paper [62] gives a representation-free characterization of the free MV-algebras using the notion of MV-partition of unity. The analytic-geometric argument needed to develop the sup-inf combination of truncated hyperplanes expounded after Lemma 2.1.21 can be found in [29] or [3]. As is well known, Mundici's functor Γ (see [29]) shows that MV-algebras are categorically equivalent to Abelian ℓ-groups with a distinguished strong unit. It follows that Theorem 2.1.2 can be seen as the specialisation to unital ℓ-groups of Baker and Beynon results on the representation of finitely generated free Abelian ℓ-groups [17], [18], [55]. Jakubík in [51] shows how to derive free MV-algebras from free Abelian ℓ-groups. Finally, it is worth noting that in the recent paper [43] Dubuc and Poveda provide a categorical duality for

MV-algebras based on suitably defined sheaves of MV-chains, and show how to derive McNaughton's Theorem as a natural consequence of their categorical constructions when they are applied to free algebras. The problem of normal forms for Łukasiewicz logic is strictly connected to the representation of free MV-algebras. See [2], [4], [42] for analysis of normal forms based on truncated hyperplanes and Lemma 2.1.21; see [10] for a normal form for finitely valued Łukasiewicz logics based on Schauder hats. The functional representation of free Wajsberg hoops is found in [1], while the results on finitely valued Łukasiewicz logics can be found in [29]: there, the study of free algebras in the corresponding varieties $\mathbb{MV}_n$ is independent of McNaughton's Theorem. Free algebras for the algebraic semantics of extensions/expansions of Łukasiewicz logics are dealt with in [60], while MV-algebras with divisors are studied in [47].

The main sources for free product algebras and free cancellative hoops are the papers [30], [31]. The functional representation theorem was partially developed in [11], and fully in [34].

Gödel algebras have been studied, under the name L-algebras, in two pioneering papers by Horn [49], [50], which characterized the prime spectrum and computed the recursive formula for the cardinality of free finitely generated algebras. It is interesting to note that the category of finite forests shown in this chapter to be dually equivalent to finitely generated Gödel algebras does not constitute a natural duality in the sense of [35]: as we have shown the product of forests has generally more points than the cartesian product of the underlying sets. A description of finitely generated free Gödel algebras and free Gödel hoops is found in [38]. The functional representation of Gödel algebras is given in [47], [46], [11]. The paper [37] is focused on computing coproducts of finitely generated Gödel algebras via their dual spectral representation as products of the corresponding finite forests. The paper [9] states the 1-set property of Lemma 4.1.6. In this chapter we have followed the section on normal forms of [9] for our presentation of the functional representation theorem. Normal forms for Gödel logic have been studied in [16] and in [13]. In this paper free algebras for some subvarieties of $\mathbb{MTL}$ are represented as algebras of maximal-length antichains over suitably constructed finite posets. Here normal forms are defined as lattice disjunctions of minterms, where a semantical minterm is the minimum (in the pointwise order of the poset) maximal-length antichain containing a prescribed element. Free n-generated algebras are then described as embeddings into products of a sufficiently large collection of n-generated chains in the variety: normal forms are used to construct the required embeddings. These poset representations allow to easily find recursive formulas for the cardinalities of free algebras. The fact that Nilpotent Minimum algebras coincides with prelinear Nelson algebras is proved in [28]. Free Nilpotent Minimum algebras with generator set of any cardinality are studied in [26], and in [13] for the finitely generated case. The first functional representation of free NM-algebras is found in [68], while Theorem 4.6.9 is stated as in [14]. The same paper contains the statement of the $0, 1$-set property of Proposition 4.6.10. The spectral duality for finite NM-algebras is developed in [8]. Poset representation and spectral duality for finite RDP-algebras are studied in [67] and [22].

The first works on free BL-algebras are by Montagna. In [58] he gives the functional representation for the singly generated case, and in [59] he characterizes free $\mathrm{BL}_\triangle$-algebras. Recently, the papers [5], [21] give the first concrete representation of free BL-

algebras. In this chapter we closely follows [5]. The paper [7] introduces spectral duality representations for locally finite subvarieties of $\mathbb{BL}$. The paper [27] describe free $\mathbb{MV}_k$-algebras with generator set of any cardinality in terms of functions from a Stone space into MV-chains. A description of the prime spectrum, which in these algebras coincides with the maximal one, is provided. When the generator set is finite this construction reduces to the spectral duality described in the chapter. Schauder-hat-like normal forms for BL have been developed in [12] and [6].

Pierce representations and weak Boolean products have been applied to determine the structure of free algebras in several subvarieties of $\mathbb{MTL}$ in a series of papers by Cignoli, Torrens and Busaniche [33], [32], [25], [26]. Our presentation is mainly based on [33]. Weak Boolean products are presented as global sections of sheaves of algebras over Boolean spaces in [24]. The universal algebraic background on directly indecomposable bounded integral residuated lattices and the Boolean Factor Congruence property can be found in [45] and [19]. The Di Nola-Lettieri Equation is introduced in [41], [40]. The variety of Nilpotent Minimum algebras without fixpoint is described, along with a classification of the subvarieties of $\mathbb{NM}$, by Gispert in [48].

BIBLIOGRAPHY

[1] Paolo Aglianò and Giovanni Panti. Geometrical methods in Wajsberg hoops. *Journal of Algebra*, 256(2):352–374, 2002.

[2] Stefano Aguzzoli. The complexity of McNaughton functions of one variable. *Advances in Applied Mathematics*, 21(1):58–77, 1998.

[3] Stefano Aguzzoli. *Geometric and Proof-Theoretic Issues in Łukasiewicz Propositional Logics*. PhD thesis, University of Siena, Siena, 1998.

[4] Stefano Aguzzoli. A note on the representation of McNaughton lines by basic literals. *Soft Computing*, 2(3):111–115, 1998.

[5] Stefano Aguzzoli and Simone Bova. The free n-generated BL-algebra. *Annals of Pure and Applied Logic*, 161(9):1144–1170, 2010.

[6] Stefano Aguzzoli and Simone Bova. Schauder hats for the two-variable fragment of BL. In *Proceedings of the International Symposium on Multiple-Valued Logic*, pages 27–32, Los Alamitos, CA, 2010. IEEE Computer Society Press.

[7] Stefano Aguzzoli, Simone Bova, and Vincenzo Marra. Applications of finite duality to locally finite varieties of BL-algebras. In *Proceedings of LFCS 2009, Symposium on Logical Foundations of Computer Science*, volume 5407/2009 of *Lecture Notes in Computer Science*, pages 1–15, 2009.

[8] Stefano Aguzzoli, Manuela Busaniche, and Vincenzo Marra. Spectral duality for finitely generated Nilpotent Minimum algebras, with applications. *Journal of Logic and Computation*, 17(4):749–765, 2007.

[9] Stefano Aguzzoli, Ottavio D'Antona, and Vincenzo Marra. Computing minimal axiomatisations in Gödel propositional logic. To appear in *Jornal of Logic and Computation*, doi:10.1093/logcom/exp054.

[10] Stefano Aguzzoli, Ottavio D'Antona, and Vincenzo Marra. Brun normal forms for co-atomic Łukasiewicz logics. In *ECSQARU*, pages 650–661, 2005.

[11] Stefano Aguzzoli and Brunella Gerla. Finite-valued reductions of infinite-valued logics. *Archive for Mathematical Logic*, 41(4):361–399, 2002.

[12] Stefano Aguzzoli and Brunella Gerla. Normal forms for the one-variable fragment of Hájek's Basic Logic. In *Proceedings of the International Symposium on Multiple-Valued Logic*, pages 248–289, Los Alamitos, CA, 2005. IEEE Computer Society Press.

[13] Stefano Aguzzoli and Brunella Gerla. Normal forms and free algebras for some extensions of MTL. *Fuzzy Sets and Systems*, 159(10):1131–1152, 2008.

[14] Stefano Aguzzoli and Brunella Gerla. Probability measures in the logic of Nilpotent Minimum. *Studia Logica*, 94(2):151–176, 2010.

[15] Stefano Aguzzoli and Vincenzo Marra. Finitely presented MV-algebras with finite automorphism group. *J. Logic Comput.*, 20:811–822, 2010.

[16] Matthias Baaz and Helmut Veith. Interpolation in fuzzy logic. *Archive for Mathematical Logic*, 38(7):461–489, 1999.

[17] Kirby A. Baker. Free vector lattices. *Canadian Journal of Mathematics*, 20:58–66, 1968.

[18] W. Meursig Beynon. Applications of duality in the theory of finitely generated lattice-ordered Abelian groups. *Canadian Journal of Mathematics*, 29(2):243–254, 1977.

[19] David Bigelow and Stanley Burris. Boolean algebras of factor congruences. *Acta Scientiarum Mathematicarum*, 54(1–2):11–20, 1990.

[20] Willem J. Blok and Don L. Pigozzi. *Algebraizable Logics*, volume 396 of *Memoirs of the American Mathematical Society*. American Mathematical Society, Providence, RI, 1989.

[21] Simone Bova. *BL-functions and Free BL-Algebra*. PhD thesis, University of Siena, Siena, 2008.

[22] Simone Bova and Diego Valota. Spectral duality for finite RDP-algebras. To appear in *Journal of Logic and Computation*.

[23] Stanley Burris and H.P. Sankappanavar. *A Course in Universal Algebra*, volume 78 of *Graduate Texts in Mathematics*. Springer-Verlag, 1981.

[24] Stanley Burris and Heinrich Werner. Sheaf constructions and their elementary properties. *Transactions of the American Mathematical Society*, 248(2):269–309, 1979.

[25] Manuela Busaniche. Free algebras in varieties of BL-algebras generated by a chain. *Algebra Universalis*, 50(3–4):259–277, 2003.

[26] Manuela Busaniche. Free Nilpotent Minimum algebras. *Mathematical Logic Quarterly*, 52(3):219–236, 2006.

[27] Manuela Busaniche and Roberto Cignoli. Free MV_n-algebras. *Algebra Universalis*, 58(3):335–340, 2008.

[28] Manuela Busaniche and Roberto Cignoli. Constructive logic with strong negation as a substructural logic. *J. Logic Comput.*, 20(4):761–793, 2010.

[29] Roberto Cignoli, Itala M.L. D'Ottaviano, and Daniele Mundici. *Algebraic Foundations of Many-Valued Reasoning*, volume 7 of *Trends in Logic*. Kluwer, Dordrecht, 1999.

[30] Roberto Cignoli and Antoni Torrens. An algebraic analysis of product logic. *Multiple-Valued Logic*, 5(1):45–65, 2000.

[31] Roberto Cignoli and Antoni Torrens. Free cancellative hoops. *Algebra Universalis*, 43(2–3):213–216, 2000.

[32] Roberto Cignoli and Antoni Torrens. Free algebras in varieties of BL-algebras with a boolean retract. *Algebra Universalis*, 48(1):55–79, 2002.

[33] Roberto Cignoli and Antoni Torrens. Free algebras in varieties of Glivenko MTL-algebras satisfying the equation $2(x^2) = (2x)^2$. *Studia Logica*, 83(1–3):157–181, 2006.

[34] Petr Cintula and Brunella Gerla. Semi-normal forms and functional representation of product fuzzy logic. *Fuzzy Sets and Systems*, 143(1):89–110, 2004.

[35] David M. Clark and Brian A. Davey. *Natural dualities for the working algebraist*, volume 57 of *Cambridge Studies in Advanced Mathematics*. Cambridge University Press, Cambridge, 1998.

[36] Paul Moritz Cohn. *Universal Algebra*, volume 6 of *Mathematics and its Applications*. D. Reidel, Dordrecht, second edition, 1981.

[37] Ottavio D'Antona and Vincenzo Marra. Computing coproducts of finitely presented Gödel algebras. *Annals of Pure and Applied Logic*, 142(1–3):202–211, 2006.

[38] Brian A. Davey. Dualities for equational classes of Browerian algebras and Heyting algebras. *Transactions of the American Mathematical Society*, 221(1):119–146, 1976.

[39] C. De Concini and C. Procesi. Complete symmetric varieties II. In R. Hotta, editor, *Algebraic groups and related topics*, volume 6 of *Advances Studies in Pure Mathematics*, pages 481–513. Kinokuniya and North-Holland, 1985.

[40] Antonio Di Nola and Ada Lettieri. Perfect MV-algebras are categorically equivalent to Abelian ℓ-groups. *Studia Logica*, 53(3):417–432, 1994.

[41] Antonio Di Nola and Ada Lettieri. Equational characterization of all varieties of MV-algebras. *Journal of Algebra*, 221(2):463–474, 1999.

[42] Antonio Di Nola and Ada Lettieri. On normal forms in Łukasiewicz logic. *Archive for Mathematical Logic*, 43(6):795–823, 2004.

[43] Eduardo J. Dubuc and Yuri A. Poveda. Representation theory of MV-algebras. *Ann. Pure Appl. Logic*, 161(8):1024–1046, 2010.

[44] Günter Ewald. *Combinatorial Convexity and Algebraic Geometry*, volume 168 of *Graduated Texts in Mathematics*. Springer-Verlag, 1996.

[45] Nikolaos Galatos, Peter Jipsen, Tomasz Kowalski, and Hiroakira Ono. *Residuated Lattices: An Algebraic Glimpse at Substructural Logics*, volume 151 of *Studies in Logic and the Foundations of Mathematics*. Elsevier, Amsterdam, 2007.

[46] Brunella Gerla. A note on functions associated with Gödel formulas. *Soft Computing*, 4(4):206–209, 2000.

[47] Brunella Gerla. *Many-Valued Logics of Continuous T-Norms and Their Functional Representation*. PhD thesis, University of Milano, Milano, 2002.

[48] Joan Gispert. Axiomatic extensions of the nilpotent minimum logic. *Reports on Mathematical Logic*, 37:113–123, 2003.

[49] Alfred Horn. Free L-algebras. *Journal of Symbolic Logic*, 34(3):475–480, 1969.

[50] Alfred Horn. Logic with truth values in a linearly ordered Heyting algebras. *Journal of Symbolic Logic*, 34(3):395–408, 1969.

[51] Ján Jakubík. On free MV-algebras. *Czechoslovak Mathematical Journal*, 53:311–317, 2003.

[52] Heldermann Keimel. Some trends in lattice-ordered groups and rings. In K. A. Baker and R. Wille, editors, *Lattice Theory and its Applications. In Celebration of Garrett Birkhoff's 80th Birthday.*, volume 23 of *Research and Exposition in Mathematics*, pages 131–161. Heldermann, Darmstadt, 1995.

[53] Corrado Manara, Vincenzo Marra, and Daniele Mundici. Lattice-ordered abelian groups and Schauder bases of unimodular fans. *Trans. Amer. Math. Soc.*, 359(4):1593–1604, 2007.

[54] Vincenzo Marra. *Non-Boolean partitions. A mathematical investigation through lattice-ordered Abelian groups and MV-algebras.* PhD thesis, University of Milano, Milano, 2003.

[55] Vincenzo Marra and Daniele Mundici. Łukasiewicz logic and Chang's MV algebras in action. In Vincent F. Hendricks and Jacek Malinowski, editors, *Trends in Logic: 50 Years of Studia Logica*, volume 21 of *Trends in Logic*, pages 145–192. Kluwer Academic Publishers, Dordrecht, 2003.

[56] Ralph N. McKenzie, George F. McNulty, and Walter F. Taylor. *Algebras, Lattices, Varieties. Vol. I.* The Wadsworth & Brooks/Cole Mathematics Series. Wadsworth & Brooks/Cole Advanced Books & Software, Monterey, CA, 1987.

[57] Robert McNaughton. A theorem about infinite-valued sentential logic. *Journal of Symbolic Logic*, 16(1):1–13, 1951.

[58] Franco Montagna. The free BL-algebra on one generator. *Neural Network World*, 10(5):837–844, 2000.

[59] Franco Montagna. Free $\mathrm{BL}_{\triangle}$ algebras. In Antonio Di Nola and Giangiacomo Gerla, editors, *Lectures on Soft Computing*, Advences in Soft Computing, pages 159–171. Springer, 2001.

[60] Franco Montagna and Giovanni Panti. Adding structure to MV-algebras. *Journal of Pure and Applied Algebra*, 164(3):365–387, 2001.

[61] Daniele Mundici. A constructive proof of McNaughton's Theorem in infinite-valued logics. *Journal of Symbolic Logic*, 59(2):596–602, 1994.

[62] Daniele Mundici. A characterization of the free n-generated MV-algebra. *Arch. Math. Logic*, 45(2):239–247, 2006.

[63] Daniele Mundici. *Advanced Łukasiewicz calculus and MV-algebras*, volume 35 of *Trends in Logics*. Springer, 2011.

[64] Vilém Novák, Irina Perfilieva, and Jiří Močkoř. *Mathematical Principles of Fuzzy Logic*. Kluwer, Dordrecht, 2000.

[65] Giovanni Panti. A geometric proof of the completeness of the Łukasiewicz calculus. *Journal of Symbolic Logic*, 60(2):563–578, 1995.

[66] Colin Patrick Rourke and Brian Joseph Sanderson. *Introduction to Piecewise-Linear Topology*. Springer, 1972.

[67] Diego Valota. Poset representation for free RDP-algebras. In Hykel Hosni and Franco Montagna, editors, *Probability, Uncertainty and Rationality*, pages 197–218. Edizioni della Normale, 2010.

[68] San-Min Wang, Bao-Shu Wang, and Xiang-Yun Wang. A characterization of truth-functions in the Nilpotent Minimum logic. *Fuzzy Sets and Systems*, 145(2):253–266, 2004.